Electoral Reform Without Representation: Voting-Rights Enforcement and Racial Discipline Gaps in California Schools

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Abstract

When voting-rights enforcement reshapes who governs, do the disparities it targets narrow? I study California school districts compelled by the California Voting Rights Act to replace at-large school board elections with single-member districts, estimating effects on racial discipline gaps with heterogeneity-robust difference-in-differences methods (222 treated and 499 comparison districts, 2011–2024). Reform did not narrow the Black–White suspension gap: the effect on the share of Black students suspended relative to White students is an informative null (+0.33 percentage points, $p = .41$) that rules out reductions larger than 0.45 points—about nine percent of the pre-treatment gap—and the gap does not close under any estimator or measurement basis. The most likely explanation is an asymmetric first stage: reform raised Hispanic board representation—the raw Hispanic share rose 9.3 percentage points—while Black representation did not change. Where enforcement empowers a different minority than the one bearing a targeted disparity, electoral reform alone need not reduce it: minority representation is not fungible across groups.

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1 Introduction

Over the past two decades, more than two hundred California school districts have been compelled to abandon at-large elections in favor of single-member district (SMD) systems for their governing boards. This wave of electoral reform was driven by the California Voting Rights Act (CVRA) of 2001, which enables legal challenges to at-large elections that dilute minority voting power. Because at-large elections require candidates to win jurisdiction-wide majorities, they can systematically disadvantage geographically concentrated minority communities. SMD elections, by contrast, create smaller electoral districts in which minority voters may constitute majorities, thereby increasing the likelihood that minority-preferred candidates win seats. The CVRA reform wave thus provides a rare opportunity to study whether changes in the racial composition of elected officials translate into changes in the policies and outcomes those officials oversee.

The theoretical prediction is straightforward. A large literature on descriptive representation argues that elected officials who share the racial or ethnic background of their constituents can be more likely to advance those constituents' interests, especially where group trust is low or interests are uncrystallized (Mansbridge 1999; Meier and England 1984); the foundational distinction between such descriptive representation and substantive representation is due to Pitkin (1967). In education specifically, minority school board members have been linked to more equitable hiring practices, reduced racial disparities in discipline, and more favorable resource allocation to minority-serving schools (Meier, Stewart, and England 1989; Leal, Martinez-Ebers, and Meier 2004; Meier and Rutherford 2016; Fischer 2023; Kogan, Lavertu, and Peskowitz 2021). Racial disparities in school discipline—particularly the disproportionate suspension of Black students—are among the most persistent and consequential inequities in American education (Skiba et al. 2002, 2011; Morris and Perry 2016). If SMD elections increase minority representation, and minority representatives reduce discipline disparities, then electoral reform should narrow racial gaps in suspensions.

This paper tests that prediction and finds that it fails—and the manner of the failure, not the failure itself, is what the paper explains.

Using a staggered difference-in-differences design with 222 treated and 499 control school districts observed from 2011 to 2024, I find that SMD adoption does not narrow the Black–White suspension gap. My primary outcome is the Black–White gap in the share of *distinct* students suspended—whether a smaller share of Black students is disciplined relative to White students. The effect on this gap is statistically indistinguishable from zero and remains so across estimators and design restrictions (Sun–Abraham +0.33 pp, SE = 0.40, $p = .41$, $n = 4{,}837$). The result is an informative null, not an underpowered one: the 95% interval $[-0.45, +1.11]$ rules out reductions larger than 0.45 pp—about 9.4% of the 4.8 pp pre-treatment gap—and no relaxation of the parallel-trends assumption produces evidence of benefit. Narrowing larger than the stated equivalence and minimum-detectable-effect margins can be ruled out under the maintained design.

The gap does not close under any measurement. Counted by suspension *incidents* rather than distinct students, the estimated Black–White gap is marginally wider (Sun–Abraham +1.61 pp, SE = 0.88, $p = .069$); this incident-based estimate is not robust across estimators—it is null under Callaway–Sant'Anna, generalized synthetic control, and synthetic DiD—so I read it as suggestive. To the extent it reflects a real pattern, the divergence between the two measures is consistent with more frequent suspension of already-disciplined Black students rather than a larger share of Black students suspended, driven by a marginal rise in Black suspension rates (+1.76 pp, $p = .10$) with flat White rates (−0.46 pp, n.s.). Overall suspension rates decline modestly in unadjusted estimates (−0.79 pp, $p = .04$; BY $q = .21$), while Hispanic–White gaps show no robust change; the targeted Black–White disparity is untouched even as discipline eases on average.

These results present a puzzle. Why would an electoral reform meant to empower minority communities fail to narrow—and, measured by incidents, appear to widen—the Black–White discipline gap? The puzzle is sharpened by recent evidence that school board incumbents face no electoral accountability for minority student outcomes—only White student performance predicts reelection (Flavin and Hartney 2017). If voters do not punish boards for racial disparities, then changing who sits on the board is the primary lever for change, and the specific racial composition of that change becomes the binding constraint.

I investigate this puzzle by tracing what the reform actually changed. The central institutional fact is that the CVRA reform wave in California primarily increased Hispanic, not Black, board representation. Among elected school board members in treated districts, the Hispanic share rose from 27.3% before SMD adoption to 36.7% afterward—an increase of 9.3 percentage points. The Black share, by contrast, declined slightly, from 6.6% to 6.3%. In the common first-stage estimation sample, the Sun–Abraham estimate is directionally consistent with this asymmetry but imprecise: Hispanic board share rises +4.6 pp (SE = 2.8; $p = .11$), Black share is flat (-0.5 pp; SE = 0.8; $p = .55$), and the formal Hispanic-minus-Black contrast is +5.1 pp under Sun–Abraham ($p = .12$; cohort-stratified bootstrap 95% CI $[-1.1, +11.5]$) and +4.3 pp under TWFE ($p = .04$)—the same significant-under-TWFE, imprecise-under-SA pattern as the levels. This pattern reflects California's demography: Latino communities are the primary beneficiaries of CVRA enforcement because they are the largest minority group with sufficient residential concentration to form SMD majorities in most districts. Within the heterogeneity-analysis sample (treated districts with valid BW-gap data and post-treatment board elections), only 18 districts gained a predicted-Black board member after switching to SMD elections, while 135 gained a predicted-Hispanic but not Black member.

Exploratory moderation analysis is consistent with a race-match interpretation, but this evidence is post-treatment, sensitive to race-prediction and multiple-testing choices, and—as Section 5 shows—not corroborated by a predetermined electoral-capacity check. The treatment's association with the *incident* gap is attenuated where the cumulative Black board share is higher (interaction -28.8; $p = .026$, though not robust to multiple testing) and, more durably, where the Black teacher share is higher. I read these patterns as suggestive, not causal.

A parallel pattern appears in the teacher workforce. SMD adoption increases minority teacher shares by 2.34 percentage points (Sun-Abraham; $p = .002$), with the gain concentrated among Hispanic teachers (+1.88 pp; $p < .001$). Black teacher representation increases modestly under TWFE (+0.21 pp; $p = .019$). In exploratory interaction models, the incident Black-White suspension-gap association is substantially smaller where Black teacher share is higher (−85.1 pp per unit share, i.e., a 1 percentage-point higher Black teacher share is associated with a 0.85 pp smaller treatment association; $p < .001$; at a one-year lag, −49.3; $p = .019$). Hispanic teachers have no significant association with Black-White gaps (+0.55; $p = .89$), consistent with the race-specificity of the pattern.

These findings make three contributions. First, I provide an *informative null* on the central outcome, made interpretable by a measurement decomposition: the share of distinct Black students suspended does not move while suspension incidents are marginally higher, and a direct duplication-ratio test is positive but statistically indistinguishable from zero. The pattern is therefore consistent with, but does not establish, more frequent discipline among already-disciplined students rather than a wider incidence of suspension. This is, to my knowledge, the first heterogeneity-robust evidence linking CVRA reform to school discipline, where racial disparities are among the largest and most consequential for affected students' life trajectories (Skiba et al. 2011; Morris and Perry 2016). Second, I document an *asymmetric first stage*: SMD reform raises minority board and teacher representation, but the gain accrues to Hispanic, not Black, constituencies (a pattern unambiguous in the raw composition and the teacher first stage, though the formal board-share contrast is imprecise under the preferred estimator)—Latinos being the group with the residential concentration to anchor trustee-area majorities in most California districts. This extends Abott and Magazinnik (2020), whose conditional finding—Latino representation gains arise where Latino voters are numerous and residentially concentrated enough to anchor trustee areas—is the same demographic-capacity logic I apply to Black representation; I document the downstream consequences over a longer panel covering the post-2016 adoption wave.

Third, as an exploratory coda, I offer a *race-match account paired with influence dilution* as a candidate boundary condition for theories of descriptive representation: the prediction that minority representation narrows disparities appears to hold only where the representative's race matches the group bearing the disparity, and voting-rights enforcement, by empowering whichever minority can form a trustee-area majority, can leave a smaller or more dispersed minority—here, Black voters—with no greater influence than before. Because this rests on endogenous post-treatment composition and is not corroborated by a predetermined check, I read it as suggestive, not causally identified. Where the group an enforcement regime empowers differs from the group bearing the disparity it targets, electoral reform alone may not reduce racial inequities.

The remainder of the paper proceeds as follows. Section 2 reviews the CVRA, the literature on electoral institutions and representation, and the theory of race-matched descriptive representation. Section 3 describes the data sources, sample construction, and identification strategy. Section 4 presents the main results, event studies, and robustness checks. Section 5 presents exploratory moderation evidence on board composition and teacher diversity. Section 6 discusses implications and limitations.

2 Background and Theory

2.1 The California Voting Rights Act

The California Voting Rights Act of 2001 (CVRA; California Elections Code §14025–14032) prohibits the use of at-large election methods when such methods impair the ability of members of a protected class to elect candidates of their choice or to influence the outcome of an election. Unlike Section 2 of the federal Voting Rights Act, the CVRA does not require plaintiffs to demonstrate that a compact, geographically defined minority community could constitute a majority in a hypothetical single-member district (Collingwood and Long 2021).

This lower evidentiary standard makes CVRA challenges substantially easier to mount than federal challenges.

A "safe harbor" provision added to the Elections Code by AB 350 in 2016 (effective January 1, 2017; Section 10010) further accelerated transitions. Under the safe harbor, a jurisdiction that passes a resolution of intent to transition to district-based elections triggers a 90-day protected window (extendable by mutual agreement) during which the jurisdiction can adopt a redistricting plan. If the jurisdiction completes the transition within this window, its reimbursement obligation to the prospective plaintiff is capped at approximately $30,000 (adjusted annually for inflation). By contrast, jurisdictions that resist and lose a contested CVRA lawsuit face attorneys' fees that routinely exceed $1 million. This asymmetry created strong incentives for districts to settle voluntarily upon receiving demand letters rather than risk litigation.

The litigation process follows a characteristic pattern. A small number of law firms specializing in CVRA enforcement identify target jurisdictions based on demographic analysis—specifically, whether a jurisdiction has sufficient residential segregation and Latino citizen voting-age population to support a viable SMD plan. The firm then sends a demand letter to the jurisdiction's governing body, notifying it of the legal vulnerability and offering to negotiate a transition to district-based elections. Faced with the near-certainty of losing a CVRA challenge and the prospect of paying the plaintiff's legal fees, the vast majority of jurisdictions settle by adopting SMD elections (Abott and Magazinnik 2020; Collingwood and Long 2021).

For causal identification, the critical feature of CVRA enforcement is that the *timing* of legal challenges is driven primarily by the capacity constraints and strategic choices of a small number of plaintiff attorneys, not by changes in the targeted jurisdictions' policies or demographic trajectories. While the *selection* of targets is related to district demographics—larger, more diverse districts with greater residential segregation are more likely to be targeted—the *sequencing* of challenges within this eligible pool is plausibly exogenous to school discipline outcomes. Some demographically similar districts were challenged in 2012 while others were not challenged until 2018 or later, a pattern driven by lawyers' caseload capacity rather than changes in the districts themselves.

2.2 Electoral Structure and Descriptive Representation

A foundational literature documents that at-large elections systematically reduce minority representation relative to single-member district systems (Engstrom and McDonald 1981; Davidson and Grofman 1994; Trebbi, Aghion, and Alesina 2008). The mechanism is well understood: at-large elections require jurisdiction-wide majorities, so under racially polarized voting, minority-preferred candidates are consistently outvoted. SMD elections create smaller electoral units in which minority communities can constitute majorities, enabling the election of minority-preferred candidates.

Abott and Magazinnik (2020) provide the most directly relevant evidence, studying the same CVRA-induced transitions I examine here. Using data on California school boards and city councils from 2002 to 2016, they study Latino officeholding and find that CVRA-induced transitions can increase Latino representation, with effects concentrated in larger and more residentially segregated jurisdictions. Trounstine and Valdini (2008) similarly show that SMD systems are associated with greater diversity on city councils, though the effects vary by group and context. Marschall, Ruhil, and Shah (2010) demonstrate that electoral structure interacts with minority population size: SMD elections increase Black representation primarily where Black populations are sufficiently large and geographically concentrated.

These findings are central to the present paper's argument. If SMD elections disproportionately increase Hispanic representation, consistent with the Latino-representation gains Abott and Magazinnik (2020) document, then the downstream effects on Black-specific outcomes may differ from the effects on Hispanic-specific outcomes.

2.3 Descriptive Representation and Policy Outcomes

Descriptive representation, which Pitkin (1967) distinguished from substantive representation, is theorized to make officials who share constituents' demographic characteristics more likely to understand and prioritize their concerns (Mansbridge 1999). In education governance, Meier and England (1984); Meier, Stewart, and England (1989) show that Black school board representation is associated with more Black teachers, fewer Black students in special education, and lower Black suspension rates—with effects interpreted through a representative-bureaucracy mechanism (hiring minority administrators and teachers who exercise discretion more equitably). Kogan, Lavertu, and Peskowitz (2021) provide modern causal evidence (from a close-election regression discontinuity) that increased minority board representation raises minority students' achievement by approximately 0.1 SD.

A critical question is why at-large boards tolerate racial disparities in the first place. Flavin and Hartney (2017) find that school board incumbents in California are electorally rewarded for White student achievement but face no accountability for Black student outcomes (formal equality test $p = .03$). Even in majority-minority districts, only White outcomes drive incumbent reelection. This structural accountability vacuum means that absent descriptive representation, there is no electoral mechanism to translate racial equity concerns into board action. Payson (2017) documents a compounding factor: performance accountability is limited to higher-turnout presidential-year elections, while the off-cycle elections that characterize most school board races are dominated by organized interests.

Fischer (2023) provides the most direct causal evidence for race-specific representation effects. Using a ballot-order instrument in California school board elections, he shows that electing a Hispanic board member causally increases per-pupil spending at above-median-Hispanic schools (approximately $93 per student), while math achievement rises district-wide ($\approx +0.04$ student-level SD) rather than specifically at Hispanic-serving schools. The spending effect is race-specific—Hispanic board members direct resources toward the schools serving their constituents—directly prefiguring the race-match pattern I document in discipline.

The teacher race-match literature provides complementary evidence. Dee (2005) finds that assignment to a non-own-race teacher raises the odds that a student is perceived as disruptive or inattentive; Gershenson et al. (2022) show that Black students assigned at least one Black teacher in grades K–3 are 9 percentage points (13%) more likely to graduate high school, with effects concentrated among Black males; and Lindsay and Hart (2017) finds that Black students exposed to same-race teachers are significantly less likely to receive exclusionary discipline.

2.4 The Race-Match Condition

The existing literature on descriptive representation typically treats "minority" representation as a single construct. Studies examine whether minority board members reduce discipline disparities for minority students, or whether diverse teacher workforces improve outcomes for students of color broadly. This framing implicitly assumes a degree of cross-racial solidarity: that a Hispanic board member will advocate for Black students, or that an increase in the overall minority share of the teaching force will benefit Black students specifically.

The condition this paper tests—what I call the *race-match condition*—holds that the beneficial effects of descriptive representation are specific to the racial group represented. A Hispanic board member may prioritize the concerns of Hispanic families—including bilingual education, immigration-related anxieties, and culturally responsive pedagogy—without necessarily attending to the distinct concerns of Black families, such as racial profiling in discipline, culturally biased behavioral expectations, and the school-to-prison pipeline. The race-match condition does not require racial animosity or indifference; it simply reflects the fact that different racial groups face different challenges, and elected officials are most effective advocates for the specific communities from which they are drawn.

Existing evidence supports this framing. Fischer's (2023) finding that Hispanic board members increase spending at Hispanic-serving schools but not at other schools is a direct empirical precedent for race-specific representation effects in the spending domain. The accountability evidence in Flavin and Hartney (2017) provides a mechanism: under SMD elections, Hispanic board members face electoral accountability from Hispanic voters who may now constitute the median voter in their trustee area, but they have no more electoral incentive to attend to Black student outcomes than the White incumbents they replaced. The accountability deficit that Flavin and Hartney document in California school board elections is thus, on my interpretation, not eliminated by SMD reform—it is relocated, persisting for whichever racial groups remain too small or too dispersed to anchor a trustee-area majority.

The race-match condition generates a prediction for the present setting. If SMD elections increase Hispanic representation (as Abott and Magazinnik 2020 show for Latino candidates), then effects on discipline could appear in Hispanic-specific outcomes—though this prediction is conditional on there being a pre-existing Hispanic-White gap large enough to reduce. In the data, the baseline HW gap is close to zero (-0.10 pp in the full sample), which may limit the scope for improvement regardless of representation gains. The Black-White gap, by contrast, may simply be left unaffected, since the reform need not change who advocates for Black students; I find no evidence of the resource diversion that could in principle widen it. The average effect across all districts will reflect the composition of treatment effects: a weighted average of any (gap-reducing) effects in districts that gained Black representation and the (null) effects in districts that gained only Hispanic representation. If the latter group vastly outnumbers the former—as California's demography ensures—then the average effect will be dominated by null effects on the Black-White gap.

3 Data and Research Design

3.1 Data Sources

I combine data from four primary sources. *Election data* come from the California Elections Data Archive (CEDA), which provides candidate-level information on school board elections from 1996 through 2023, including candidate names, vote totals, incumbency status, and whether each election was conducted at-large or by trustee area (California Elections Data Archive 2024). The trustee area field is the basis for identifying SMD elections.

Discipline data come from the California Department of Education (CDE), which publishes annual district-level counts of suspensions and expulsions disaggregated by race/ethnicity for the 2010–11 through 2023–24 academic years (California Department of Education 2024). For each race I construct two rates per 100 enrolled students: the *unduplicated* count of distinct students suspended (my primary measure) and the total count of suspension *incidents* (a secondary, event-count measure that can exceed the distinct-student count when students are suspended more than once). From these I form the Black-White and Hispanic-White suspension-rate gaps, each in percentage points; the *primary outcome is the unduplicated Black-White gap*, with the incident-based gap reported as a measurement decomposition.

The discipline outcomes are observed against a statewide reform backdrop. During the panel, California repeatedly narrowed the use of exclusionary discipline and changed the school-finance environment: AB 1729 (2012) clarified and expanded age-appropriate alternatives and other means of correction before suspension or expulsion; AB 420 (2014) eliminated expulsion recommendations, and K–3 suspensions, for willful defiance; SB 419 (2019) extended willful-defiance suspension restrictions to grades 4–5 and, through 2024–25, grades 6–8; and the 2013–14 Local Control Funding Formula (LCFF) replaced much of the prior K–12 finance system with grade-span base grants plus supplemental and concentration grants tied to high-need pupils.[1] These statewide changes are absorbed by year fixed effects in the DiD design, but they matter for interpretation: they likely contribute to the secular decline in exclusionary discipline over the panel, leaving the central question as whether CVRA-induced governance changes narrowed racial gaps beyond that statewide policy environment.

[1]AB 1729, 2011–2012 Reg. Sess. (Cal. 2012); AB 420, 2013–2014 Reg. Sess. (Cal. 2014); SB 419, 2019–2020 Reg. Sess. (Cal. 2019); California Department of Education, Local Control Funding Formula overview.

Teacher workforce data also come from the CDE staff files, available for 2011–12 through 2016–17 (StaffDemo, individual-level) and 2021–22 through 2023–24 (Staff by Race/Ethnicity, aggregate), with a gap around 2018–2020 when CDE transitioned staff-data products. I calculate the share of teachers in each district who are Black, Hispanic, and minority (non-White) in each available year.

Validation data come from the Lawyers' Committee for Civil Rights of the San Francisco Bay Area (LCCR), which maintains a database of 102 California school districts that transitioned to trustee-area elections through November 2013 (Lawyers' Committee for Civil Rights of the San Francisco Bay Area 2014). I use this database to validate and correct the CEDA-based treatment identification.

To predict the race/ethnicity of school board candidates—who are not asked to self-report race—I apply Bayesian Improved Surname Geocoding (BISG) using the `wru` package (Imai and Khanna 2016). BISG combines surname-based race probabilities with county-level demographic data from the Census to produce posterior probabilities of each candidate belonging to each racial/ethnic group. BISG predictions are available for all candidates in two tiers: candidates whose surnames match the Census surname list receive full surname-plus-geography predictions, while the remainder receive county-distribution-only predictions with no individual surname signal. The first, full-BISG tier covers 90% of board candidates and 91% of elected members; the remaining roughly 10% rely on the county prior alone. The county-only fallback biases measured Black and Hispanic shares toward the county-majority race in low-diversity counties; in these data, no county-only candidate is classified as Black. To the extent this misclassification is non-differential—uncorrelated with district discipline outcomes and treatment timing—it understates rather than overstates measured Black representation and so works against, not toward, the race-match moderation; because the county-only candidates cluster in low-diversity counties that may differ systematically, differential misclassification could in principle bias the interaction in either direction. Because these predictions feed the first-stage and board-composition moderators, the resulting estimates should nonetheless be read with this measurement limitation in mind.

3.2 Treatment Identification

Treatment—the transition from at-large to SMD elections—is identified from the CEDA `area` field. A district-year observation is coded as "SMD" if any election in that district in that year includes a non-missing trustee area identifier. The first year in which a district holds an SMD election is coded as the treatment year.[2] Districts that always used at-large elections during the sample period serve as controls; districts that always used SMD elections are excluded.

I validate this identification against the LCCR database, which provides an independent record of CVRA-induced transitions. Of the 102 LCCR records, I match 94 unique districts to CEDA within county. The validation flags a set of districts as correction candidates not cleanly captured by the CEDA area field; after adjudication, one in-window correction is incorporated (Sulphur Springs Union Elementary), bringing the full-CEDA treated count from 244 to 245, while the remaining candidates are dispositioned out under explicit rules (13 as undatable and 2 as pre-window) rather than silently substituted. After merging with CDE discipline data and applying the analysis-window and validity requirements, the discipline panel contains 222 treated districts.

A residual concern is whether LCCR-confirmed adopters that the CEDA area field misses enter the never-treated control pool. The disposition rules drop such cases rather than reclassifying them: the 13 undatable and 2 pre-window LCCR districts are excluded from the panel, not retained as controls. A direct check against the analysis panel confirms this, finding at most two LCCR-flagged districts (Lakeside Union Elementary and Pioneer Union Elementary) remaining among the controls; dropping them leaves the primary estimate

[2] The code maps election years to discipline years by their shared calendar-year label. CDE discipline files are keyed to the first calendar year of the academic year (`academic_year` is parsed as its first four digits), so an SMD election in calendar year Y is joined to the CDE district-year labeled Y, i.e., the Y–$Y+1$ school year. Section 4.4 reports timing-shift checks that bound alternative adoption-versus-election-year conventions.

unchanged (+0.34 vs. +0.33). Control-pool contamination by missed adopters is therefore negligible.

The treatment rule codes a district as SMD from its first election carrying a trustee-area identifier, so a stray single-election area value could create false positives. As a sensitivity check, 65 of 221 treated districts (29%) carry a trustee-area identifier in only one election. Restricting the treated set to districts whose trustee-area coding persists across at least two elections—a stricter rule that also removes genuine recent switchers with only one observed post-transition election—yields a primary ATT of +0.70 ($p = .07$), if anything a wider gap. This check does not isolate false positives from cohort-composition changes, but it indicates that single-election coding cases are not generating the absence of a narrowing.

As a further external check on treatment timing, I compare my CEDA-derived first-SMD years to those independently coded by Abott and Magazinnik (2020). The primary null is robust to this recoding: re-estimating the primary unduplicated gap on the overlapping districts using A&M's transition dates yields +0.36 ($p = .35$), unchanged from the +0.34 obtained with my own dates on the same sample. Raw agreement between the two codings is nonetheless modest—52% within ± 1 year for the 84 districts dated in both sources—with some disagreement plausibly reflecting adoption-versus-election-year timing conventions (14 of the 40 date disagreements are exactly two years). Two caveats bound this check: A&M's panel ends in 2016, so the 62% of my treated districts that adopt in the 2018–2022 cohorts lie outside their validation window and rest on the CEDA coding (validated against LCCR through 2013); and the modest concordance means individual transition years are measured with error. That the null is invariant to substituting A&M's dates where they differ indicates it is not an artifact of treatment misdating for the cohorts A&M covers; for the later cohorts that A&M cannot check, accurate timing rests on the CEDA coding and remains a measurement limitation I cannot fully rule out.

The discipline-analysis panel contains 222 treated and 499 control school districts. Treated districts adopt SMD between 2011 and 2023 (Figure 1); the largest cohorts are 2022

(51 districts, 23%), 2020 (47, 21%), and 2018 (39, 18%), and the bulk of the identifying variation comes from the 2016–2022 CVRA-era cohorts. The remaining treated districts fall in edge cohorts with limited estimation windows.[3] Board-composition analyses (Section 5) use the CEDA-linked election panel, a subset of 220 treated and 386 control districts with matched pre- and post-treatment board elections; the heterogeneity tables in Appendix E draw from a further-restricted subsample of treated districts with both valid BW-gap data and post-treatment elections.

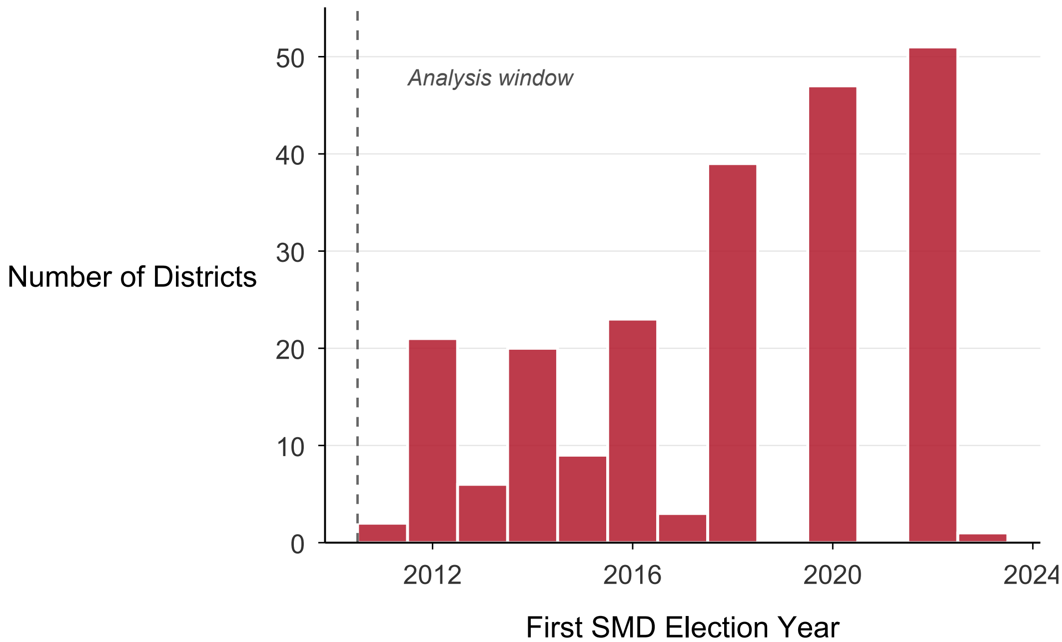

Figure 1: Treatment Timing: Staggered Adoption of Single-Member District Elections

Notes: Distribution of the 222 treated school districts in the discipline-DiD panel by year of first SMD election. The core CVRA-era cohorts (2012–2020) contain the bulk of identifying variation. Pre-window early adopters predate the analysis window and are excluded; late adopters have limited post-treatment data.

[3]Eleven CEDA treatment entities with known SMD adoption before the 2011–2024 analysis window have no observable pre-treatment period and are excluded from the causal sample (Cutoff B); forty additional CEDA entities with no clean pre-treatment period or reverse transitions are dispositioned out separately under explicit rules. The 2022 cohort has limited post-treatment observation; leave-one-cohort-out analysis (Section 4.4) confirms that excluding any single cohort does not change the primary BW-gap result qualitatively.

3.3 Outcome Variables

The unduplicated Black-White suspension-rate gap is the primary outcome; the Hispanic-White gap and the total suspension rate are comparison outcomes, each measured in percentage points. The gaps are measured as the difference in suspensions per 100 enrolled students between each minority group and White students. Racial gaps are the natural estimand because the theoretical prediction concerns disparities, not discipline levels per se: descriptive representation is hypothesized to reduce differential treatment across racial groups (Meier and England 1984; Skiba et al. 2011). Gaps also difference out district-wide policy changes (e.g., a universal shift toward restorative practices) that affect all students equally, isolating race-specific treatment effects. I therefore pair gap estimates with race-specific rate estimates in the main results, so the analysis distinguishes relative disparity effects from absolute changes in each group's suspension risk. The Black-White gap is valid only for district-years in which both Black and White enrollment exceed a minimum threshold of 10 students—a stricter analysis filter than the California Department of Education's raw suppression of cells with fewer than five students—yielding approximately 57% coverage of the analysis panel (465 districts: 203 treated and 262 control; 4,849 non-COVID district-years, of which the Sun-Abraham estimate uses 4,837 after fixed-effect singletons are dropped). The Hispanic-White gap has approximately 92% coverage, reflecting the larger Hispanic population in California. I also examine the total suspension rate as a secondary outcome.

Academic years 2019–20 and 2020–21 are excluded from all analyses due to the COVID-19 pandemic, which substantially disrupted both school operations and discipline practices. The final analysis panel contains 8,439 non-COVID district-year observations from 721 districts.

3.4 Identification Strategy

Identification relies on a staggered difference-in-differences design. The baseline specification is:

$$Y_{dt} = \alpha_d + \lambda_t + \beta \cdot \text{SMD}_{dt} + \varepsilon_{dt} \tag{1}$$

where Y_{dt} is the discipline outcome for district d in year t, α_d and λ_t are district and year fixed effects, SMD_{dt} is an indicator equal to one for treated districts in post-treatment years, and ε_{dt} is the disturbance. Under homogeneous treatment effects the coefficient β identifies the average treatment effect on the treated (ATT); under staggered adoption with effects heterogeneous across cohorts and event time, β is instead a possibly negatively weighted combination of group-time effects (Goodman-Bacon 2021)—which is why I rely on the heterogeneity-robust estimators introduced below rather than on this two-way fixed-effects specification directly. Standard errors are clustered at the district level throughout.

Identification requires that, conditional on district and year fixed effects, treated and control districts would have followed parallel trends in the absence of treatment. I assess this assumption through event study specifications:

$$Y_{dt} = \alpha_d + \lambda_t + \sum_{k \neq -1} \gamma_k \cdot \mathbf{1}[\text{rel\_time}_{dt} = k] + \varepsilon_{dt} \tag{2}$$

where rel_time_{dt} is the number of years relative to treatment ($k = 0$ is the first treatment year), and the omitted category is $k = -1$. Under parallel trends, the pre-treatment coefficients γ_k for $k < -1$ should each be zero. I report event study estimates from the heterogeneity-robust estimators described below; each modifies the aggregation of cohort-specific relative-time effects to avoid the negative-weighting problem inherent in the TWFE version of Equation 2.

The exogeneity of treatment timing is the key identifying assumption. CVRA litigation timing is driven by plaintiff attorneys' capacity constraints and case-selection strategies, not by changes in districts' discipline policies. While the *pool* of eligible districts is selected on demographics (size, diversity, segregation), the *sequence* of challenges within this pool is plausibly orthogonal to discipline trends. The identifying assumption is one of parallel *trends*, not parallel *levels*: two groups can have identical levels but diverging trajectories. I therefore assess this assumption primarily by examining pre-treatment dynamics in the event study (Section 4). Table 1 additionally shows that treated and control districts have broadly similar pre-treatment discipline levels, though I note that similar levels are neither necessary nor sufficient for parallel trends, and that the normalized differences are non-trivial on enrollment and on the BW gap itself.

3.5 Estimators

Recent econometric work has demonstrated that the TWFE estimator in Equation 1 can be biased when treatment effects are heterogeneous across cohorts or over time (Goodman-Bacon 2021; de Chaisemartin and D'Haultfœuille 2020). The bias arises because TWFE implicitly uses already-treated units as comparisons for newly-treated units, potentially creating "negative weights" on some treatment effects.

I therefore supplement the TWFE estimates with five alternative estimators. The *Sun-Abraham* (SA) estimator uses interaction-weighted averages of cohort-specific treatment effects, restricting comparisons to never-treated and not-yet-treated units (Sun and Abraham 2021). I designate this as the preferred estimator. The *Callaway-Sant'Anna* (CS) estimator constructs group-time average treatment effects for each cohort-period pair, then aggregates to an overall ATT (Callaway and Sant'Anna 2021). The *Borusyak-Jaravel-Spiess* (BJS) imputation estimator first estimates the counterfactual outcomes for treated observations using only untreated data, then takes the average difference between observed and imputed outcomes as the ATT (Borusyak, Jaravel, and Spiess 2024). I also implement Synthetic Difference-in-Differences (SDID), which combines the unit-weight approach of synthetic control with the time-differencing of DiD (Arkhangelsky et al. 2021). Finally, I report estimates from the Generalized Synthetic Control (GSC) method, which models untreated potential outcomes using interactive fixed effects (Xu 2017). SDID and GSC serve as robustness checks under alternative identifying assumptions; because they absorb interactive (unit-specific) trends, they may over-correct if the true data-generating process does not contain such trends.

For a unified treatment of these estimators and the design choices that distinguish them, see Baker et al. (2026).

All standard errors are clustered at the district level. I apply Benjamini-Yekutieli corrections for multiple testing across the main reported outcome family (Benjamini and Yekutieli 2001).[4]

4 Results

4.1 Descriptive Statistics and Balance

Table 1 presents summary statistics for the analysis sample. Treated districts are substantially larger than control districts (mean enrollment 11,182 vs. 2,576; normalized difference = 1.14), reflecting the CVRA's focus on larger, more diverse jurisdictions. Treated districts also have somewhat larger Black-White suspension gaps (9.04 vs. 6.10 percentage points; normalized difference = 0.25) and Hispanic-White gaps (0.56 vs. -0.36; normalized difference = 0.18). These all-years descriptive means are distinct from the 4.8 pp baseline used to scale the informative-null bound in Section 4, which is the pre-treatment mean BW gap in the validity-filtered estimation sample (districts with ≥ 10 Black *and* ≥ 10 White students), not the descriptive universe shown here. Total suspension rates are modestly higher in treated districts (6.37 vs. 4.97; normalized difference = 0.22). White suspension rates are similar across groups (5.92 vs. 5.17; normalized difference = 0.11), while race-specific rates for Black students (14.80 vs. 11.42; normalized difference = 0.23) and Hispanic students (6.47 vs. 4.81; normalized difference = 0.28) are elevated in treated districts.

The normalized differences, while non-trivial for enrollment, are generally moderate for discipline outcomes, suggesting that the two groups are reasonably comparable conditional on district and year fixed effects. The enrollment imbalance is a mechanical consequence of

[4] Unless otherwise noted, all reported p-values are two-tailed. The randomization-inference p-value reported in the robustness section is computed as the two-sided share of placebo assignments whose absolute estimate is at least as large as the observed estimate.

Table 1: Summary Statistics and Balance

	Full Sample		By Treatment Status		
Variable	Mean	SD	Control	Treated	Norm. Diff.
Enrollment (total)	5,273	8,539	2,576	11,182	1.14
Total suspension rate	5.41	6.45	4.97	6.37	0.22
Black suspension rate	13.06	14.85	11.42	14.80	0.23
Hispanic suspension rate	5.36	6.04	4.81	6.47	0.28
White suspension rate	5.42	7.01	5.17	5.92	0.11
BW suspension gap (pp)	7.53	11.79	6.10	9.04	0.25
HW suspension gap (pp)	−0.05	5.11	−0.36	0.56	0.18
Sample sizes					
Districts	721		499	222	
District-year obs (non-COVID)	8,439				

Notes: Suspension rates are suspensions per 100 enrolled students. BW gap = Black rate − White rate. HW gap = Hispanic rate − White rate. Normalized difference = (Treated mean − Control mean) / $\sqrt{(s_T^2 + s_C^2)/2}$. Sample restricted to non-COVID years (excluding 2019–20 and 2020–21). Full-sample Mean/SD are from the descriptive summary; treatment/control means and normalized differences are non-COVID descriptive statistics computed directly from the final analysis panel (not the pre-treatment balance table). The descriptive universe is 721 districts (222 treated, 499 control; 8,439 non-COVID district-years); outcome-specific estimation samples are smaller because of validity filters such as the Black/White enrollment requirement. BW gap available for ≈60% of observations due to minimum Black enrollment requirements.

the CVRA targeting large, diverse districts and does not threaten identification so long as treatment timing within the eligible pool is exogenous, which I argue above.

Figure 2 displays the raw Black-White suspension gap over time for treated and control districts. Before the modal treatment window (2016–2020, shaded), the two groups track each other closely, with treated districts maintaining a consistently higher BW gap of approximately 2–3 percentage points. After adoption, the treated and control gaps continue to move together, with no visible divergence—consistent with the null average effect estimated below. The visual parallel in pre-treatment trends motivates the DiD design. For comparison, Figure A13 in Appendix B shows race-specific suspension rates by treatment group.

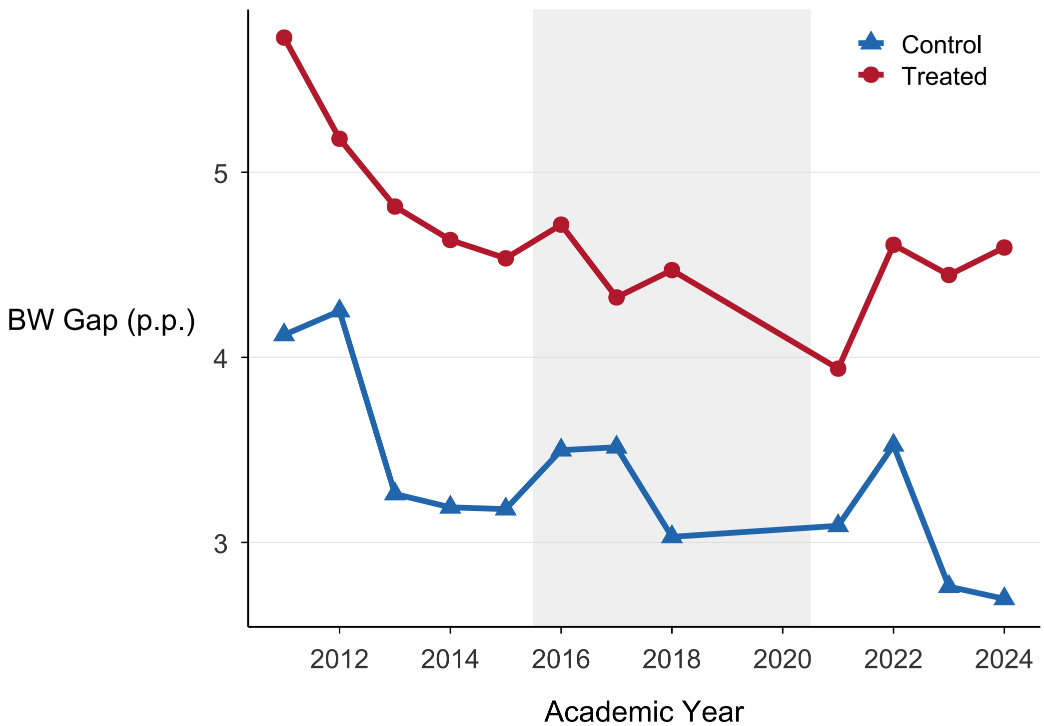

Figure 2: Raw Trends: Black-White Suspension Rate Gap

Notes: Annual mean Black-White suspension rate gap (percentage points) for treated and control districts. The shaded band marks the modal treatment window (2016–2020). COVID years (2019–20, 2020–21) are excluded.

4.2 Main Effects

Table 2 presents the main difference-in-differences estimates. My primary outcome is the Black-White gap in the share of *distinct* students suspended (column 1). The preferred Sun-Abraham estimate is +0.33 percentage points (SE = 0.40; $p = .41$), and the remaining estimators are likewise indistinguishable from zero (Table 2).[5] No estimator distinguishes the primary gap from zero, and the Benjamini-Yekutieli adjusted q-value is 0.85. This is an informative null. The 95% interval $[-0.45, +1.11]$ rules out reductions beyond 0.45 pp (about 9% of the 4.8 pp pre-treatment gap) and widenings beyond 1.11 pp (about 23%). A formal equivalence test sharpens what is ruled out: under a two-one-sided-tests procedure

[5]Effective sample size for cohort-aggregated estimators (SA, CS, BJS) is smaller than the nominal observation count because cohort-specific weighting concentrates identification on cohorts with both pre- and post-treatment coverage.

the effect is statistically equivalent to zero within a margin of ± 1.0 pp at the 5% level (the 90% interval is $[-0.33, +0.99]$), and the minimum detectable effect at 80% power is 1.1 pp. This ± 1.0 pp margin is benchmarked against the discipline effects this literature treats as substantively meaningful—for example, exposure to a same-race teacher reduces exclusionary discipline for Black students by roughly 1–2 pp in some subgroup estimates (Lindsay and Hart 2017)—and is approximately one-fifth of the 4.8 pp baseline gap. Equivalence within it therefore rules out reductions at or above a discipline-scale magnitude the design can resolve, while smaller effects—below roughly 20% of the gap—remain below the design's resolution.

Counted by suspension *incidents* rather than distinct students, the estimated gap is marginally wider (Sun-Abraham +1.61; SE = 0.88; $p = .069$), but the other estimators are null and the estimate collapses under enrollment weighting (Section 4.4). It also does not survive Benjamini-Yekutieli correction ($q = .21$). Here TWFE (+0.59) is attenuated relative to SA, consistent with the negative-weighting bias that arises when already-treated units serve as controls for late adopters (Goodman-Bacon 2021). The divergence between the two measures—null in distinct students, marginal in incidents—is consistent with more frequent suspension of already-suspended Black students rather than a larger share of Black students suspended; a direct test of the Black-White gap in the suspension duplication ratio (incidents per suspended student) is positive in the predicted direction but statistically indistinguishable from zero (Sun-Abraham +0.05, $p = .34$; log-ratio +0.04, $p = .12$), so I treat the repeat-suspension reading as suggestive rather than established.

The Hispanic-White gap (column 2) shows no robust change. TWFE yields -0.21 (SE = 0.11; $p = .05$) and SA -0.45 (SE = 0.25; $p = .074$), with the remaining estimators negative but insignificant (Table 2). The point estimates are directionally consistent—negative under every estimator—but, by the same standard applied to the marginal incident-BW estimate, nothing survives multiple-testing correction (BY $q = .21$), and I read the gap as unchanged.

The total suspension rate (column 3) *declines*: SA = -0.79 (SE = 0.38; $p = .04$), and the remaining estimators are all negative (Table 2). The decline is significant in unadjusted estimates but does not survive Benjamini-Yekutieli correction ($q = .21$); I read it as a modest easing of overall discipline that, importantly, does not narrow the targeted Black-White disparity. For the Hispanic-White gap, the minimum detectable effect at 80% power (based on SA standard errors) is approximately 0.7 pp, so the Hispanic-White null is informative rather than simply underpowered.

Race-specific rate regressions show only a marginal rise in Black suspension rates (+0.79 pp under TWFE, n.s.; +1.76 pp under SA, $p = .10$), with White rates flat (-0.35 TWFE; -0.46 SA; both n.s.) and Hispanic rates lower (-0.75 TWFE, $p < .01$; -0.83 SA, $p = .05$).[6] The larger Black-rate increase that appeared before the final data-integrity pass is not present in the certified analysis panel. The race-match structure nonetheless anticipates the moderation analysis in Section 5: within the heterogeneity-analysis subsample, only 18 treated districts gained a predicted-Black board member after SMD adoption, while 135 gained a predicted-Hispanic but not Black member—so the new representation accrued overwhelmingly to a group other than the one bearing the disparity.

Expulsions, the severe margin. Expulsions are far rarer than suspensions—the baseline Black-White expulsion gap is only 0.10 pp—but they are the more consequential end of the exclusionary-discipline distribution. SMD adoption has no detectable effect on the Black-White expulsion gap (Sun-Abraham +0.05 pp, SE = 0.06, $p = .45$, $n = 4{,}505$; Callaway-Sant'Anna -0.06, $p = .73$, $n = 4{,}519$), and the Hispanic-White expulsion gap is at most marginally negative (SA -0.04, $p = .11$; CS -0.08, $p = .06$); neither survives Benjamini-Yekutieli correction (smallest $q = .44$). The minimum detectable effect at 80% power for the Black-White expulsion gap is 0.17 pp—comparable to the baseline gap itself—so the null is informative against effects on the order of the gap's own magnitude, though not against very small ones. The discipline null thus holds across both margins: SMD adoption did not narrow (or widen) the Black-White gap in either suspensions or expulsions.

[6] The race-specific rate regressions are estimated on the full analysis panel, while the BW gap regression is restricted to district-years with at least 10 Black *and* 10 White students. The two samples are not directly comparable, and the race-specific rates should not be read as an exact arithmetic decomposition of the BW gap estimate.

Figure 3 provides an at-a-glance comparison of all five estimators across the three primary outcomes, illustrating the consistency of the BW gap finding, the absence of robust change in the HW gap, and the modest, non-robust decline in the total rate.

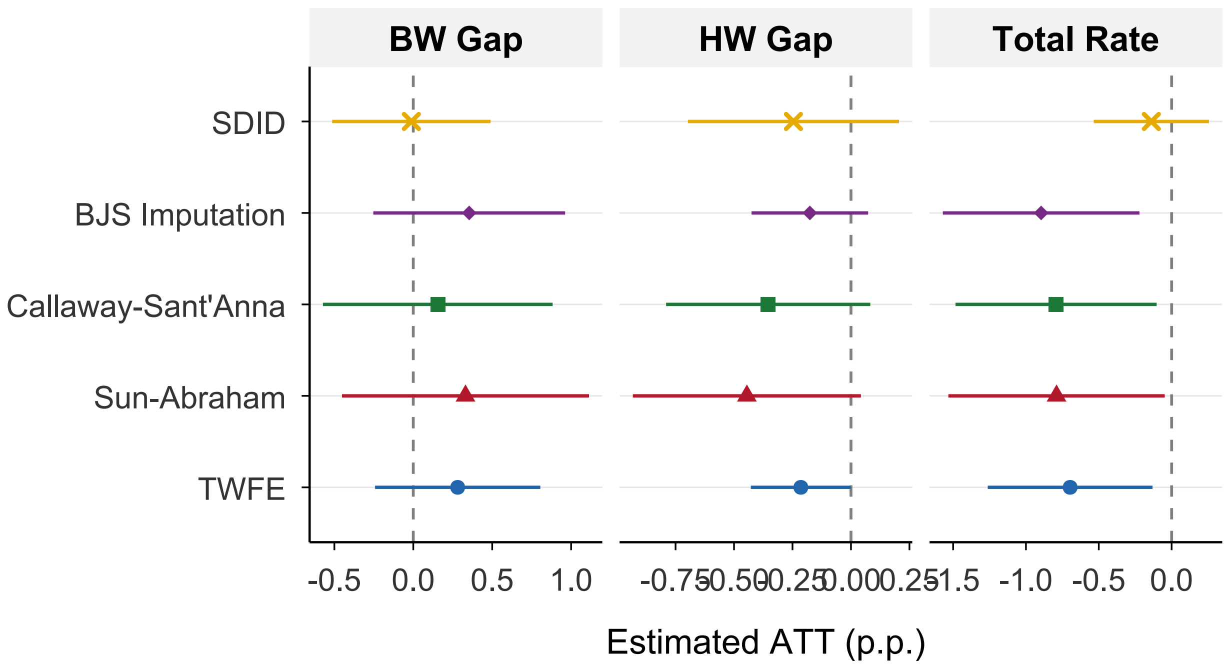

Figure 3: ATT Estimates Across Estimators and Outcomes

Notes: Point estimates with 95% confidence intervals for each estimator-outcome combination. TWFE = two-way fixed effects; SA = Sun-Abraham; CS = Callaway-Sant'Anna; BJS = Borusyak-Jaravel-Spiess imputation; SDID = Synthetic DiD. All specifications include district and year fixed effects and exclude COVID years.

Interactive fixed effects methods. The SDID estimate for the primary BW gap (-0.01; SE = 0.26; p = .96) and the Generalized Synthetic Control estimate (+0.05; p = .89; nonparametric bootstrap with 1,000 replications) are likewise null, consistent with the DiD-family null rather than indicating an effect masked by differential trends (see Appendix C for trajectories and diagnostics). Cross-validation selected zero latent factors for the BW and HW gap outcomes ($r^* = 0$) and one factor for the total rate ($r^* = 1$); the $r^* = 0$ result indicates that the data do not support interactive fixed effects beyond standard unit and time

Table 2: Main Difference-in-Differences Results

	Unduplicated BW Gap (1)	Unduplicated HW Gap (2)	Total Rate (3)
Panel A: Estimators			
TWFE	0.28	−0.21	−0.70
	(0.27)	(0.11)	(0.29)
Sun-Abraham	0.33	−0.45	−0.79
	(0.40)	(0.25)	(0.38)
Callaway-Sant'Anna	0.16	−0.35	−0.79
	(0.37)	(0.22)	(0.35)
BJS Imputation	0.35	−0.18	−0.90
	(0.31)	(0.13)	(0.34)
SDID	−0.01	−0.25	−0.14
	(0.26)	(0.23)	(0.20)
Panel B: Inference			
BY q-value (SA)	0.85	0.21	0.21
N (obs)	4,837	7,769	8,327

Notes: Each cell reports the ATT estimate from the indicated estimator with standard errors clustered at the district level in parentheses. Unduplicated BW Gap = Black-White gap in the share of distinct students suspended (the primary outcome); the incident-based (event-count) BW gap is reported in the text as a measurement decomposition (SA +1.61, $p = .069$, not robust across estimators). Unduplicated HW Gap = Hispanic-White gap in the share of distinct students suspended, matching column 1; the incident-based HW gap is likewise not robust (SA −0.47, $p = .31$). Total Rate = overall suspension rate per 100 students. All specifications include district and year fixed effects and exclude COVID years (2019–20, 2020–21). BY q-values are Benjamini-Yekutieli adjusted across the primary family; no outcome survives the correction.

effects for the primary outcomes. SA remains the preferred estimator for reasons detailed in Section 4.4.[7]

[7]GSC removes treated units with fewer than $r^* + 2$ pre-treatment periods; all removed units are treated, while control units are observed in all years. Under GSC, the HW gap shows a marginally significant negative estimate (−0.66; SE = 0.26; $p = .010$ nonparametric), suggesting a possible reduction in the Hispanic-White gap that the DiD-family estimators do not detect. This result should be interpreted cautiously given the substantial sample attrition.

4.3 Event Study

Figure 4 presents the Sun-Abraham event study for the primary (unduplicated) Black-White suspension gap over a $[-5, +5]$ window. The pre-treatment coefficients are small and non-monotonic: $t{-}5 = +0.08$, $t{-}4 = +0.46$, $t{-}3 = -0.27$, $t{-}2 = -0.03$ (relative to $t{-}1 = 0$). None is individually significant, showing no systematic pre-trend.

For the preferred Sun-Abraham primary event study, post-treatment coefficients are small and statistically indistinguishable from zero throughout: $t0 = +0.27$, $t{+}1 = +0.16$, $t{+}2 = +0.28$, $t{+}3 = +0.65$, $t{+}4 = +1.00$, $t{+}5 = +0.31$. The primary event study is flat both before and after adoption, consistent with the null average effect in Table 2.

Figure 5 presents event studies for all primary outcomes, comparing SA, CS, and BJS estimators. The estimators produce similar flat patterns for the unduplicated BW gap. The HW gap is mostly flat to marginally negative—one CS lead at $t = -5$ and two post coefficients (SA at $t{+}4$, BJS at $t0$) cross the 5% threshold, but the average HW effect does not survive multiple-testing correction—and the total rate drifts downward after adoption, consistent with the average effects in Table 2.

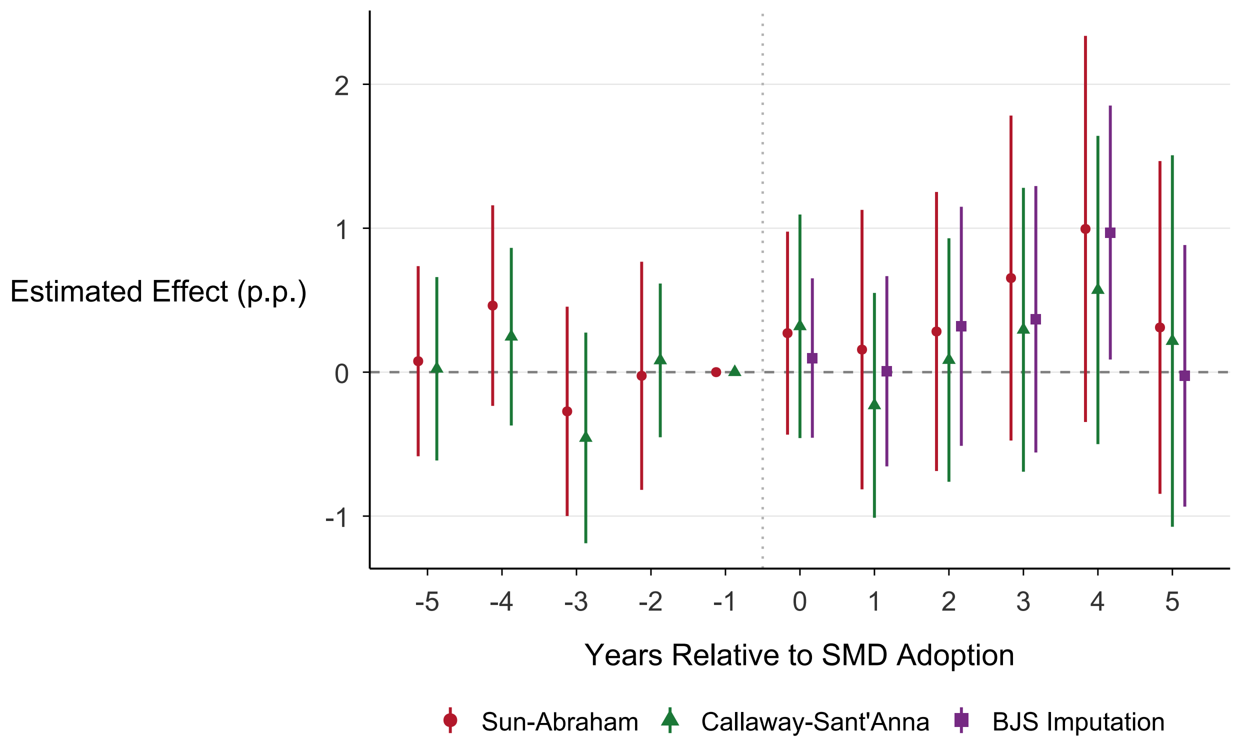

Figure 4: Event Study: Effect of SMD Adoption on Black-White Suspension Gap

Notes: Coefficients from Sun-Abraham, Callaway-Sant'Anna, and BJS imputation event study specifications. The dependent variable is the Black-White suspension rate gap in percentage points. The omitted period is $t-1$. Whiskers show 95% confidence intervals based on district-clustered standard errors. COVID years (2019–20, 2020–21) are excluded.

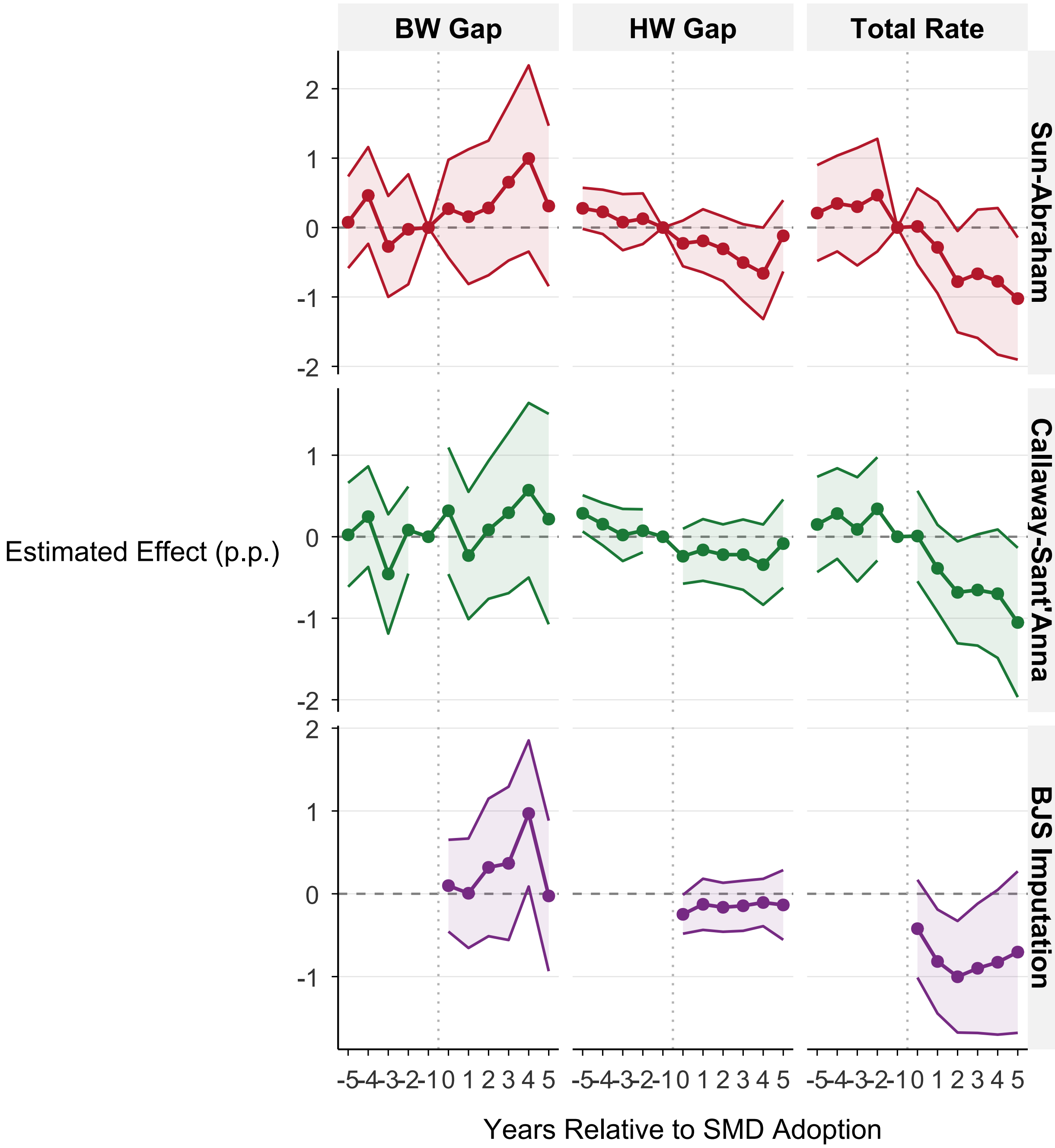

Figure 5: Event Studies: All Primary Outcomes

Notes: Event study coefficients for the unduplicated BW gap, HW gap, and total suspension rate across three heterogeneity-robust estimators (Sun-Abraham, Callaway-Sant'Anna, BJS Imputation). Shaded bands show 95% confidence intervals. The omitted period is $t-1$. The unduplicated BW gap is flat across all estimators; the HW gap is mostly flat to marginally negative, with isolated 5%-level coefficients but no multiple-testing-robust average effect; and the total rate drifts downward, consistent with the average effects in Table 2. COVID years excluded.

4.4 Robustness

The main threat to this design is not sampling error but differential pre-treatment movement in the Black-White suspension gap between treated and control districts. I address this concern through design-based inference, specification and cohort stability checks, falsification tests, and explicit sensitivity bounds—all summarized in Figure 7 and detailed in Appendix D, with the pre-trends discussion consolidated below.

Design-based inference. Because the primary gap is an informative null, the design-based checks bear mainly on the one positive signal, the incident-based gap. Randomization inference on the incident gap—reassigning treatment timing 500 times—yields $p = .078$ (marginal), and a triple-difference specification that differences out district-level trends common across racial groups produces a small, insignificant estimate (+1.69; SE = 1.07; $p = .12$). Both are consistent with, at most, a fragile incident-based effect and no effect on the distinct-student gap.

Specification stability. The primary unduplicated gap is a stable null across specifications (Figure 7): SA is +0.33 at baseline, +0.27 pre-COVID, +0.27 excluding the top-10 largest districts, and +0.33 for clean switchers only—none significant—and is similarly null under controls for poverty (FRPM percentage), English Learner shares, ACS income/poverty, SAIPE child poverty, and when COVID years are retained with year fixed effects (Appendix D). Incident BW estimates remain positive but fragile: SA is +1.61 at baseline ($p = .069$), +1.57 excluding the top 10 districts ($p = .074$), +1.58 at the 20-student threshold ($p = .053$), +2.22 in the pre-COVID sample ($p = .104$), and +0.87 when COVID is retained with year fixed effects ($p = .246$); CS, GSC, SDID, and enrollment-weighted variants are null.

Estimand weighting. Because the district-level estimators weight districts equally, I also report enrollment-weighted ATTs. Weighting by Black-and-White enrollment leaves the primary unduplicated gap null (SA +0.27; CS −0.07) and drives the incident estimate toward zero (SA +0.76; CS +0.14), confirming that the marginal incident signal is concentrated in small districts. A fully doubly-robust specification is infeasible on this panel because of sparse cohort-period cells; a covariate-adjusted (regression) conditional-trends specification likewise leaves the primary gap null (+0.19; $p = .61$) and renders the incident estimate insignificant (+1.20; $p = .14$).

Placebo and falsification tests. Most placebo outcomes are null under the preferred Sun-Abraham estimator: total enrollment (SA +6, n.s.; significant only under TWFE, −344, reflecting negative-weighting bias rather than a substantive enrollment effect), the White and Asian suspension rates, and standardized achievement gaps (SEDA 6.0 (Reardon et al. 2025); W-H +0.015 SD, $p = .26$; W-B +0.004 SD, $p = .78$; Appendix H). One placebo warrants caution: the percent-Black-enrollment placebo is negative and significant under both TWFE (−0.24; $p = .02$) and SA (−0.21; $p = .03$), indicating that SMD adoption is associated with a modest decline in measured Black enrollment share. Because the Black-White gap conditions on Black enrollment, this compositional shift counsels caution in interpreting Black-specific outcomes. Three checks suggest that this composition shift does not explain the primary null. First, the decline is small—about 8% of the 2.7% mean Black enrollment share—and appears to reflect enrollment dilution in large CVRA-targeted districts rather than loss of Black enrollment below the analysis threshold: treatment is associated with a *higher* probability that a district-year remains BW-valid (+0.024, $p = .02$), the opposite of the attrition pattern that would mechanically remove treated observations from the BW-gap sample. Second, the 127 validity-switching districts (22 treated, 105 control) are small, threshold-marginal units—median 10 Black students, 0.7% Black share, 1,152 total enrollment, and smaller baseline gaps (1.5 vs. 3.8 pp)—and are disproportionately control-side rather than treated-side. Third, re-estimating the gap on a validity-stable panel (+0.31; SE 0.36; $p = .39$) and on a specification that fixes Black and White enrollment denominators at their pre-treatment means (+0.81; SE 0.54; $p = .14$) leaves it insignificant either way; because the fixed-denominator adjustment *raises* the estimate, observed enrollment composition attenuates a wider-gap estimate rather than manufacturing the null from an underlying narrowing. These checks narrow the threat rather than eliminating every composition concern, but they do not support the specific selection channel the placebo raises.

Treatment-timing robustness. Treatment timing is coded from the CEDA trustee-area field and cross-checked against LCCR (through 2014) and Abott–Magazinnik (through 2016). One concern is that timing error could attenuate estimated effects toward zero. I probe this in three ways. First, restricting to the LCCR-window cohorts (first SMD year $\leq$ 2014; 41 treated), whose timing is externally checkable, leaves the primary estimate essentially unchanged at +0.32 (SE 0.86; $p = .71$); this subsample is small and imprecise, but it gives no indication that cleaner timing reveals a hidden gap reduction. Second, randomly shifting treated districts' adoption years by ± 1 year over 200 draws moves the SA ATT within $[-0.76, +0.76]$ (mean +0.07), with one of 200 draws producing a significant narrowing; widening the perturbation to ± 2 years gives a range of $[-0.36, +0.99]$ (mean +0.28) and none. Third—because much of the disagreement with external codings is the systematic adoption-versus-election-year gap rather than random noise—I shift *every* treated district's year by ± 2 years: both directions move the estimate *away* from zero in the positive, gap-*widening* direction (+0.58, $p = .17$ for +2; +0.76, $p = .03$ for -2), not toward a narrowing. These are sensitivity exercises rather than a complete model of every dating error, but across symmetric, larger, and systematic perturbations, plausible timing error does not produce evidence of a policy-relevant reduction in the Black–White gap; if anything, systematic mis-dating biases the estimate toward a wider gap.

Cohort stability. Leave-one-cohort-out analysis (Figure 6) shows the primary unduplicated gap remains small and insignificant when any single treatment cohort is excluded (SA ATT between +0.09 and +0.66; all $p > .15$): no single cohort drives the null. The interactive fixed-effects estimators (SDID -0.01; GSC +0.05) agree.

Pre-trends and sensitivity. I address the pre-trend rejection directly rather than minimizing it. The disaggregated pre-trend test rejects ($F = 7.50$, $p < 2 \times 10^{-23}$, 22 cohort-by-lead terms), even though the Sun-Abraham event study is flat pre-treatment. Decomposing

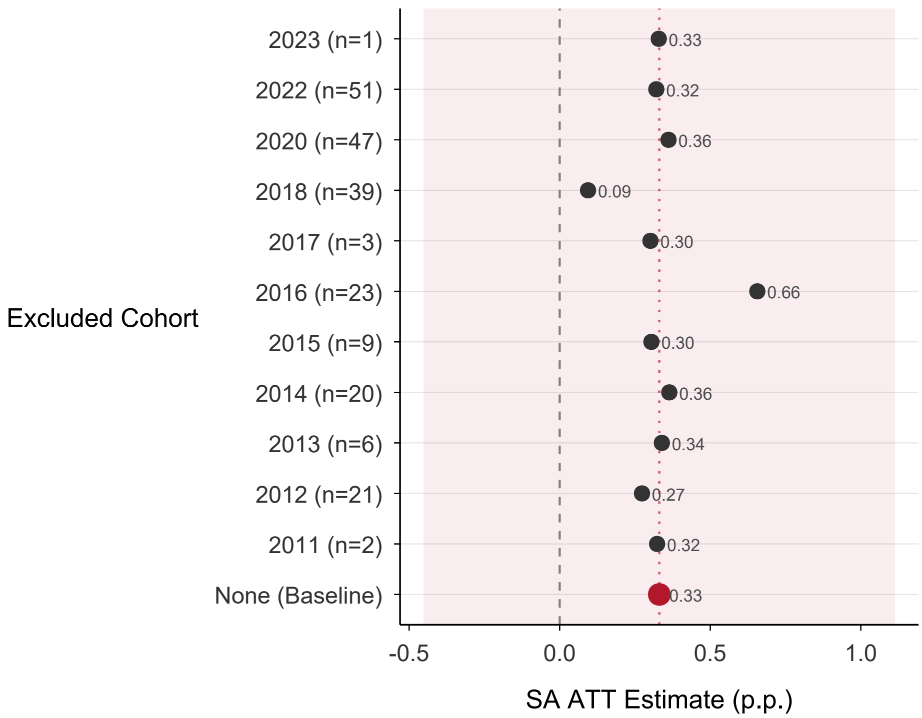

Figure 6: Leave-One-Cohort-Out: BW Suspension Gap (Sun-Abraham)

Notes: SA ATT estimates for the BW suspension gap when each treatment cohort is excluded in turn. The "None" row shows the baseline estimate including all cohorts. The vertical dashed line marks the baseline estimate. Cohort sizes reflect the BW-gap-valid subsample and may differ slightly from the full-sample counts reported in Section 3.

the test clarifies the source of the tension. The 22 cohort-specific leads are substantively small on average (mean -0.02 pp, SD 0.98, range $[-2.11, +1.47]$), sign-offsetting, and only 3 are individually significant. The largest deviations are concentrated largely in smaller cohorts—2016 ($n = 22$), 2014 ($n = 17$), 2015 ($n = 9$), 2017 ($n = 2$), and the single-district 2023 cohort—whereas the larger supported cohorts carrying much of the identifying variation (2022, $n = 44$; 2018, $n = 37$) have leads no larger than about 0.8–0.9 pp. Under the SA aggregation, which places more weight on larger supported cohorts, these cohort-specific deviations do not appear as a systematic pre-trend: the aggregated leads are +0.08, +0.46, -0.27, and -0.03 pp at $t = -5$ through $t = -2$ (all $p > .19$). The rejecting joint test therefore remains a real identification caution, but it is better understood as high-powered detection of small cohort-specific deviations than as evidence of a systematic differential trend in the aggregated event study.

A linear fit through the aggregated leads gives a slope of -0.10 pp per event year (SE 0.17; 95% CI $[-0.43, +0.22]$; $p = .53$), consistent with no systematic linear pre-trend.[8] This does not validate parallel trends; it narrows the concern. The rejecting disaggregated test shows real cohort-specific pre-period deviations, but the aggregated leads, the leave-one-cohort-out checks below, and the HonestDiD bounds all indicate these deviations are unlikely to mask a policy-relevant gap reduction: turning the $+0.33$ pp null into even a -1 pp narrowing would require more than a percentage point of unobserved relative improvement immediately after adoption, despite flat aggregated leads. A null can still be biased, so I interpret the result under maintained-design tension rather than as assumption-free.

Appendix D reports formal sensitivity analyses following Rambachan and Roth (2023). For the primary unduplicated gap the robust confidence set includes zero at every smoothness restriction $M \in [0, 0.5]$, including the $M = 0$ benchmark—which bounds the post-treatment *change* in the differential trend at zero, allowing the estimated linear pre-trend to extrapolate into the post-period rather than assuming parallel trends outright. The null therefore survives even when a linear differential trend is permitted to continue; relaxing the restriction further only widens an interval already centered near zero.

The design-based checks point the same way. Randomization inference and the triple-difference (above) find at most a fragile incident-based signal and no distinct-student effect. Leave-one-cohort-out analysis (Figure 6) leaves the primary null unchanged (SA between $+0.09$ and $+0.66$, all insignificant), and the specification curve (Figure 7) places the primary SA ATT near zero throughout. No outcome in the primary family survives Benjamini-Yekutieli correction (unduplicated-gap $q = .85$; incident, HW, and total all $q = .21$). Appendix D also reports a matched-sample pre-treatment balance diagnostic showing that PanelMatch

[8] The diagnostic uses only four estimated lead coefficients and should be read as descriptive rather than as a high-powered test.

matching reduces the raw pre-period BW-gap imbalance by 38 to 90 percent and produces a matched difference at $t = -1$ that is effectively zero.

The moderation evidence in Section 5 is consistent with the race-match interpretation, but it is not an independent identification strategy: it relies on post-treatment composition and the same panel structure. A predetermined-moderator check using pre-treatment Black voting-age share (Section 5.5) does not corroborate it, reinforcing that the moderation is suggestive rather than causally identified.

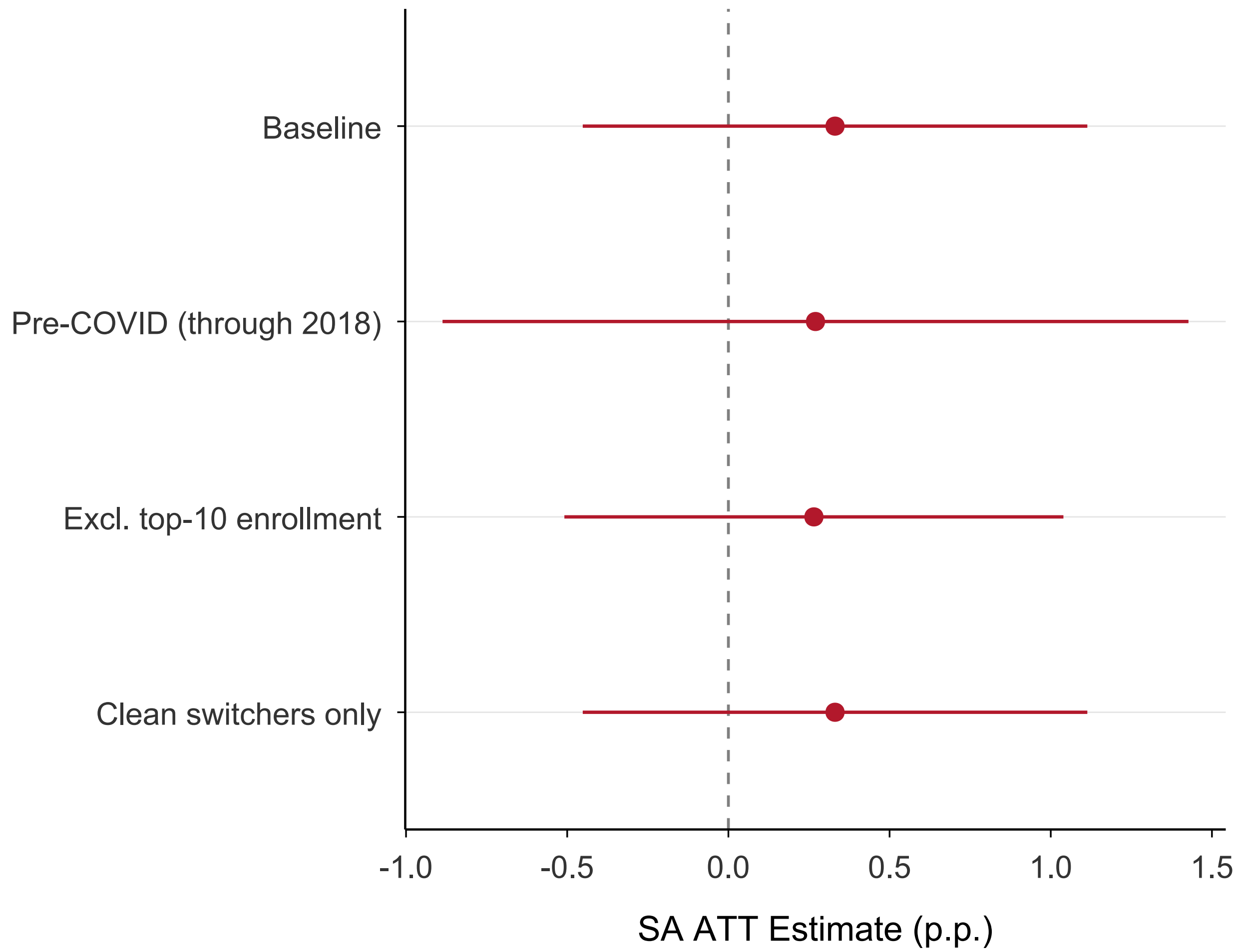

Figure 7: Robustness: Sun-Abraham ATT Estimates Across Specifications

Notes: Each row shows the Sun-Abraham ATT estimate for the primary (unduplicated) BW suspension gap under a different specification. Points are estimates; whiskers show 95% confidence intervals. The vertical dashed line marks zero. All estimates are small and statistically indistinguishable from zero.

5 Exploratory Moderation Evidence: Race-Matched Patterns

The average effect on the primary (unduplicated) gap is null, while the incident-based gap rises only marginally. I use the secondary incident signal to ask where any gap-association concentrates: the board-composition first stage, board-composition moderation, and teacher-workforce moderation. This section is exploratory, not causal. Board and teacher racial composition are realized after treatment—conditioning on them risks post-treatment (collider) bias—and the moderated outcome is the fragile incident gap rather than the primary distinct-student gap. The goal is therefore descriptive localization: only the contemporaneous Black-teacher-share interaction survives familywise correction, the board-composition interaction does not, and the predetermined Black voting-age-share check below fails to corroborate the pattern.

5.1 First Stage: SMD Elections and Board Composition

Table 3 documents the effect of SMD adoption on elected board composition. The raw pre/post shift among treated districts is unambiguous: minority probability rises from 47.6% to 57.1% (+9.5 pp), Hispanic share from 27.3% to 36.7% (+9.3 pp), and Black share is flat to slightly lower, 6.6% to 6.3% (−0.3 pp), across 222 pre-SMD and 220 post-SMD districts. The DiD estimates net out secular representation gains in control districts and are correspondingly smaller. Under Sun-Abraham, minority share rises +2.6 pp (SE 2.4), Hispanic share +4.6 pp (SE 2.8), and Black share −0.5 pp (SE 0.8); under TWFE, the corresponding estimates are +3.4 pp ($p = .047$), +4.4 pp ($p = .019$), and +0.1 pp (n.s.). A formal Hispanic-minus-Black contrast shows the same significant-under-TWFE, imprecise-under-SA pattern: +4.3 pp ($p = .04$) under TWFE and +5.1 pp (SE 3.3; $p = .12$; cohort-stratified bootstrap 95% CI $[-1.1, +11.5]$) under Sun–Abraham. The Black first-stage null is informative: the Sun–Abraham 95% CI of $[-1.9, +1.0]$ rules out Black board-share gains larger than about 1.0

Table 3: First Stage: Electoral Reform and Board Composition

	Minority Share	Hispanic Share	Black Share
Panel A: DiD Estimates			
TWFE	0.034	0.044	0.001
	(0.017)	(0.019)	(0.005)
Sun-Abraham	0.026	0.046	−0.005
	(0.024)	(0.028)	(0.008)
Panel B: Pre-Post Means			
Pre-SMD	0.476	0.273	0.066
Post-SMD	0.571	0.367	0.063
Difference	+0.095	+0.093	−0.003
N candidates (Pre / Post)		3,632 / 1,229	
N districts (Pre / Post)		222 / 220	

Notes: Panel A reports DiD estimates of the effect of SMD adoption on the BISG-predicted racial composition of elected board members. Standard errors clustered at the district level in parentheses. Sun-Abraham is the preferred estimator (as elsewhere in the paper); its first-stage estimates are insignificant for all three outcomes, reflecting the smaller never-treated comparison group and the candidate-level BISG measurement on which board composition is built. The asymmetric *direction*—a Hispanic gain with Black representation flat—is consistent across both estimators (Panel A) and the raw pre/post means (Panel B), so the first-stage conclusion does not hinge on the TWFE–SA significance difference. Panel B reports raw mean BISG-predicted race probabilities for elected candidates before and after SMD adoption.

pp—roughly a sixth of the 6.6% pre-treatment Black share. These estimates echo Abott and Magazinnik (2020), who find Latino representation gains concentrated in larger and more segregated districts, while adding the Black-specific null they do not estimate.

Figure 8 shows the same timing: minority and Hispanic shares rise at $t = 0$ and persist through t+5, while Black shares remain flat.

5.2 Board Composition and Discipline Outcomes

If the race-match condition holds, the BW gap should decline in districts that gain Black board members and be unaffected in districts that gain only Hispanic members. I test this with

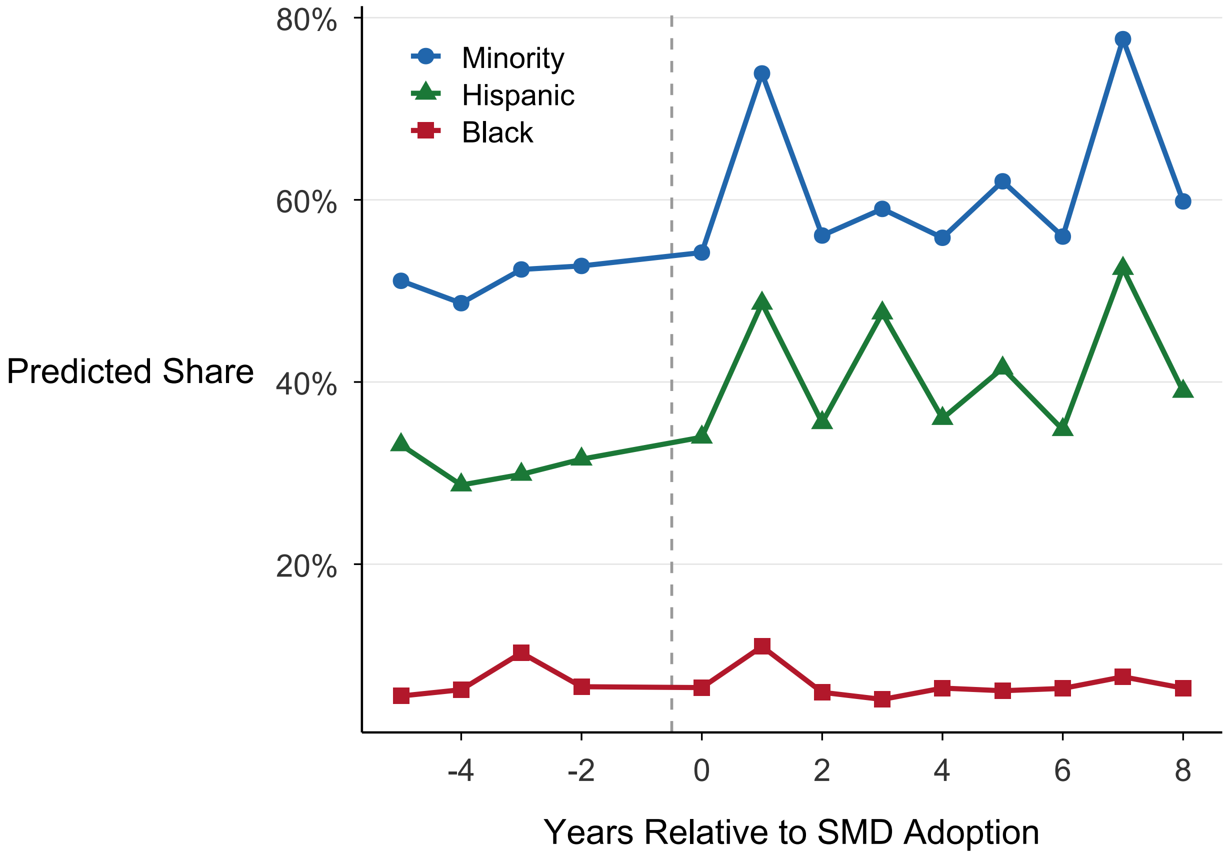

Figure 8: Board Composition Dynamics Around SMD Adoption

Notes: Mean BISG-predicted racial shares of elected board members by event time relative to SMD adoption. The dashed vertical line marks the first SMD election. Minority and Hispanic shares increase sharply at adoption; Black shares remain flat.

treatment-by-board-composition interactions; because board composition is post-treatment, these results are exploratory moderation evidence rather than causal mechanism estimates.

Table 4, Panel A, uses the *incident* BW gap, since the primary unduplicated gap has no average effect to moderate. The key specification interacts treat_post with cumulative Black board share, the running mean of BISG-predicted Black probability among elected members from the first SMD election onward. The base ATT at zero Black board share is +2.79 ($p = .008$), and the interaction is -28.8 (SE = 12.9; $p = .026$): a 10-percentage-point higher cumulative Black board share is associated with about a 2.9-pp smaller incident-gap treatment association, crossing zero near 9.7% Black board share. The Hispanic-board-share mismatch test is less precise and marginal (-5.10; $p = .07$), with the same sign as the

Black-share interaction rather than the null specificity the race-match prediction implies. The HW gap is not significantly moderated by Hispanic board share (-0.90; $p = .58$) or Black board share ($+2.72$; $p = .52$), and the total suspension rate is not significantly moderated by minority board share (-1.58; $p = .15$). The board interaction is therefore suggestive but not familywise-robust.

Figure 9 visualizes the same pattern. At zero Black representation the incident-gap treatment effect is positive (+2.8 pp), declines with Black board share, and crosses zero near 9.7%; the 95% band excludes zero only at low Black representation, where the gap widens, and includes zero where Black representation is higher and sparse (only 18 treated districts gained a Black board member).

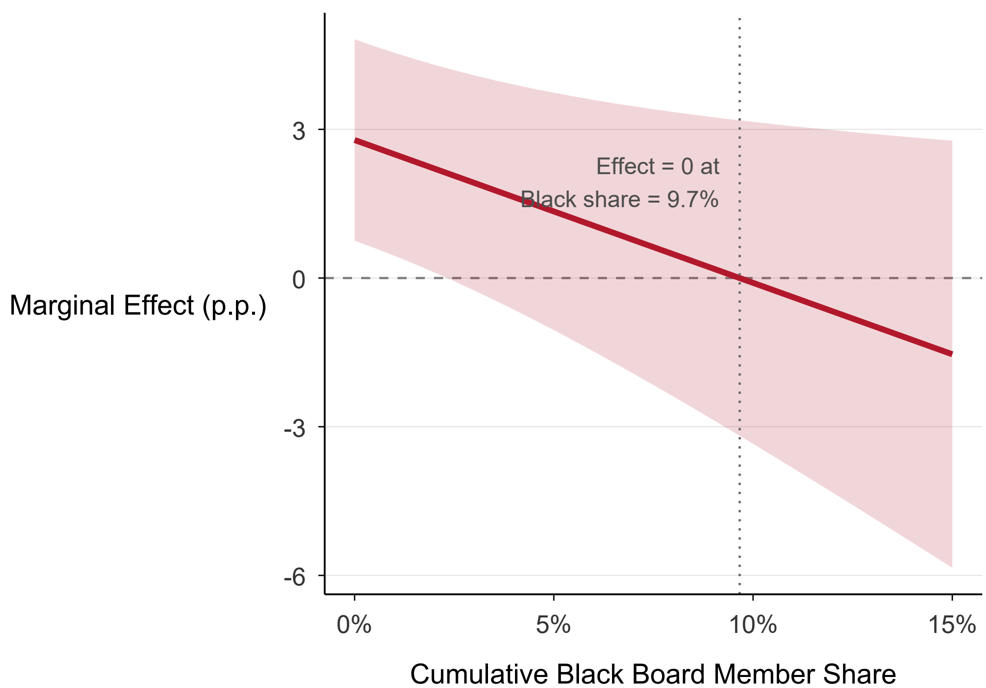

Figure 9: Race-Match Moderation: Marginal Effect of SMD on the Incident BW Gap by Black Board Share

Notes: Marginal effect of SMD adoption on the *incident* Black-White suspension gap (p.p.) as a function of cumulative Black board member share, from the moderation model in Table 4; the primary distinct-student gap is null on average and is not the object of this exploratory moderation. The shaded band shows 95% confidence intervals. The dotted vertical line marks the Black share at which the marginal effect equals zero.

Binary subgrouping is a blunt check. Within the heterogeneity sample, 18 districts gained a predicted-Black member, 135 gained a predicted-Hispanic but not Black member, and 42 gained no minority member. The subgroup SA estimates are individually insignificant and statistically similar: +1.44 for gained-Black ($p = .20$), +1.39 for gained-Hispanic-not-Black ($p = .20$), and +2.99 for gained-no-minority ($p = .17$; Figure A33). At face value this is a generic "any minority" pattern ($\approx +1.4$) versus "none" ($\approx +3.0$), not a Black-specific binary channel. It also reconciles with the continuous interaction: most gained-Black districts gained one member on a five- or seven-member board, leaving cumulative Black share below the approximately 9.7% zero-crossing point, so the negative interaction attenuates but usually does not offset the average incident-gap association.

5.3 Teacher Workforce Diversity

CDE staff data show a parallel workforce first stage. SMD adoption increases minority teacher share by 2.34 percentage points (Sun-Abraham; $p = .002$), concentrated among Hispanic teachers (+1.88 pp; $p < .001$). Black teacher share is small and insignificant under SA (+0.10 pp; $p = .36$) but larger under TWFE (+0.21 pp; $p = .019$), a divergence that may reflect the same weighting problem discussed above.[9] The student-teacher Black mismatch gap falls by 0.46 pp under TWFE ($p = .002$), but this is not clean workforce diversification: with Black teacher share up +0.21 pp, the identity $\Delta(\text{mismatch}) = \Delta(\text{student share}) - \Delta(\text{teacher share})$ implies a -0.25 pp Black-student-share change, essentially the significant %Black-enrollment placebo in Section 4.4. Roughly half the apparent alignment is therefore fewer Black students, not more Black teachers.

Teacher moderation is stronger than board moderation, but because teacher composition is post-treatment it remains correlational rather than causally identified. The Black teacher share interaction with the incident BW gap is -85.1 (SE = 23.4; $p < .001$; base

[9]The divergence between TWFE and SA for Black teacher share parallels the suspension-gap pattern. For this outcome, TWFE appears to amplify rather than attenuate the estimate, consistent with the direction-ambiguity of Goodman-Bacon contamination.

ATT = +2.83); at a one-year lag it is −49.3 (SE = 20.9; $p = .019$; base ATT = +1.75). A one-standard-deviation increase in Black teacher share (about 2.5 pp) is associated with about a 2.2-pp smaller incident BW gap. The teacher mismatch test is null for Hispanic teacher share (+0.55; $p = .89$). The total suspension rate is moderated by lagged minority teacher share (−2.47; $p = .04$), though not robust to multiple testing.

Multiple testing. The moderation family contains 21 interaction tests. Applying Benjamini-Yekutieli correction, only the contemporaneous Black teacher share × incident BW-gap interaction survives at $q < .05$ ($p = .0003$; $q_{BY} = .023$; $q_{BH} = .006$). The one-year-lagged Black-teacher interaction remains raw-significant but does not survive correction ($p = .019$; $q_{BY} = .36$; $q_{BH} = .098$). The student-teacher Black-gap and board cumulative-Black interactions ($q_{BY} = .36$ each), and the minority-teacher total-rate interactions, are raw-significant in several specifications but robust to neither correction. Teacher diversity provides the only familywise-robust moderation signal.

5.4 Assessing Spending as a Moderator

Although SMD adoption modestly increases per-pupil spending in log terms (+2.6% under SA; $p < .001$; Appendix Figure A38), the level estimate is negative (−$498 per pupil; $p = .01$), so the spending first stage is ambiguous. Spending does not moderate the discipline gaps: the log per-pupil-spending interaction is −0.68 ($p = .67$) for the incident BW gap. In a joint model, spending remains insignificant (+0.17; $p = .92$) while Black teacher share remains strongly significant (−88.4; $p < .001$), suggesting that detectable moderation in these data is concentrated in representation and personnel rather than resource reallocation. This contrasts with Fischer (2023), who shows race-specific board effects on spending, and is consistent with Jackson and Mackevicius (2024), whose meta-evaluation finds that aggregate spending effects depend heavily on deployment.

Table 4: Race-Matched Moderation: Board Composition and Teacher Diversity

Outcome	Moderator	Base ATT	Interaction	p-value
Panel A: Board Composition				
Incident BW gap	Cum. Black board share	2.79	−28.80	.026
HW gap	Cum. Hispanic board share	0.03	−0.90	.584
Incident BW gap	Cum. Hispanic board share [mismatch test]	2.12	−5.10	.074
Total rate	Cum. minority board share	0.93	−1.58	.148
Panel B: Teacher Diversity				
Incident BW gap	Black teacher share	2.83	−85.10	<.001
Incident BW gap	Black teacher share (L1)	1.75	−49.27	.019
Incident BW gap	Hispanic teacher share [mismatch test]	0.57	+0.55	.895
Total rate	Minority teacher share (L1)	0.54	−2.47	.043
Incident BW gap	St.-teacher Black gap	1.62	−32.20	.016

Notes: Each row reports results from a model interacting treat_post with the indicated moderator; the BW-gap outcome is the incident (event-count) measure, since the primary unduplicated gap shows no average effect to moderate. "Base ATT" is the estimated treatment effect when the moderator is zero. "Interaction" is the coefficient on the interaction term. "L1" indicates a one-year lag of the moderator. Teacher lags are calendar-adjacent: because CDE teacher data are unavailable for 2018–2020, the lagged specification gives 2018 its 2017 lag but leaves the 2021 seam missing rather than carrying stale 2017 values across the gap. All models include district and year fixed effects with district-clustered standard errors.

5.5 A Predetermined Electoral-Capacity Check

As a design-based check that avoids post-treatment composition, I interact SMD treatment with each district's pre-treatment Black voting-age population share, measured from the 2010 Census (table P10, race for the population 18 and over) at the school-district level. This proxy captures pre-existing Black electoral capacity before any sample district adopted SMD; unlike board or teacher share, it is fixed before treatment. Because decennial SF1 does not report citizenship by race at this geography, it is a districtwide VAP proxy rather than a CVAP measure or trustee-area concentration index. The outcome is the primary unduplicated gap, and the moderator is available for 98% of the estimation sample (458 of 465 districts).

If race-match operated through electoral empowerment, the primary gap should decline more in districts with larger pre-existing Black voting-age presence. The point estimate is in the predicted direction but fragile: -9.3 per unit share ($p = .068$) in the raw linear specification, -8.9 ($p = .16$) winsorized at the 95th percentile, -1.3 ($p = .24$) under a rank transform, and -3.4 ($p = .75$) when the highest-capacity 5% of districts are dropped. A median split shows no gradient: the Sun-Abraham treatment effect is $+0.49$ ($p = .24$) above the median and $+0.51$ ($p = .59$) below it, both statistically indistinguishable from the $+0.31$ full-sample proxy estimate. (The proxy-sample $+0.31$ sits just below the $+0.33$ headline because seven districts lack a 2010 VAP match; both halves lie slightly above it because a median split alters the never-treated comparison within each half.)[10]

This is the cleanest test of the mechanism precisely because it is predetermined and carries no collider risk, but it is still coarse and low-powered: the proxy is districtwide VAP, not CVAP or trustee-area concentration, and the average gap effect is itself null. Its failure to reproduce the moderation pattern therefore does not prove race-match is absent, but it removes the one design here that could have supported the mechanism without conditioning on post-treatment composition.

5.6 Interpreting the Average Effect

Figure 10 synthesizes the evidence: board composition dynamics, board interactions, teacher moderation, and the lag structure of the board association.

The average effect is best understood as SMD adoption in a setting where Hispanic representation usually changes and Black representation rarely does. The first stage is race-specific: Hispanic board representation rises in raw pre/post terms (+9.3 pp) and directionally in DiD estimates (Sun–Abraham +4.6 pp, SE = 2.8, $p = .11$; TWFE +4.4 pp, $p = .019$), while Black representation is flat (Sun–Abraham -0.5 pp, SE = 0.8, $p = .55$).

[10]An analogous interaction using pre-treatment Black *enrollment* share is statistically significant (-10.2 per unit share, $p = .004$), but it does not survive substituting the cleaner voting-age measure, and enrollment share is partly mechanical with the Black-White gap it moderates—it shares components with the outcome. I therefore do not rely on it.

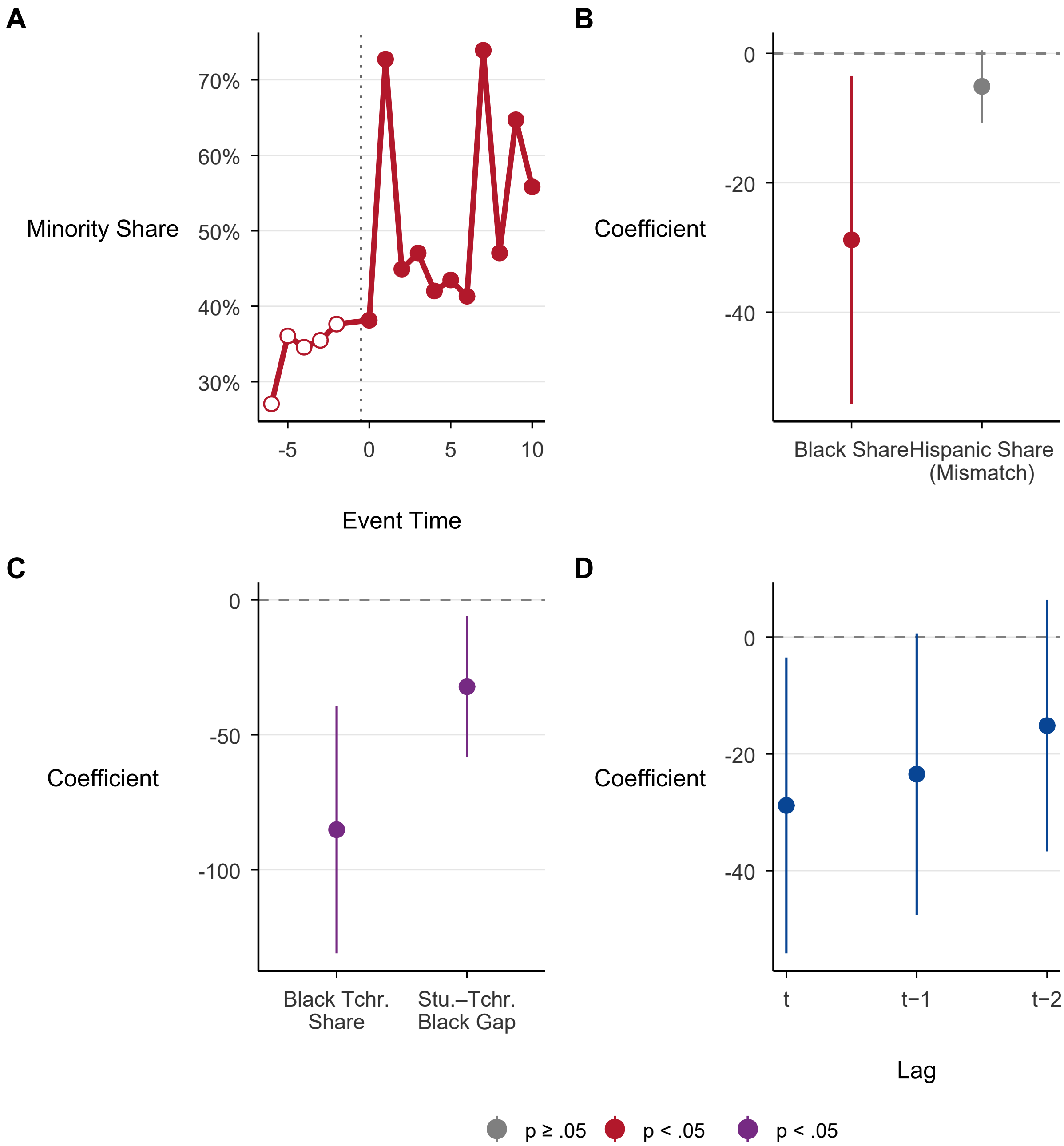

Figure 10: Race-Match Summary: Board Composition, Moderation, and Teacher Moderation

Notes: Panels B–D report exploratory moderation of the *incident* BW gap (the primary distinct-student gap is null on average). Panel A: minority share of elected board members by event time relative to SMD adoption (hard classification). Panel B: board race-match interaction coefficients—Black and Hispanic board share interacted with SMD treatment on the incident BW gap, with 95% CIs. Panel C: teacher moderation—Black teacher share and student-teacher Black gap interacted with SMD treatment on the incident BW gap (note different scale from Panel B). Panel D: lag structure of the Black board share × incident BW gap interaction at contemporaneous, one-year, and two-year lags, with 95% CIs.

The Hispanic-minus-Black contrast is +5.1 pp on the common sample (SA, $p = .12$; cohort-stratified cluster-bootstrap 95% CI $[-1.1, +11.5]$), reflecting California's demography: Latino populations are often large enough to anchor SMD majorities, while Black populations—about 5–6% of the state—rarely are.

The subgroup and composition arithmetic underscores the limited quantitative bite of the race-match account. Districts that gained a Black member (18), gained a Hispanic but not Black member (135), or gained no minority member (42) all have small, individually insignificant incident-gap effects (+1.44, +1.39, +2.99). Weighting those subgroup estimates by their treated-district shares (w = .09, .69, .22) yields a composition-implied average of +1.74 pp, close to the pooled incident-gap estimate of +1.61. But the counterfactual is nearly flat: even if every treated district gained Black representation, the predicted gap would be +1.44 pp, because the gained-Black and gained-Hispanic estimates are nearly identical. The framework reproduces the average but does not support a strong race-match counterfactual.

Two non-exclusive interpretations fit the null. One is attention displacement: newly empowered Hispanic constituencies do not necessarily shift administrative attention toward the Black-White disparity, especially when boards face weak accountability for minority student outcomes (Flavin and Hartney 2017) and Hispanic board members may direct resources toward their own constituents' schools (Fischer 2023). A second is that SMD reform changed who governs but not who benefits from governance: with Black representation rarely produced, the reform did not counteract differential secular trends in Black versus White suspension rates. The present design cannot adjudicate between these accounts.

6 Discussion and Conclusion

This paper provides the first systematic evidence linking CVRA-induced electoral reform to racial discipline gaps. Under the maintained parallel-trends design, the CVRA-mandated transition to single-member district elections—which increased minority school board representation primarily among Hispanic candidates rather than Black candidates—did not narrow the Black-White suspension gap. In the share of distinct students suspended (my primary outcome), the effect is an informative null—statistically equivalent to zero within ± 1.0 pp (+0.33 pp; $p = .41$); counted by suspension incidents it is marginally wider (+1.61; $p = .069$) but not robust across estimators, and none of the main treatment-effect estimates (gap, incident, total rate) survives multiple-testing correction. Overall suspension rates declined modestly, while the targeted Black-White disparity did not move. That total-rate decline should be read against the statewide reform floor introduced above: California was already pushing districts toward less exclusionary discipline and toward greater support for high-need students, so the policy-relevant null is not whether discipline fell statewide but whether CVRA-induced governance changes closed the Black-White gap beyond that background. The moderation analysis is consistent with a *race-match condition*: the incident-gap treatment association is substantially smaller in districts with higher Black teacher shares (-85 contemporaneously and -49 with a one-year lag per unit Black teacher share; the contemporaneous association survives BY multiple-testing correction). A similar pattern appears for board racial composition (-28.8 per unit Black board share; $p = .026$), though this result is sensitive to the BISG race-classification specification and does not survive multiple-testing correction. Because these moderation patterns rest on endogenous post-treatment composition—and because a predetermined check interacting treatment with pre-treatment Black voting-age share does not reproduce them—I read them as suggestive rather than causally identified. "Minority" representation is not fungible across racial groups.

Implications for voting rights. The CVRA and similar state-level voting rights statutes are powerful tools for increasing minority representation, but their downstream effects depend on which minority groups benefit. In California, where the Latino population is large and geographically concentrated, CVRA enforcement operates primarily through Latino representation—the raw Hispanic board share rises +9.3 pp pre/post and prior work finds significant Latino representation effects (Abott and Magazinnik 2020); my Sun–Abraham first-stage estimate is directionally consistent but imprecise (+4.6 pp, $p = .11$), while the Black first stage is flat (-0.5 pp, $p = .55$). For Black communities—which are smaller and less residentially concentrated in California than in the South—the CVRA-induced transition to SMD elections may not generate the representation gains needed to reduce Black-specific disparities. The challenge is compounded by the weak accountability environment in school board elections: Payson (2017) shows that incumbents face performance accountability only in higher-turnout presidential-year elections, while in the off-cycle elections that characterize most school board races, organized interests dominate and performance signals are muted. Combined with the racial asymmetry in accountability documented by Flavin and Hartney (2017)—where only White student outcomes predict reelection—the baseline for translating representation into equitable outcomes is low. Voting rights advocates should consider complementary strategies, such as targeted candidate recruitment and support, community organizing, and direct policy advocacy, to ensure that electoral reform benefits all minority groups.

A final interpretive lens comes from the selection logic of electoral barriers. Anzia and Berry (2011) show that when a group faces a higher barrier to office, the candidates who clear it are positively selected on the qualities voters value—and outperform once in office. The CVRA operates on exactly this margin, in reverse: replacing at-large with trustee-area elections lowers the barrier for whichever group is geographically positioned to win. Two implications follow. First, representation produced by barrier removal should carry a smaller selection premium than the heavily selected minority representation studied in much of the descriptive-representation literature, so the downstream effects of reform-induced representation may be systematically smaller than that literature predicts—a consideration that applies even to the Hispanic representation gains documented here. Second, a lowered barrier helps only groups positioned to clear it: where Black candidates and constituencies remain below the electoral-capacity threshold, reform can raise descriptive representation for one minority group without producing Black representation at all. The selection frame thus complements the asymmetric first stage: it speaks both to why representation rose for one group and not the other, and to why even realized representation gains need not translate into gap reductions.

Implications for education policy. The teacher-diversity patterns are consistent with race-matched teacher hiring as a possible lever for reducing exclusionary discipline (Lindsay and Hart 2017), part of a broader literature on the benefits of same-race teachers (Dee 2005; Gershenson et al. 2022). The BW-gap treatment association is substantially smaller in districts with higher Black teacher shares, both contemporaneously and at a one-year lag. This pattern is consistent with a growing literature showing that same-race teachers reduce exclusionary discipline for Black students, plausibly through more culturally responsive behavioral expectations and stronger teacher-student relationships. My spending results *contrast* with Fischer's (2023) finding that same-race board members shift spending toward their group's schools: per-pupil spending does not moderate the Black-White discipline gap here, so any detectable moderation operates through personnel and representation rather than resource reallocation, and the board-composition pattern is itself fragile (BISG-specification-sensitive and not robust to multiple testing). The smaller Black-White gap in higher Black-teacher-share districts is therefore consistent with, but does not establish, a benefit of Black teacher recruitment; like the board result it rests on endogenous post-treatment composition and is not causally identified, so it is best read as a correlational pattern warranting further study rather than a demonstrated policy lever.

Limitations. This study faces several important limitations that qualify the conclusions. First, the parallel-trends assumption is not unambiguously validated: the joint F-test on pre-treatment coefficients rejects. This concern bears less heavily on a null than on a positive estimate—for the primary unduplicated gap, the formal sensitivity analysis reported in Appendix D shows the robust confidence set includes zero at every smoothness restriction, so the null is not an artifact of the parallel-trends assumption. The design-based checks corroborate the null: leave-one-cohort-out leaves it unchanged, and the only positive signal—

the incident-based gap—is itself fragile, with a triple-difference of +1.69 ($p = .12$) and randomization inference of $p = .078$. Readers should nonetheless weigh the pre-trends rejection in interpreting the marginal incident estimate.

Second, the BISG race predictions for board candidates, while the best available method, introduce measurement error that affects the race-match moderation results. BISG mean confidence for Black candidates is only 0.477, compared to 0.821 for Hispanic and 0.726 for White candidates. To assess sensitivity, I re-estimated the BW gap $\times$ Black board share interaction under five alternative BISG specifications: continuous posterior probabilities (the BISG-sensitivity continuous-posterior spec, which uses a different estimation panel from the headline race-match specification reported in Table 4), hard argmax classification, posterior thresholds of 0.3 and 0.5, and a high-confidence-only filter. The interaction is significant only under the continuous-posterior spec (-20.7; $p = .04$); under all four alternatives, the interaction is insignificant ($p > .50$). This sensitivity indicates that the race-match moderation result for board composition depends on the BISG specification and should be interpreted with caution. The teacher diversity results, which use administratively reported race data rather than BISG predictions, are not subject to this limitation and provide stronger evidence for race-matched moderation.

Third, the study is specific to California, a state with distinctive racial demography (large Hispanic, small Black population), a particular legal framework (CVRA), and a specific political culture. The race-match condition may manifest differently in states with larger Black populations, different patterns of residential segregation, or federal Voting Rights Act coverage. Extensions to Southern states, where SMD transitions have been driven by federal law and Black populations are larger, would provide a valuable test of external validity.

Fourth, the relatively small Black population in many California school districts limits the precision and coverage of the BW gap estimates ($\approx$57% of observations). While this coverage is sufficient for the main analysis, it means that the estimates are drawn disproportionately from the subset of California districts with larger Black communities.

Relatedly, the design observes electoral rules and their outcomes but not political participation: whether the Black-side first-stage null reflects only the structural constraint of residential concentration, or also behavioral margins—registration and turnout in trustee-area elections, which the pre-treatment voting-age measure in Section 5.5 cannot capture—is unobservable in these data, and distinguishing the two is a natural question for future work.

Fifth, the panel cannot begin before 2011. California's universal, race-disaggregated district discipline series starts with CALPADS in 2011–12, and direct inquiry to the California Department of Education confirmed that the department holds no earlier suspension or expulsion data.[11] The federal Civil Rights Data Collection offers no remedy: before its first universal collection in 2011–12 it was sample-based and weighted toward large districts, which would induce coverage differential by treatment status—my treated districts are far larger than controls—and is therefore unsuitable for a balanced treated-versus-control panel. This 2011 floor shortens the pre-treatment window for the earliest cohorts and limits overlap with the earlier CVRA wave studied by Abott and Magazinnik (2020), whose 2002–2016 representation panel my discipline-constrained data cannot reproduce; the analysis accordingly cannot speak to discipline dynamics during the first decade of CVRA enforcement.

Conclusion. The question "does descriptive representation reduce racial disparities?" has no single answer. It depends on whose representation increases and whose disparities are measured. The lesson is not that descriptive representation fails, but that its policy consequences depend on which groups gain representation and which outcomes are at stake. In the California CVRA setting, electoral reform increased Hispanic representation but did not generate comparable Black representation, and the Black-White discipline gap did not move—no detectable change in the distinct-student gap, and at most a fragile, non-robust incident-based signal. Electoral institutions can change who governs, but equitable

[11] Personal communication, CDE Data Request Team, April 2022: the department confirmed it holds no district discipline data prior to 2011–12, and the older UMIRS incident-reporting series is administered through SAMHSA and likewise begins in 2011. The 2009–10 federal CRDC was a sample that force-included districts above 3,000 students and selected the remainder by stratified sampling; the CRDC first achieved universal district coverage in 2011–12 (OCR CRDC data notes, 2009–10 and 2011–12).

downstream outcomes appear to depend on whether those changes align with the communities experiencing the disparity.

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Online Appendix

A Data Construction Details

This appendix documents the construction of the analysis dataset, with detailed coverage statistics and data quality diagnostics for each data source.

A.1 CEDA Election Data

The California Elections Data Archive (CEDA) provides candidate-level records for school board elections from 1996 through 2023. I load annual Excel files, harmonize variable names across years, and standardize district names for cross-referencing with CDE data. The `area` field, which records trustee area identifiers for district-based elections, is the basis for identifying SMD systems.

Table A1: Overview of All Data Sources Used in Analysis

Data Source	Years	N Observations	N Districts	Key Variables
CEDA Election Data	1998–2024	38,340	1,437	Candidates, electoral system, winners
CDE Suspension Data	2011–2024	542,998	1,089	Suspension rates by race, gaps
CDE Expulsion Data	2011–2024	514,592	1,089	Expulsion rates by race, gaps
Analysis Dataset	2011–2024	9,844	721	Merged outcomes + treatment

Note: Suspension and expulsion data have different district coverage because CDE reporting requirements differ across discipline categories.

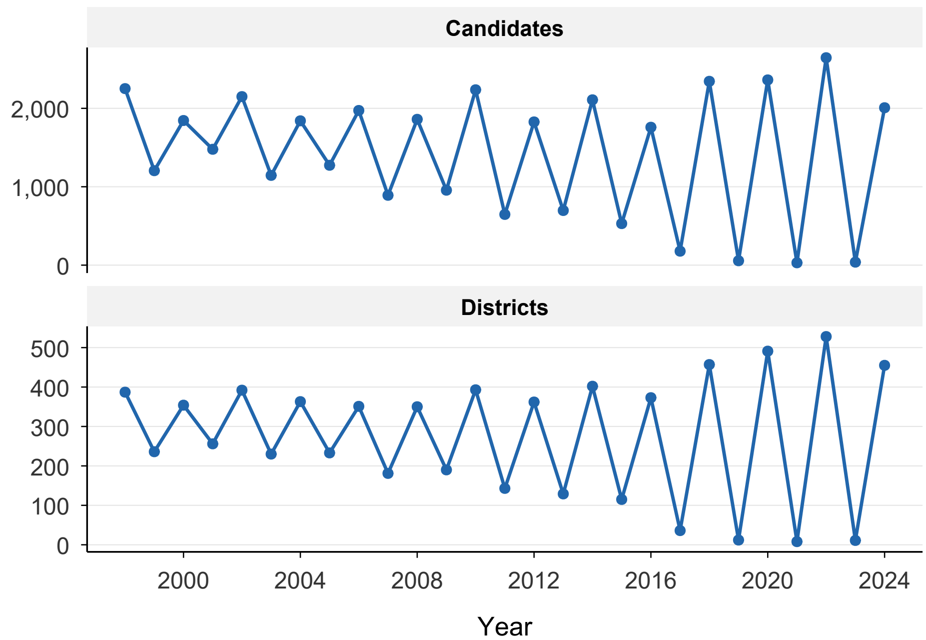

Figure A1: CEDA Election Data Coverage Over Time

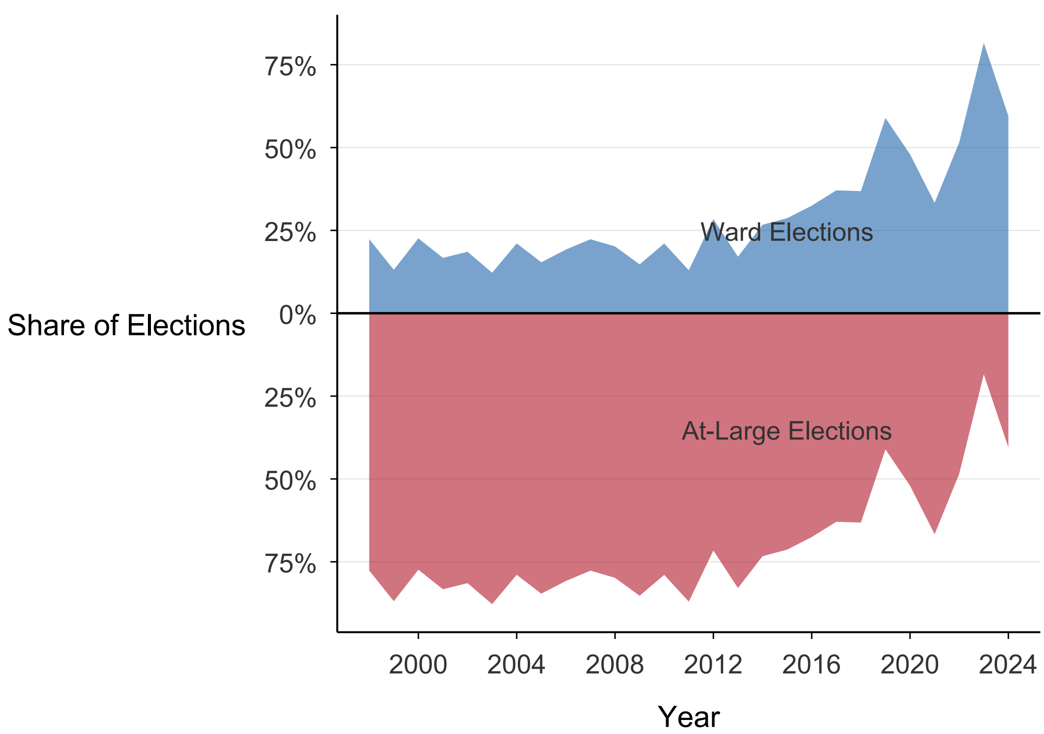

Figure A2: Electoral System Adoption Trends

Notes: Ward (SMD) elections shown above axis; at-large elections below axis. Each series shows the share of all CEDA-recorded school board elections in that year using the given electoral system.

Table A2: CEDA Election Data Coverage by Year

Year	Districts	Candidates	Avg. Cand./Dist.	Ward Elections	% Ward
1998	387	2,252	5.8	504	22.4
1999	236	1,205	5.1	158	13.1
2000	354	1,845	5.2	417	22.6
2001	256	1,478	5.8	247	16.7
2002	392	2,150	5.5	399	18.6
2003	230	1,145	5.0	140	12.2
2004	363	1,841	5.1	388	21.1
2005	233	1,274	5.5	196	15.4
2006	351	1,973	5.6	379	19.2
2007	181	891	4.9	199	22.3
2008	350	1,860	5.3	375	20.2
2009	190	956	5.0	141	14.7
2010	393	2,237	5.7	471	21.1
2011	143	647	4.5	84	13.0
2012	362	1,827	5.0	519	28.4
2013	129	697	5.4	119	17.1
2014	402	2,109	5.2	563	26.7
2015	115	530	4.6	152	28.7
2016	373	1,759	4.7	571	32.5
2017	36	178	4.9	66	37.1
2018	457	2,345	5.1	864	36.8
2019	12	56	4.7	33	58.9
2020	491	2,363	4.8	1,133	47.9
2021	8	30	3.8	10	33.3
2022	528	2,646	5.0	1,364	51.5
2023	11	38	3.5	31	81.6
2024	455	2,008	4.4	1,195	59.5

A.2 District Characteristics

The table below summarizes the CEDA district universe and electoral system classifications. Of the districts observed in CEDA, a substantial majority used at-large elections throughout the sample period, while a smaller share transitioned to single-member districts. Figure A3 shows the distribution of years covered per district; the broader CEDA universe is sparsely covered (median about 4 years), whereas the analysis panel of 721 districts is nearly balanced over the 2011–2024 window.

Table A3: CEDA District Summary Statistics

Category	Value
Total CEDA districts	1,437
Classified districts	1,568
Years of coverage (median)	4.0
Years of coverage (mean)	5.2
Districts with 10+ years	301
Districts with 20+ years	0
Ever used ward elections	590
Always ward elections	238
Always at-large elections	978
Switched electoral systems	352

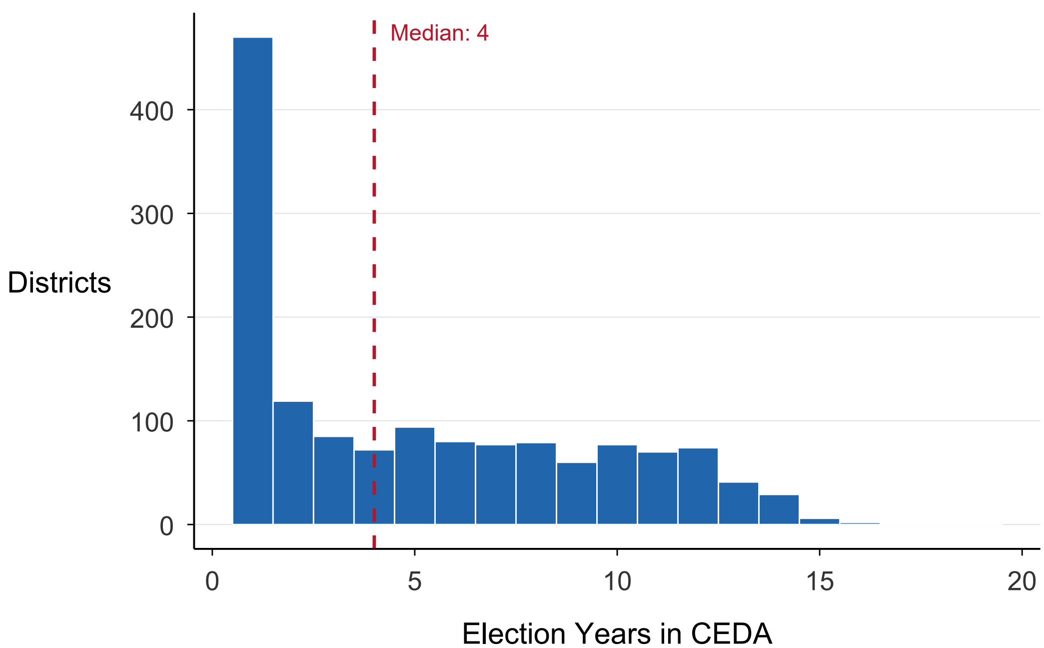

Figure A3: Distribution of Years Covered per District

A.3 Candidate Race/Ethnicity Predictions

I predict candidates' race/ethnicity using Bayesian Improved Surname Geocoding (BISG) via the `wru` package (Imai and Khanna 2016).

Table A4: BISG Race/Ethnicity Predictions for School Board Candidates

Race/Ethnicity	Candidates	%	Mean Confidence
White	25,334	66.1	0.726
Hispanic	10,708	27.9	0.821
Asian	1,676	4.4	0.848
Black	569	1.5	0.477
Other	52	0.1	0.578
Total	38,340	100.0	0.754

Notes: Race/ethnicity predicted via Bayesian Improved Surname Geocoding (BISG) using the `wru` package (Imai and Khanna 2016). Confidence is the maximum posterior probability across racial categories (argmax classification). Mean predicted minority probability: 48.6%.

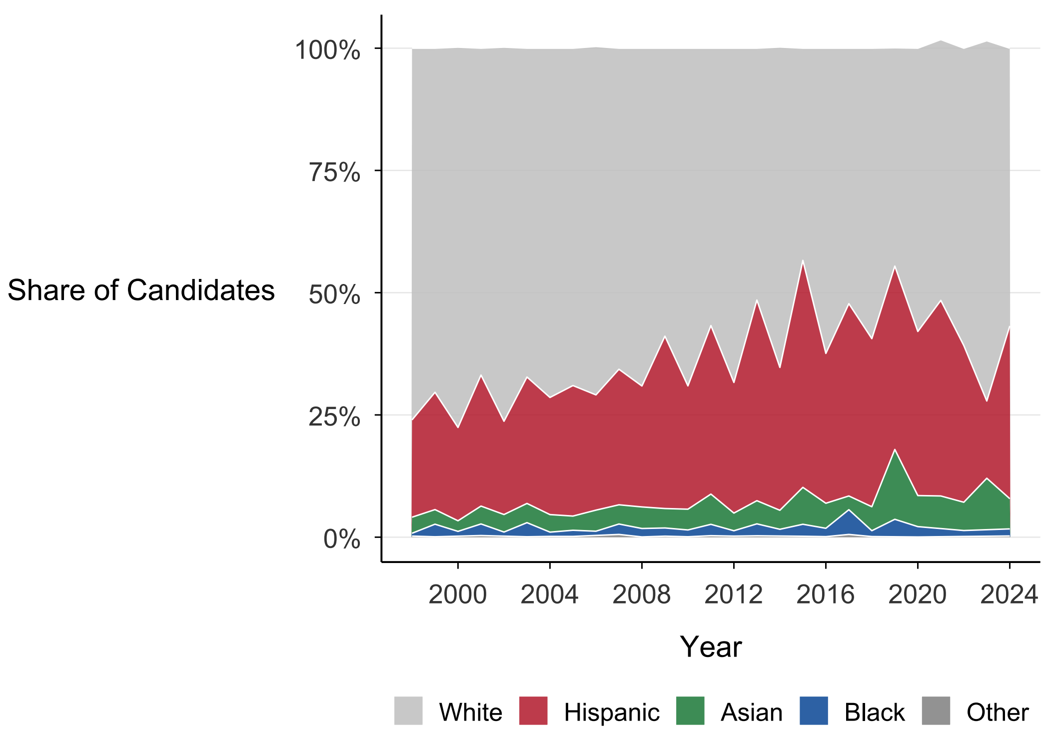

Figure A4: Racial/Ethnic Composition of Candidates Over Time

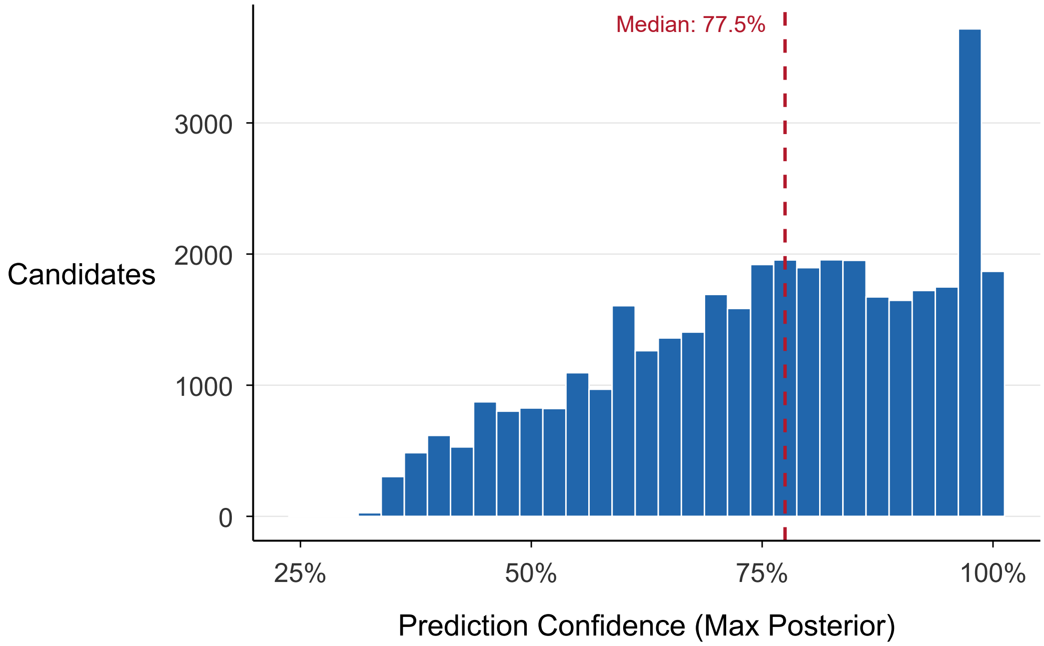

Figure A5: Distribution of BISG Prediction Confidence

A.4 CDE Suspension Data

Suspension data are downloaded from the California Department of Education for academic years 2010–11 through 2023–24. The coverage table below reports the number of districts with suspension data in each year; coverage is stable at approximately 700 districts per year. Figure A6 plots the mean BW and HW suspension rate gaps over time, separately for treated and control districts, while Figure A7 shows the cross-sectional distribution of gaps.

Table A5: Suspension Data Coverage by Year

Year	N	Valid Gap		Mean Rate (p.p.)		
		BW	HW	BW Gap	HW Gap	Black
2011	716	406	627	10.6	0.3	19.3
2012	711	409	639	9.9	0.1	17.3
2013	706	418	653	8.7	0.1	15.3
2014	705	413	648	7.9	-0.1	13.6
2015	705	406	651	6.9	-0.4	12.6
2016	703	414	653	7.4	-0.5	13.0
2017	703	409	654	7.2	-0.0	11.8
2018	700	404	652	6.3	-0.0	10.9
— *COVID years 2019–20, 2020–21 excluded* —						
2021	700	399	650	6.0	0.2	10.1
2022	698	392	651	6.8	0.1	11.5
2023	695	391	650	6.5	-0.1	11.0
2024	697	388	650	5.9	-0.3	9.9

Notes: Non-COVID district-year observations from the analysis panel. N = districts observed. Valid Gap = districts with sufficient enrollment to compute gap. Gaps and rates in percentage points.

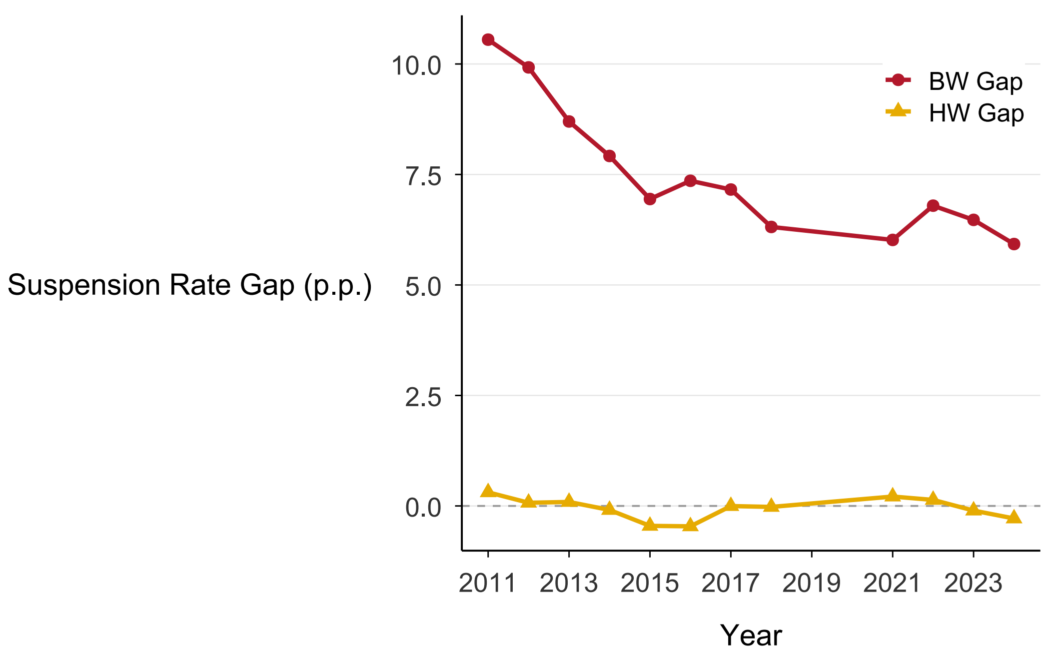

Figure A6: Mean Suspension Rate Gaps Over Time

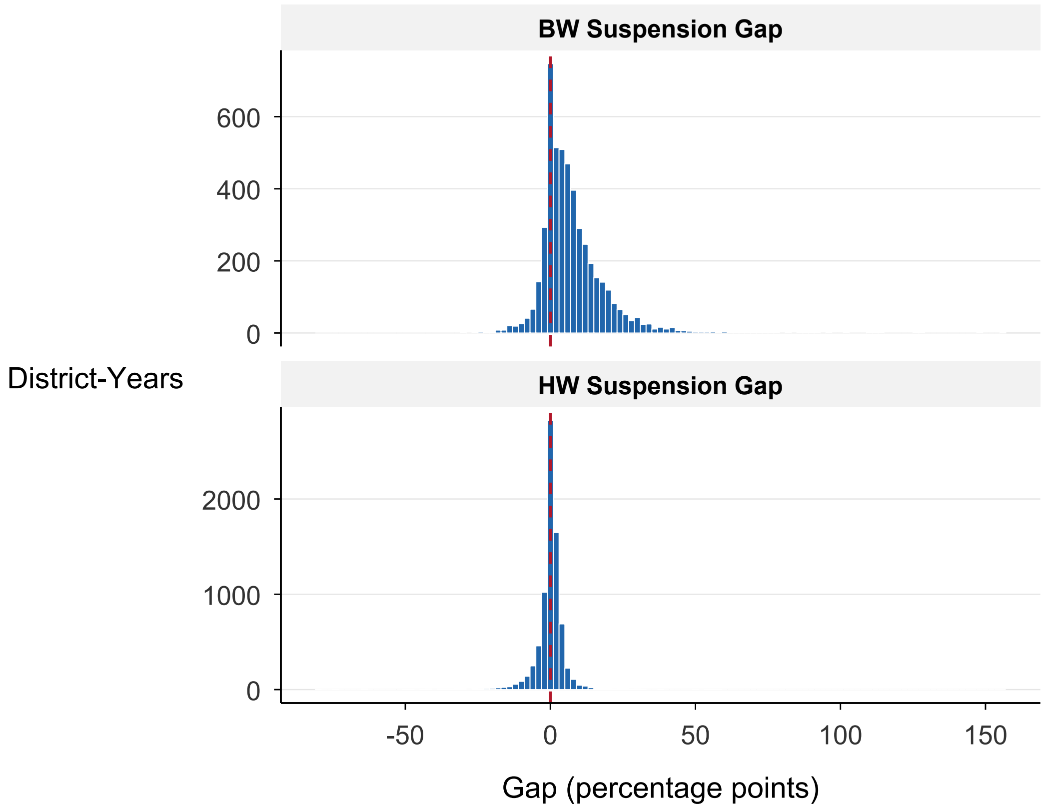

Figure A7: Distribution of Racial Suspension Gaps

Table A6: Summary Statistics: Suspension Rates by Race/Ethnicity

Race	N	Mean	SD	Median	25th	75th
Black	4,860	13.1	14.9	9.1	3.4	17.9
Hispanic	7,957	5.4	6.0	4.0	1.5	7.2
White	8,083	5.4	7.0	3.4	1.2	7.2
Asian	5,188	2.1	4.3	0.9	0.0	2.5
Total	8,353	5.4	6.5	3.7	1.5	7.3

Notes: Suspension rates in percentage points (suspensions per 100 enrolled students). N = non-COVID district-year observations with non-missing enrollment for the given group. Total includes all students regardless of race.

A.5 Expulsion Data

Expulsion data come from the same CDE source and cover the same 2011–2024 window. Coverage is stable at approximately 1,020 districts per year throughout the sample period. Expulsion rates are an order of magnitude smaller than suspension rates and are used as a secondary outcome. Figure A8 shows the relationship between suspension and expulsion BW gaps at the district-year level.

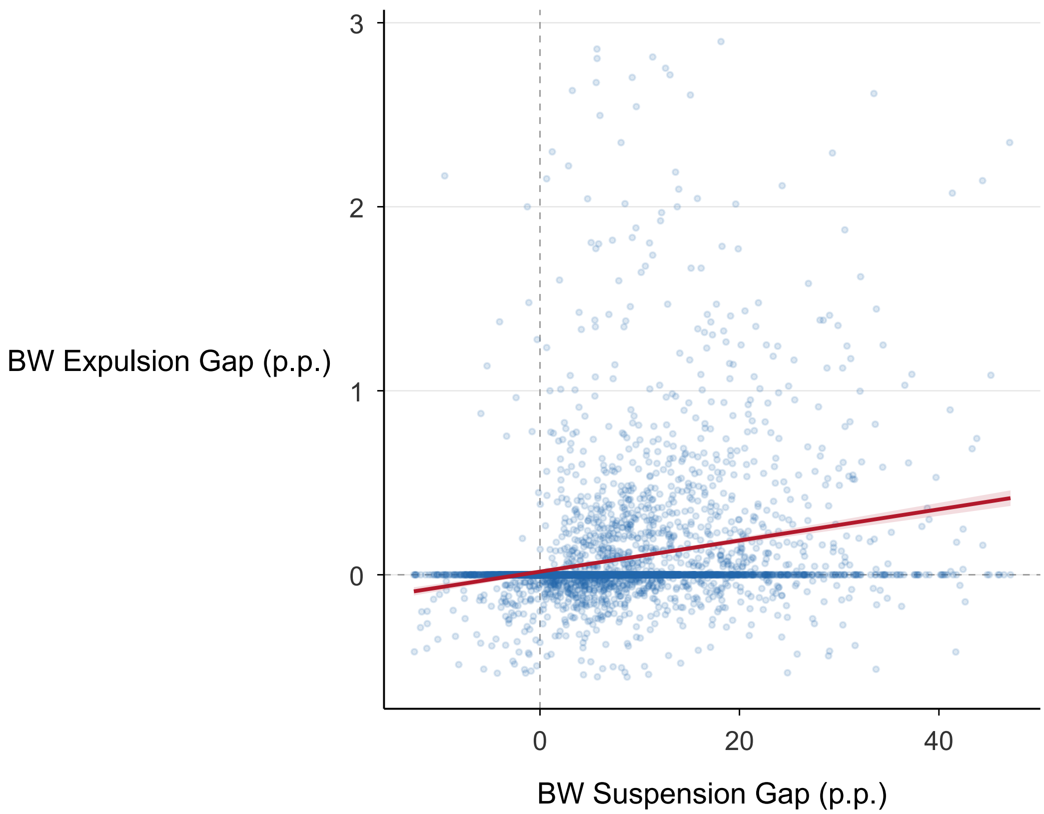

Figure A8: Relationship Between Suspension and Expulsion Gaps

Table A7: Expulsion Data Coverage by Year

Year	Districts with Data
2011	1,044
2012	1,038
2013	1,028
2014	1,022
2015	1,024
2016	1,023
2017	1,027
2018	251
2019	1,033
2020	1,029
2021	1,020
2022	1,019
2023	1,010
2024	1,015

A.6 Treatment Timing

The table below reports the distribution of treatment cohorts—the year in which each district first held a single-member district election. The largest cohorts are 2022 (51 districts), 2020 (47), and 2018 (39), reflecting the acceleration of CVRA-induced transitions in the late 2010s. Figure A9 visualizes the treatment stagger across districts and years, and Figure A10 shows the distribution of post-treatment exposure (number of post-treatment years observed).

Table A8: Distribution of Treatment Timing (Ward Election Adoption)

Cohort Year	Districts	%	Cum. %
2011	2	0.9	0.9
2012	21	9.5	10.4
2013	6	2.7	13.1
2014	20	9.0	22.1
2015	9	4.1	26.1
2016	23	10.4	36.5
2017	3	1.4	37.8
2018	39	17.6	55.4
2020	47	21.2	76.6
2022	51	23.0	99.5
2023	1	0.5	100.0
Total	222	100.0	

Notes: Cohort year = year of first single-member district election. Cohorts before 2011 adopted SMD prior to the analysis window; 0 of 222 treated districts fall in this group. The analysis window covers 2011–2024 (excluding COVID years 2019–20, 2020–21).

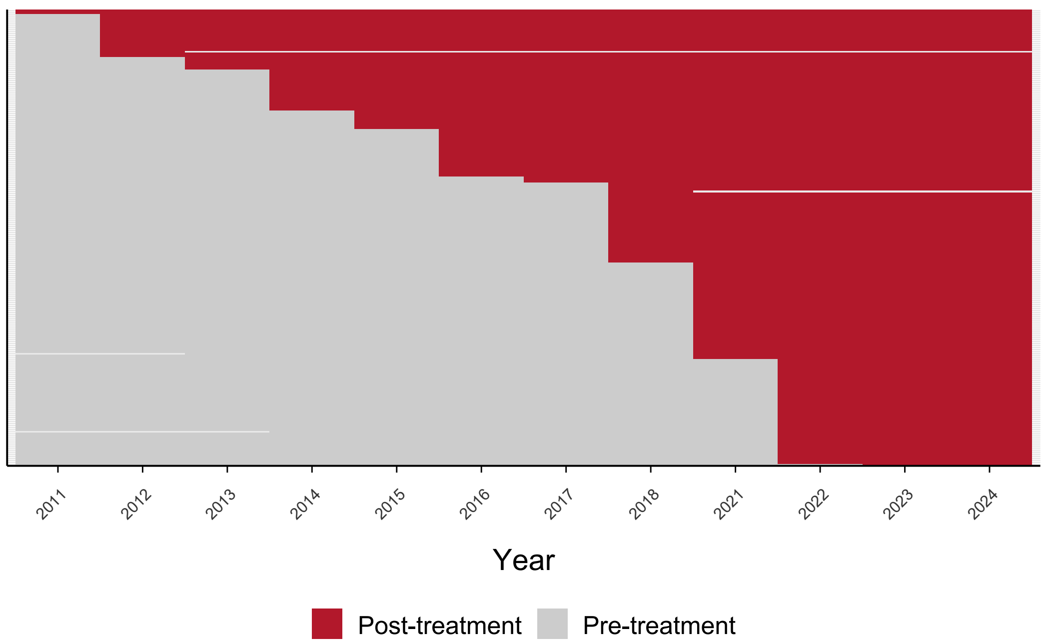

Figure A9: Treatment Stagger Visualization

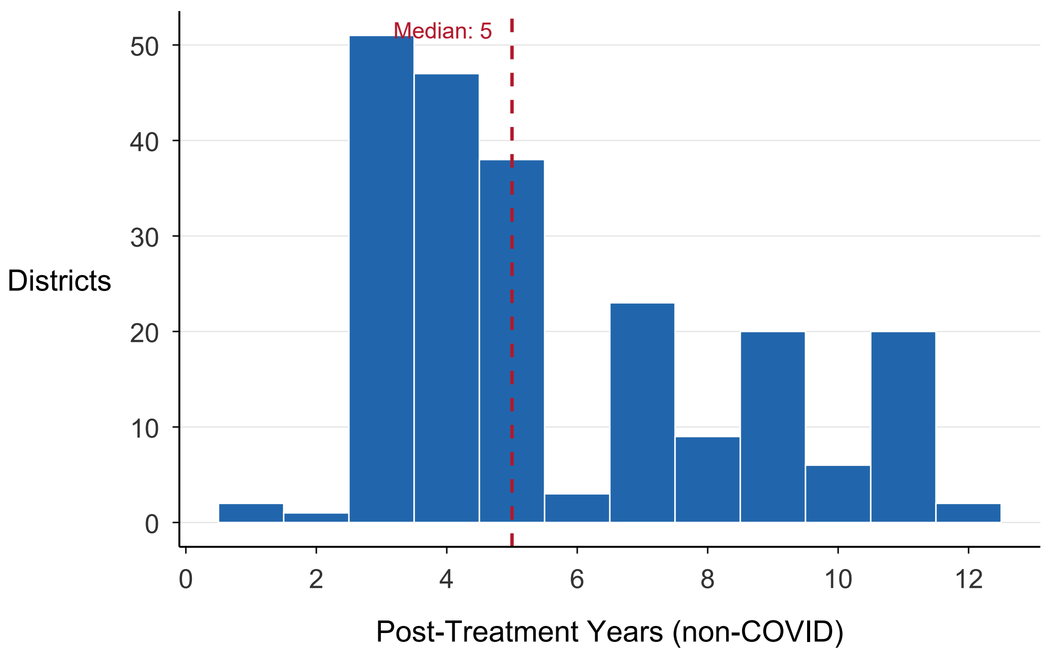

Figure A10: Distribution of Post-Treatment Exposure

A.7 Analysis Sample

The tables below summarize the composition of the final analysis panel after merging treatment and outcome data and excluding COVID years. The panel covers 721 districts over 2011–2024 (8,439 non-COVID district-year observations). The balance table compares pre-treatment means for treated and control districts; treated districts are larger, more diverse, and start with higher discipline gaps. Figure A11 maps sample coverage across California counties. The final table documents missing data patterns; the BW gap is available for approximately 57% of observations (due to small Black enrollment in many California districts), while the HW gap has approximately 92% coverage.

Table A9: Final Analysis Sample Composition

Category	Value
Total observations	8,439
Unique districts	721
Years covered	2011 - 2024
Treated district-years ($SMD_{dt} = 1$)	1,270
Untreated district-years ($SMD_{dt} = 0$)	7,169
Pre-treatment observations	7,169
Post-treatment observations	1,270
Districts with valid BW gap	466
Districts with valid HW gap	690
Avg years per district	11.7

Table A10: Balance Table: Pre-Treatment Characteristics

	Control		Treated		
Variable	Mean	SD	Mean	SD	Norm. Diff
Suspension rate (total)	4.97	6.83	6.84	6.30	0.28
Suspension rate (Black)	11.42	15.85	15.75	15.09	0.28
Suspension rate (Hispanic)	4.81	6.43	7.02	5.66	0.36
Suspension rate (White)	5.17	7.52	6.09	5.84	0.14
BW suspension gap	6.10	12.52	9.66	11.61	0.30
HW suspension gap	-0.36	5.50	0.94	3.85	0.27
Expulsion rate (total)	0.05	0.19	0.10	0.17	0.29
BW expulsion gap	0.10	0.75	0.15	0.54	0.08
HW expulsion gap	0.01	0.32	0.04	0.15	0.12
Enrollment (total)	2,576	5,469	11,111	10,606	1.01

Notes: Pre-treatment means for control districts (all years) and treated districts (pre-treatment years only). Rates and gaps in percentage points. Normalized difference = $(\bar{x}_T - \bar{x}_C)/\sqrt{(s_T^2 + s_C^2)/2}$. Values > $|0.25|$ suggest meaningful imbalance.

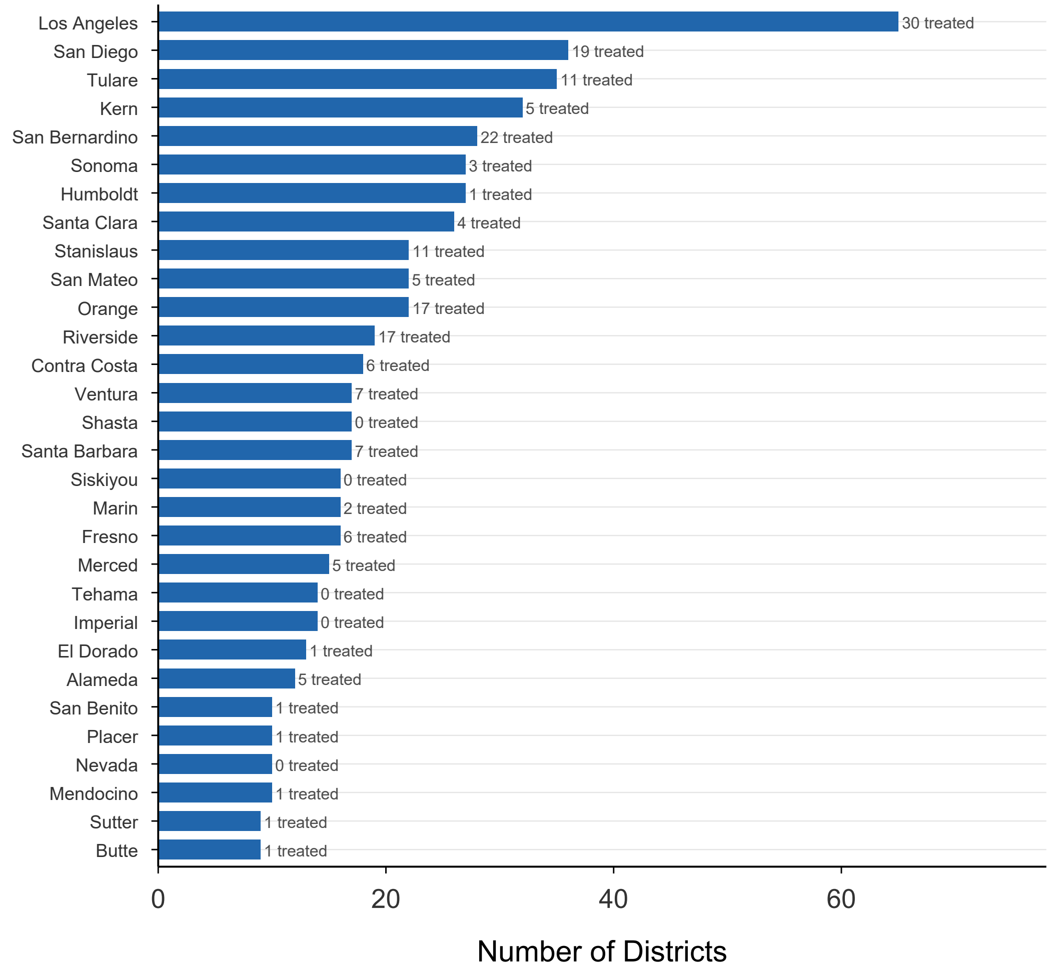

Figure A11: Sample Coverage by County

Table A11: Missing Data Patterns in Analysis Dataset

Variable	N Valid	N Missing	% Missing
Suspension rate (total)	8,353	86	1.0
Suspension rate (Black)	4,860	3,579	42.4
Suspension rate (Hispanic)	7,957	482	5.7
Suspension rate (White)	8,083	356	4.2
Suspension rate (Asian)	5,188	3,251	38.5
BW suspension gap	4,849	3,590	42.5
HW suspension gap	7,778	661	7.8
Expulsion rate (total)	7,829	610	7.2
BW expulsion gap	4,578	3,861	45.8
HW expulsion gap	7,295	1,144	13.6
Enrollment (total)	8,439	0	0.0

Notes: Out of 8,439 non-COVID district-year observations. BW gap missingness is driven by small Black enrollment: CDE suppresses race-level counts where a group's cumulative enrollment is 10 or fewer (verified against the raw files: the smallest visible race-cell enrollment is 11 in every vintage), and the analysis additionally requires at least 10 enrolled Black and White students for a valid gap. HW gap has higher coverage due to California's large Hispanic population.

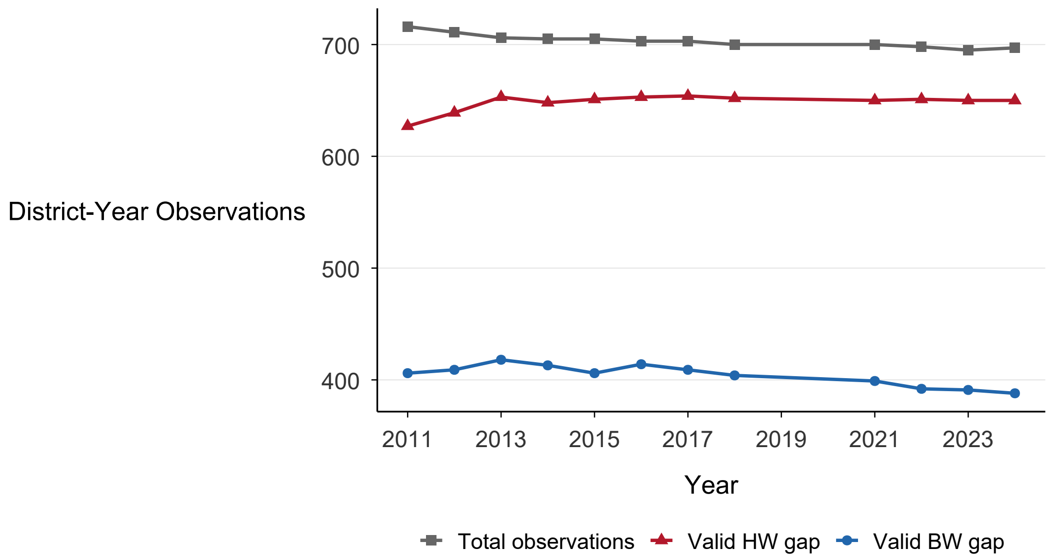

Figure A12: Data Availability Over Time

B Additional Descriptive Figures

This appendix presents supplementary descriptive evidence on suspension rate trends. Figure A13 plots race-specific suspension rates (Black, Hispanic, White) separately for treated and control districts, showing that the treated–control divergence is concentrated among Black students. Figure A14 presents raw HW gap trends, which show no clear divergence—consistent with the null DiD estimates. Figures A15 and A16 restrict the sample to pre-treatment years for each cohort, providing a direct visual check of the parallel trends assumption.

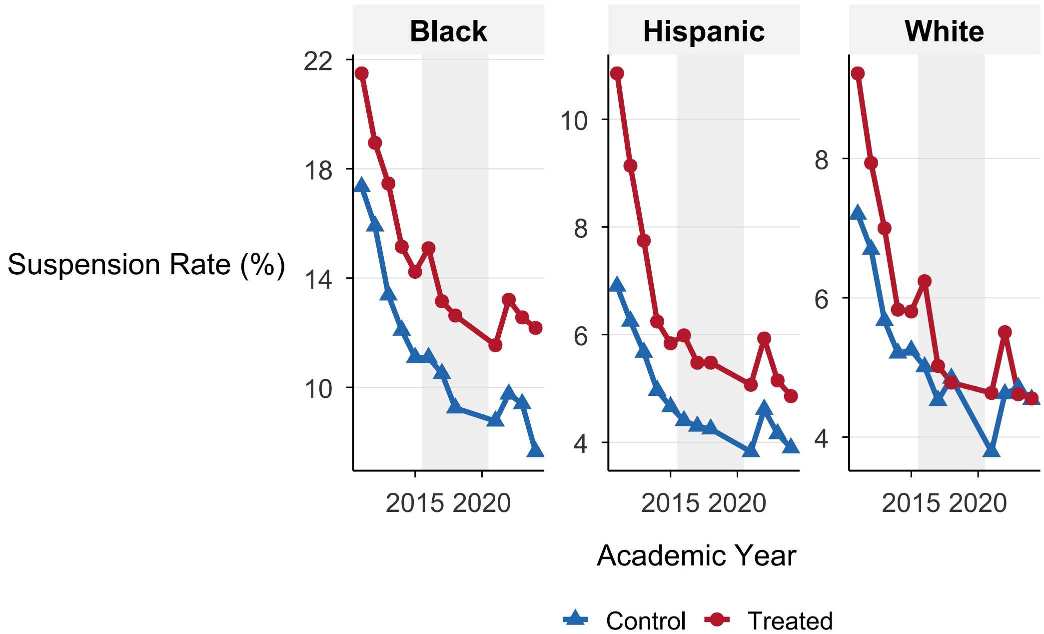

Figure A13: Race-Specific Suspension Rates by Treatment Group

Notes: Mean suspension rates by race (Black, Hispanic, White) for treated (solid) and control (dashed) districts. The shaded band marks the modal treatment window. Black rates are highest and show the largest treated-control divergence. COVID years excluded.

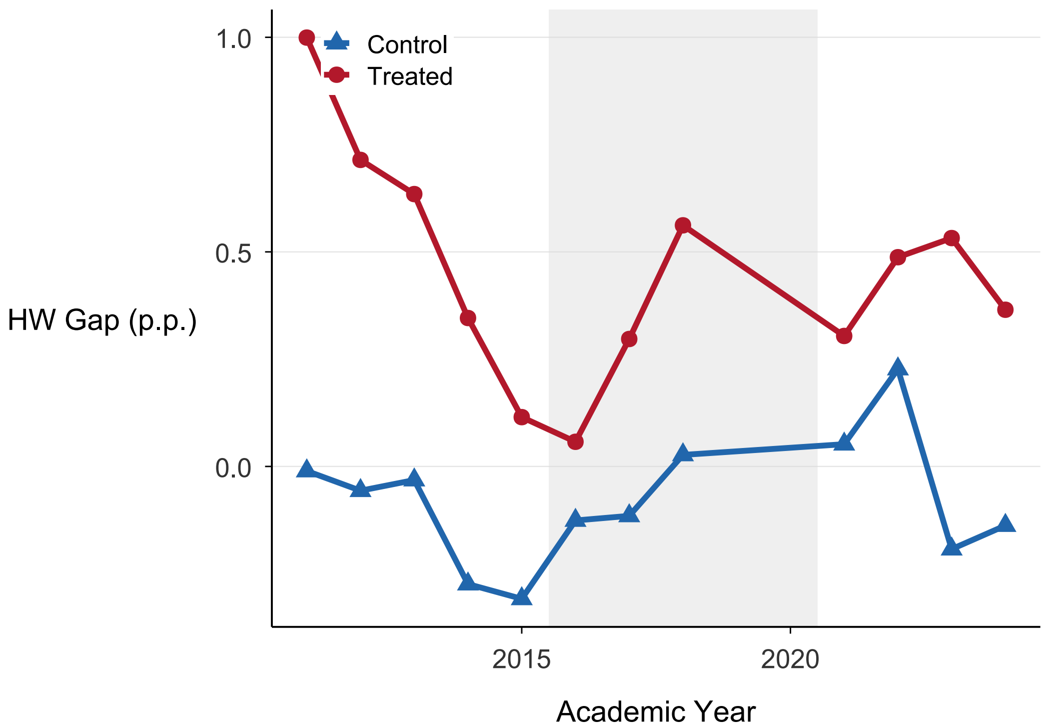

Figure A14: Raw Trends: Hispanic-White Suspension Rate Gap

Notes: Annual mean Hispanic-White suspension rate gap (percentage points) for treated and control districts. Unlike the BW gap (Figure 2), the HW gap shows no clear divergence after the modal treatment window, consistent with the null DiD estimates.

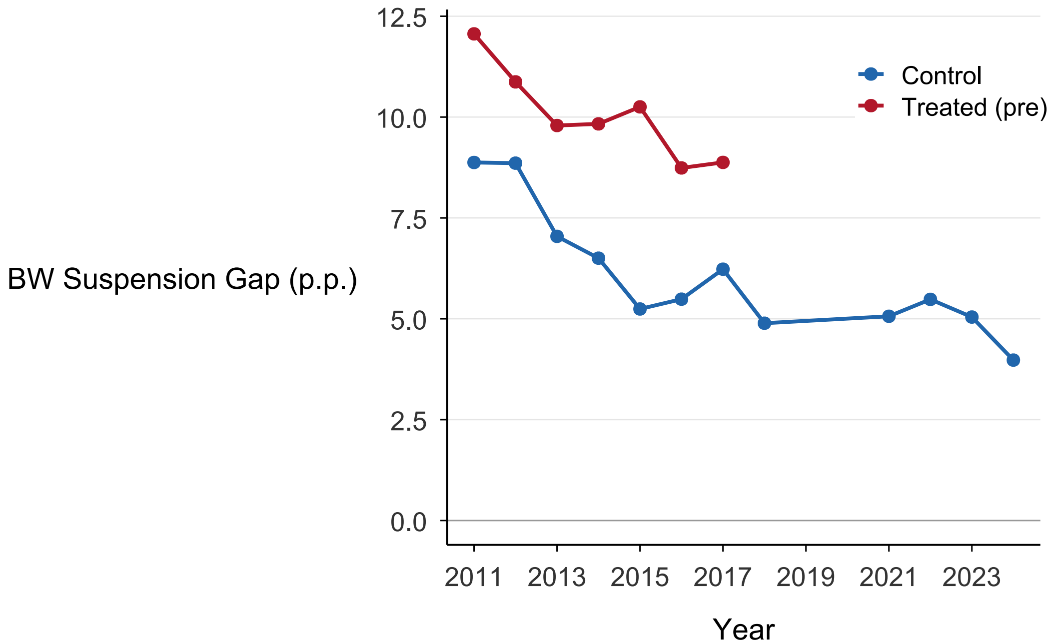

Figure A15: Pre-Treatment Trends: BW Suspension Gap

Notes: Pre-treatment BW suspension gap trends for treated and control districts, restricted to years before each cohort's first treatment year. Both groups show declining gaps in the early period, with no systematic divergence.

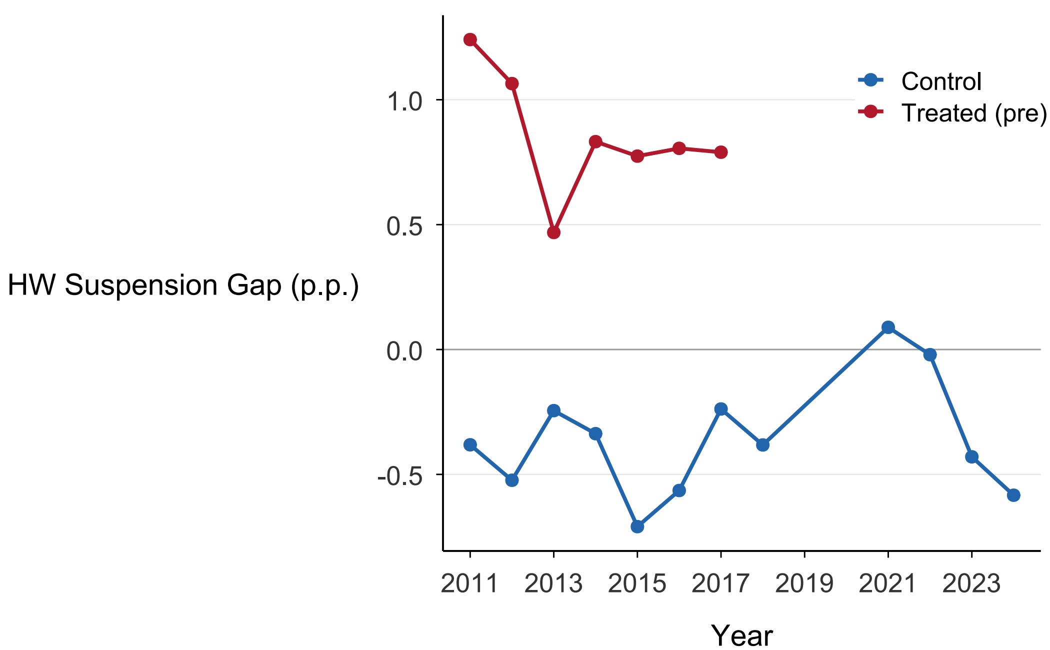

Figure A16: Pre-Treatment Trends: HW Suspension Gap

Notes: Pre-treatment HW suspension gap trends for treated and control districts.

C Additional Event Studies

This appendix reports event study estimates for outcomes not shown in the main text. Figure A17 presents the incident (event-count) HW gap event study across all three heterogeneity-robust estimators. Post-treatment coefficients are all indistinguishable from zero; the only exceptions pre-treatment are two positive leads at the most distant horizon ($t = -5$, SA and CS), where cohort composition is thinnest—the same distant-lead pattern noted for the unduplicated HW facet in Figure 5. Figure A18 shows the same for the total suspension rate. Figures A19 and A20 provide diagnostic plots for the synthetic DiD estimator, including the period-by-period gap between observed and synthetic control units and cohort-specific SDID estimates.

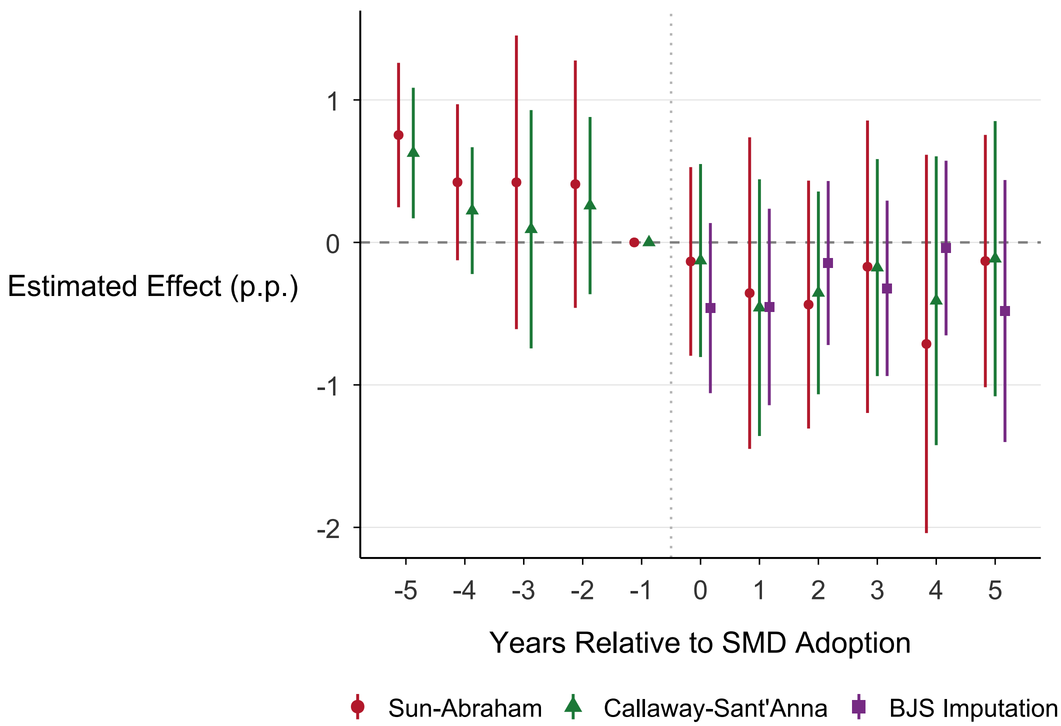

Figure A17: Event Study: Hispanic-White Suspension Gap

Notes: Sun-Abraham, Callaway-Sant'Anna, and BJS event study estimates for the incident (event-count) Hispanic-White suspension gap; the main text's unduplicated HW variant appears in Figure 5. The dependent variable is in percentage points. Whiskers show 95% confidence intervals. COVID years excluded.

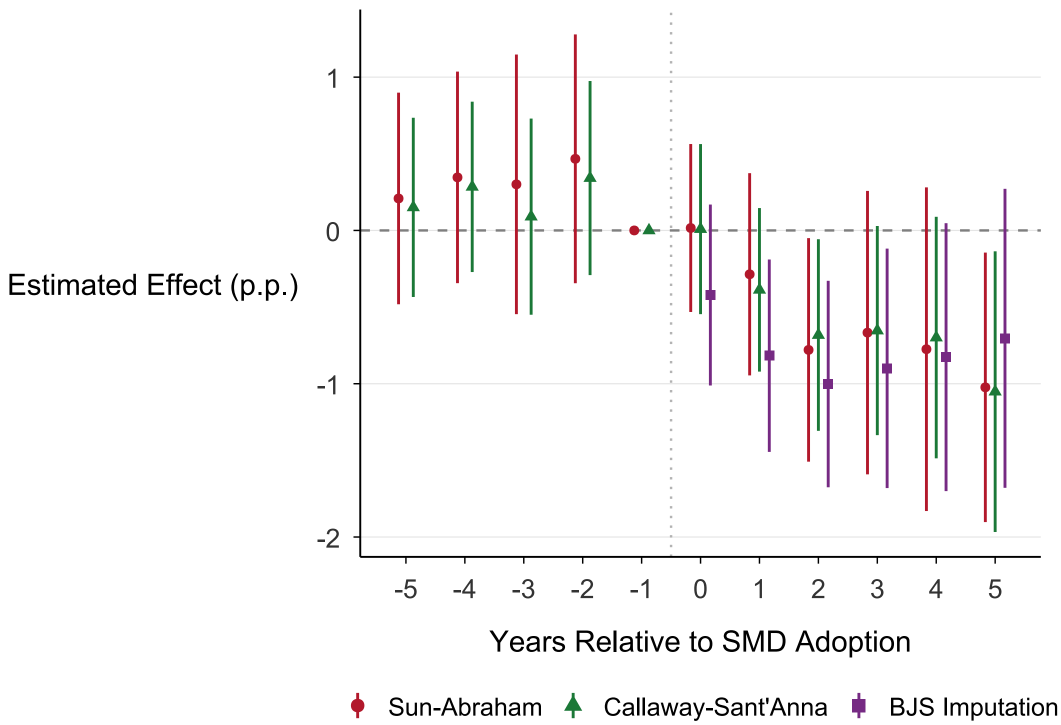

Figure A18: Event Study: Total Suspension Rate

Notes: Sun-Abraham, Callaway-Sant'Anna, and BJS event study estimates for the total suspension rate per 100 students. Whiskers show 95% confidence intervals. COVID years excluded.

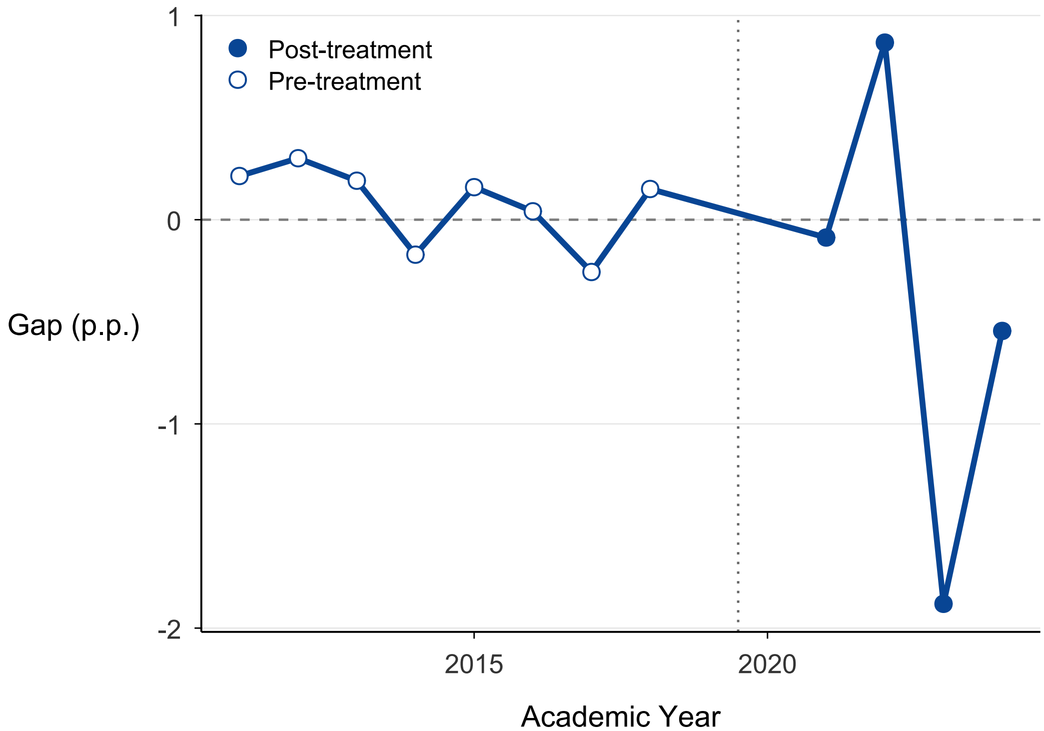

Figure A19: SDID Gap Plot: BW Suspension Gap (Diagnostic)

Notes: Period-by-period difference between observed and SDID synthetic control for the BW gap (largest treatment cohort, 2020). Pre-treatment residuals exhibit meaningful fluctuation (± 0.8 p.p.), reflecting the difficulty of constructing a close synthetic match for the BW gap—consistent with the smaller, statistically insignificant SDID point estimate relative to DiD estimators. Filled points are post-treatment periods.

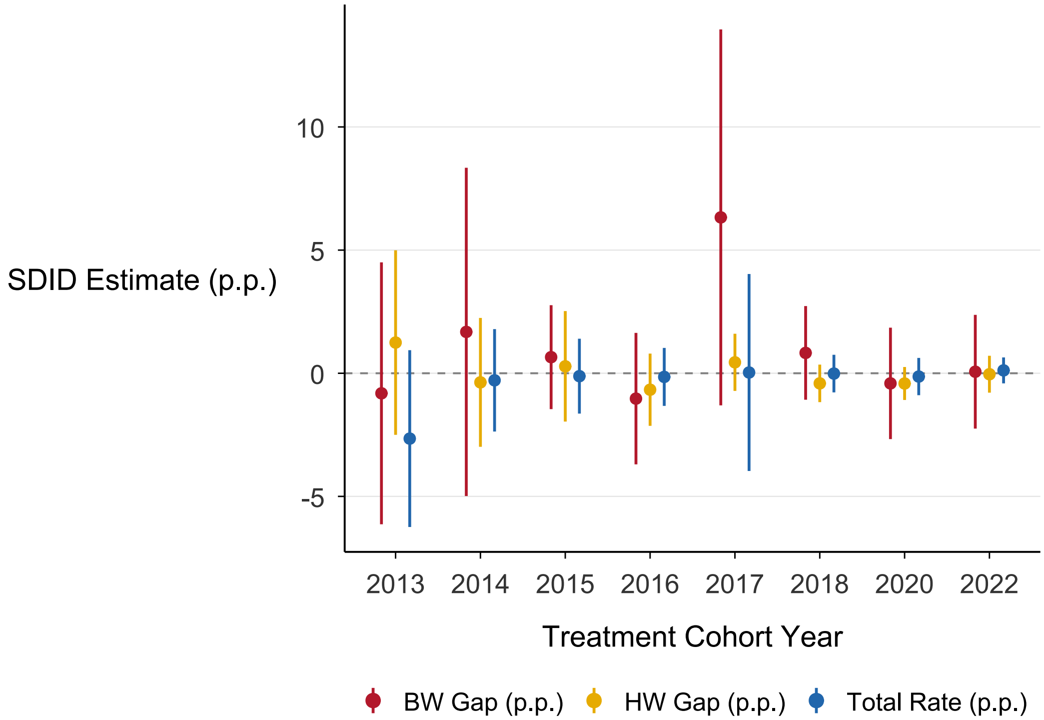

Figure A20: SDID Estimates by Treatment Cohort

Notes: Cohort-specific SDID ATT estimates with 95% bootstrap confidence intervals for BW gap, HW gap, and total suspension rate. Most cohort-specific estimates for the BW gap are positive, though only the largest cohorts reach conventional significance. HW gap and total rate estimates are centered near zero.

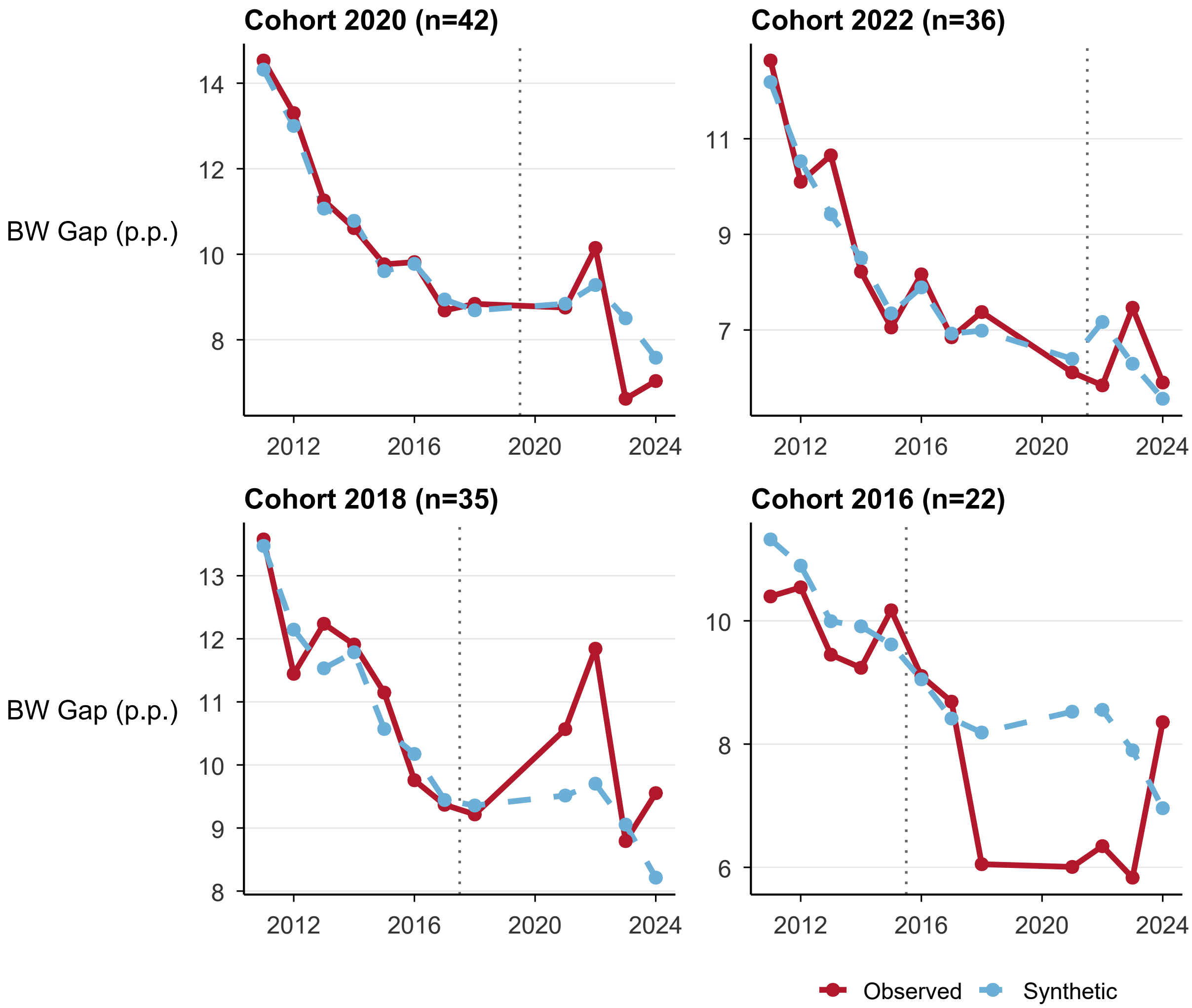

Figure A21: SDID Counterfactual Trajectories: BW Suspension Gap

Notes: Observed treated-group mean vs. SDID synthetic control for the four largest treatment cohorts. Each panel shows the outcome trajectory in levels. The dotted vertical line marks the cohort's first treatment year. Following Lal and Thompson (2024), no error bars are shown on the trajectory plot.

D Robustness Details

This appendix provides additional detail on the robustness checks summarized in Section 4.4. Across alternative specifications the Sun-Abraham ATT for the primary (unduplicated) BW gap is small and statistically insignificant—+0.33 at baseline, +0.27 pre-COVID, +0.27

excluding the ten largest districts, and +0.33 for clean switchers only (Figure 7 in the main text). Figures A22 and A23 report the Rambachan and Roth (2023) sensitivity analysis under smoothness (ΔSD) and relative-magnitudes (ΔRM) restrictions, respectively. Under ΔSD, the robust 95% CI is $[-0.34, +1.49]$ at $M = 0$—which bounds the post-treatment change in the differential trend at zero, a linear-extrapolation benchmark weaker than assuming parallel trends—and includes zero at every $M \in [0, 0.5]$; under ΔRM, the robust CI includes zero at every $\bar{M} \in [0.5, 2]$. The primary null is therefore not an artifact of the parallel-trends assumption—relaxing it only widens an interval already centered near zero. Figure A24 presents the randomization-inference distribution; on the marginal incident gap the RI p-value is .078, consistent with a fragile incident-based signal, while the primary distinct-student gap is null. Figures A25–A30 report placebo outcome tests (including the significant %Black-enrollment placebo discussed in Section 4.4), the CS not-yet-treated comparison, leave-one-cohort-out stability, and the full specification curve.

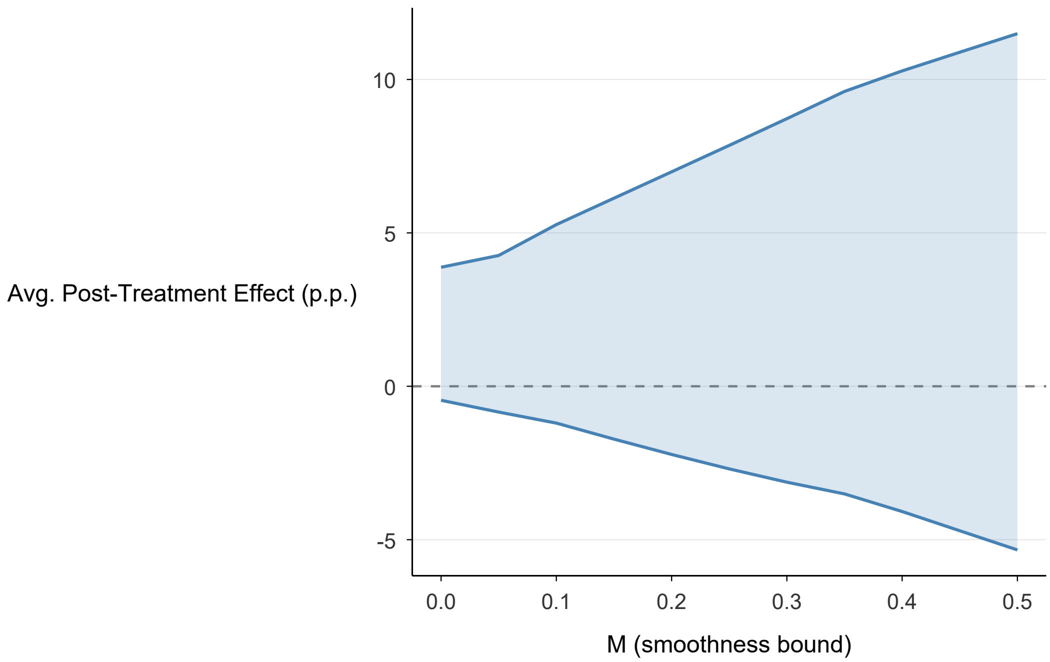

Figure A22: HonestDiD: Smoothness Sensitivity (ΔSD)

Notes: Robust confidence sets from Rambachan and Roth (2023) for the primary (unduplicated) BW suspension gap effect under smoothness restrictions (ΔSD). The x-axis shows the smoothness parameter M, which bounds the maximum change in the trend violation between consecutive periods relative to the maximum observed pre-trend. At $M = 0$, the post-treatment change in the differential trend is bounded at zero (the Rambachan-Roth benchmark), permitting the estimated linear pre-trend to extrapolate forward—a restriction weaker than, not equivalent to, assuming parallel trends; the robust CI includes zero at every $M \in [0, 0.5]$, including this $M = 0$ benchmark.

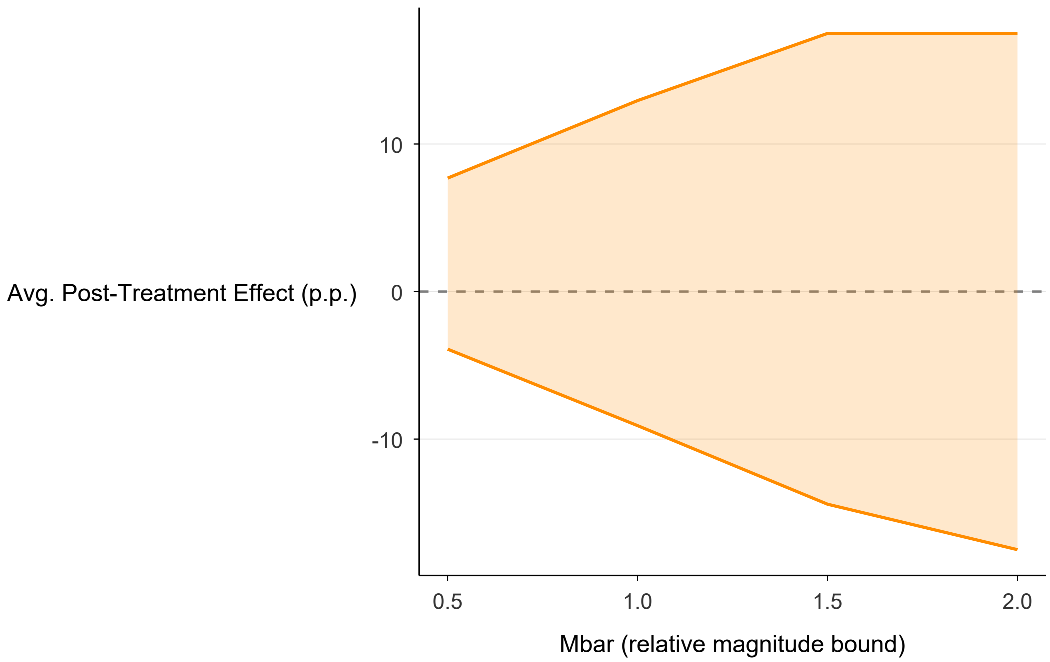

Figure A23: HonestDiD: Relative Magnitudes Sensitivity (ΔRM)

Notes: Robust confidence sets from Rambachan and Roth (2023) under relative magnitudes restrictions (ΔRM) for the primary (unduplicated) BW gap effect. The x-axis ($\bar{M}$) parameterizes the maximum ratio of the post-treatment violation of parallel trends to the largest pre-treatment violation. The robust CI includes zero at every $\bar{M} \in [0.5, 2]$, so the primary null is robust to relative-magnitude relaxations.

Matched-sample pre-treatment balance. Sun-Abraham's identifying assumption is parallel trends, not parallel levels; the raw BW suspension gap is on average higher in treated than in never-treated districts before SMD adoption. Table A12 provides a complementary balance diagnostic from the PanelMatch matched-DiD estimator (Section 4.4): the average difference in BW gap between each treated district and its matched controls across the four pre-treatment years available to PanelMatch ($t = -4$ to $t = -1$), alongside the contemporaneous raw difference between treated districts and the unmatched never-treated control pool. Matching reduces the pre-period imbalance by 38% at $t = -4$ and by 90% at $t = -1$, leaving a matched difference at $t = -1$ that is effectively zero (+0.20 pp; SE 0.50). This diagnostic indicates that the primary null is not masked or manufactured by pre-treatment level imbalance between treated and never-treated districts.

Table A12: Pre-Treatment BW Gap Balance: Matched vs. Raw

Lag	Raw difference	(SE)	Matched difference	(SE)
$t = -4$	+2.56	(2.30)	+1.58	(1.07)
$t = -3$	+1.61	(1.75)	+0.69	(0.56)
$t = -2$	+1.12	(1.49)	+0.43	(0.49)
$t = -1$	+2.10	(1.43)	+0.20	(0.50)

Notes: "Raw" difference is the mean BW suspension gap among treated districts minus the contemporaneous mean among never-treated districts. "Matched" difference is the mean BW gap among treated districts minus the weighted mean among their PanelMatch matched controls (lag = 4, Mahalanobis refinement, size match = 5). Standard errors are cross-set standard deviations divided by $\sqrt{n}$ where n is the number of contributing treated sets at each lag.

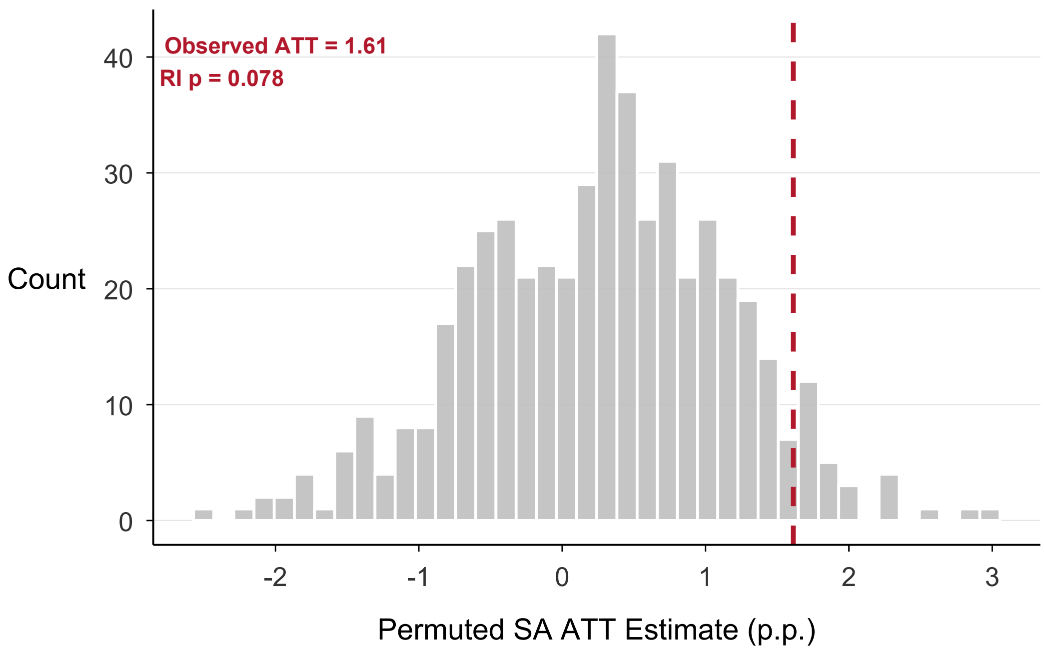

Figure A24: Randomization Inference: Incident BW Gap

Notes: Distribution of placebo Sun-Abraham ATT estimates for the *incident* Black-White gap from 500 random permutations of treatment timing. The vertical red line marks the observed estimate (+1.61). The randomization-inference p-value is .078, consistent with the marginal, non-robust incident signal; the primary unduplicated gap is a null and is not the object of this test.

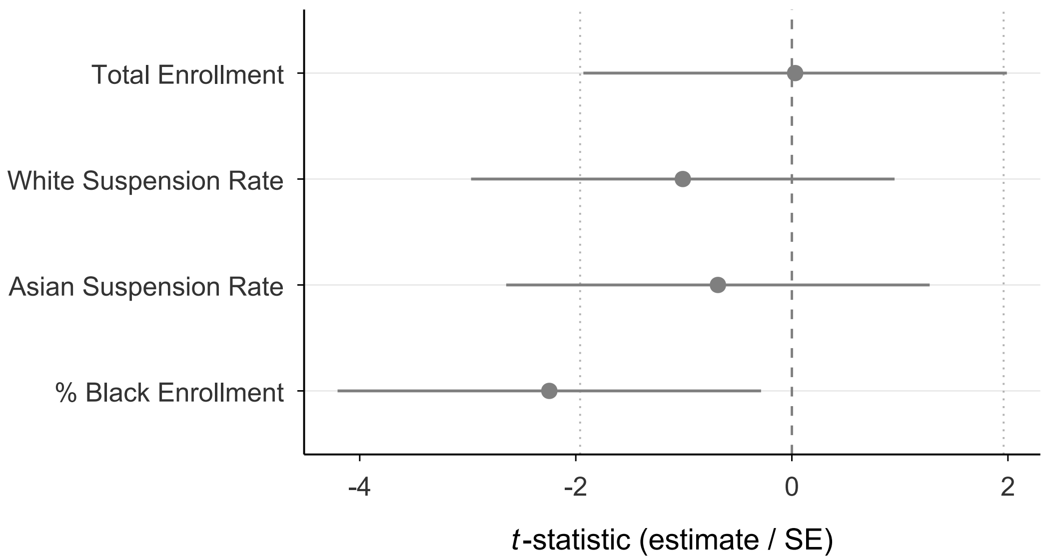

Figure A25: Placebo Outcome Tests

Notes: Sun-Abraham ATT t-statistics (estimate/SE) for placebo outcomes that should not be affected by SMD adoption. Dotted lines mark the ± 1.96 critical values. Most estimates are small and statistically insignificant; the percent-Black-enrollment placebo is a significant exception (negative under both TWFE and SA), addressed with a composition-stable robustness check in Section 4.4.

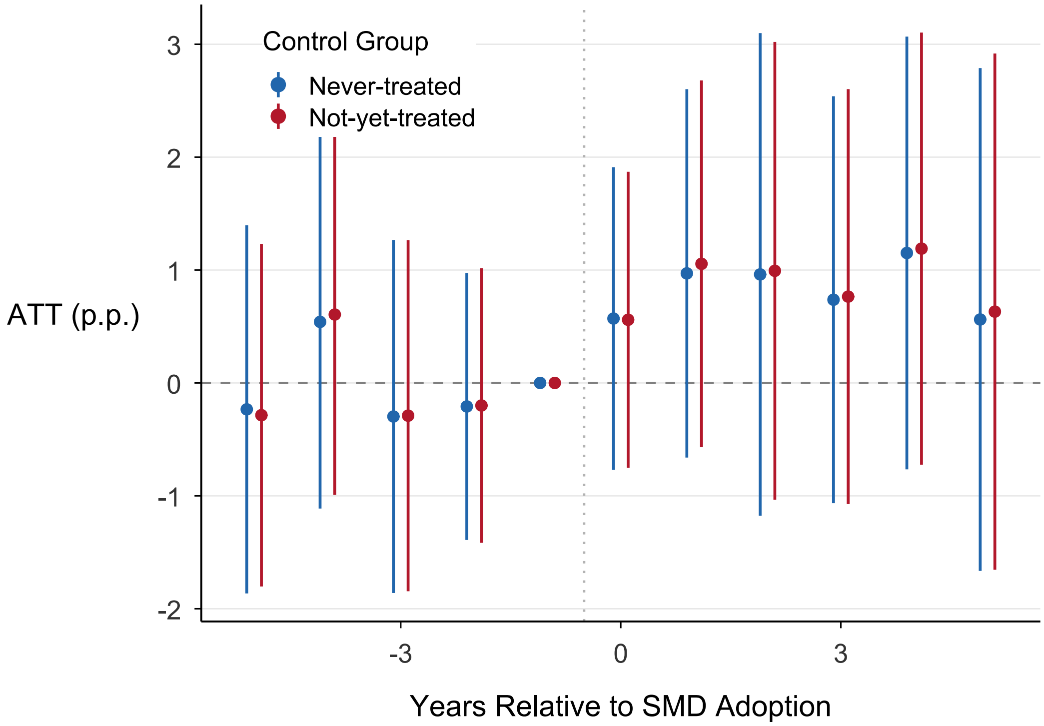

Figure A26: Callaway-Sant'Anna Event Study: Not-Yet-Treated Comparison

Notes: Callaway-Sant'Anna event study using only not-yet-treated units as the comparison group, rather than never-treated units. Results are similar to the baseline CS specification.

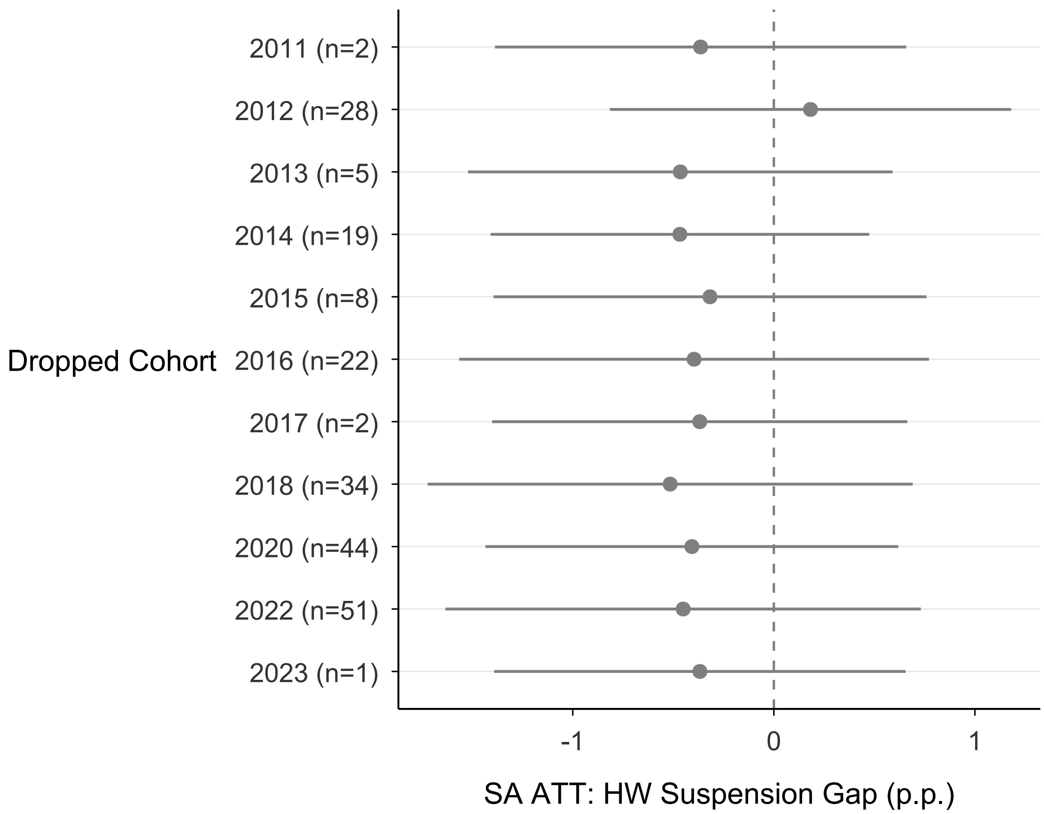

Figure A27: Leave-One-Cohort-Out: HW Suspension Gap

Notes: SA ATT estimates for the HW suspension gap when each treatment cohort is excluded. All estimates remain close to zero and statistically insignificant, confirming the null result for the HW gap.

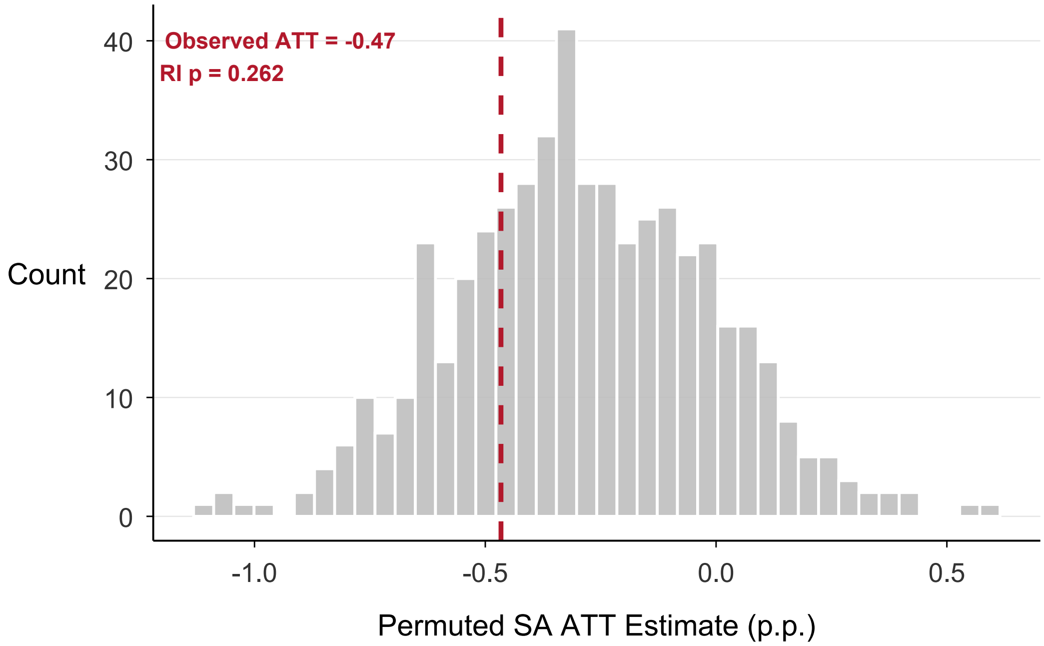

Figure A28: Randomization Inference: HW Gap

Notes: Distribution of placebo SA ATT estimates for the HW gap from 500 random permutations of treatment timing. The observed estimate falls well within the placebo distribution ($p = .73$), consistent with a null effect.

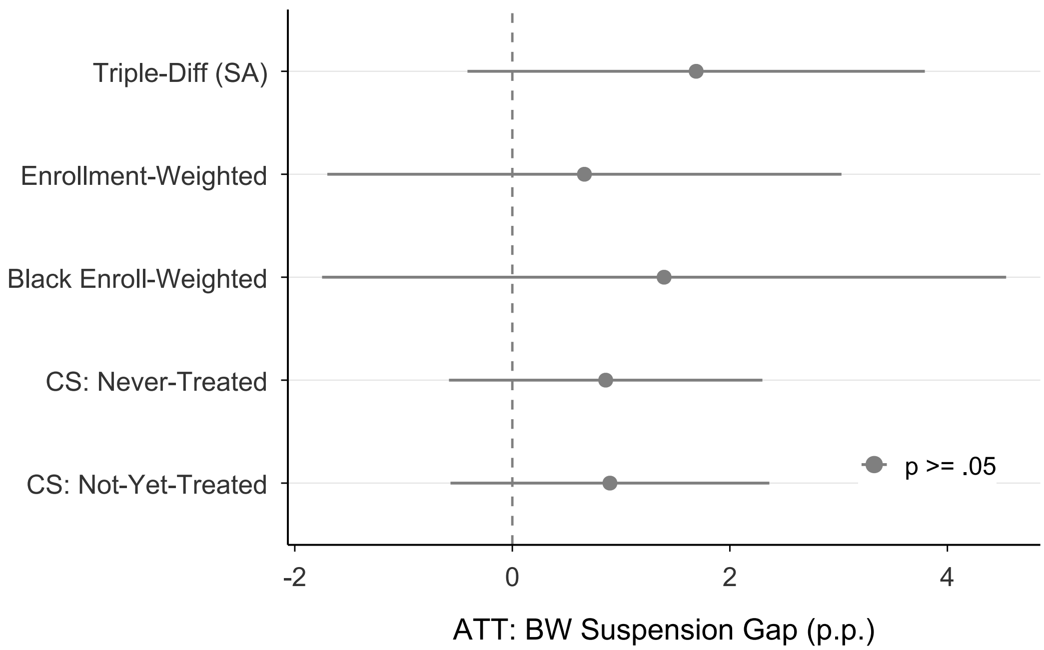

Figure A29: Additional Robustness Checks: Summary

Notes: Summary of additional robustness checks including triple-difference, enrollment weighting, CS not-yet-treated control group, and placebo outcomes. The primary unduplicated BW-gap null is stable across specifications; the incident-based gap is positive but fragile (significant only under Sun-Abraham at baseline).

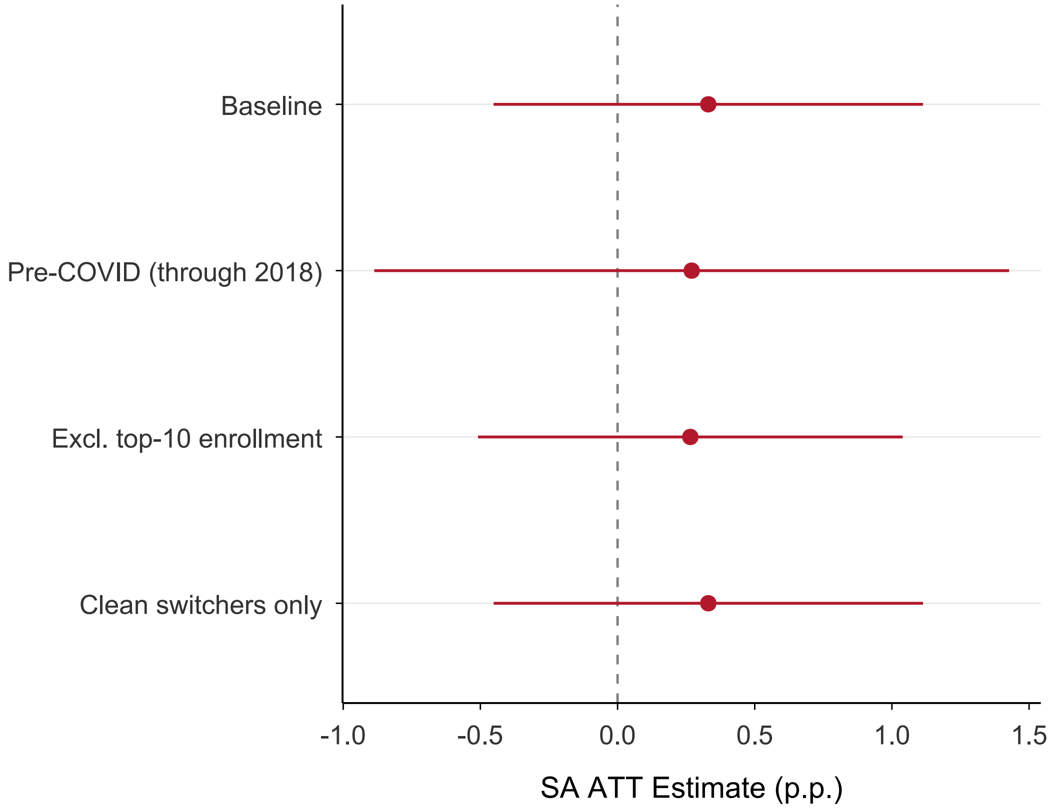

Figure A30: Robustness: SA BW Gap Across All Specifications

Notes: Sun-Abraham ATT estimates for the BW suspension gap across all robustness specifications including the baseline, pre-COVID only, excluding top-10% enrollment, minimum enrollment 20, clean switchers, and log/asinh transformations. Points are estimates; whiskers show 95% confidence intervals.

E Heterogeneity

This appendix presents the full subgroup heterogeneity results summarized in Section 4. These are secondary, exploratory patterns on the *incident* BW gap; the primary distinct-student gap is null on average. Figure A31 reports SA ATT estimates for the incident BW suspension gap across subgroups defined by district size, minority student share, Black student share, Hispanic student share, and pre-treatment gap level. The incident BW-gap subgroup effect is larger in districts with low Black student shares (+6.0 pp vs. +1.1 pp in high-share districts), consistent with small-denominator rate volatility (districts with few Black students exhibit larger swings in race-specific suspension rates) and threshold-driven sample selection (the minimum-enrollment criterion selects for districts where small fluctuations produce large rate changes). Figure A32 shows the corresponding HW gap estimates, which are uniformly null. Figure A33 disaggregates by board composition gains after SMD adoption, showing that districts gaining a predicted-Black board member exhibit a similar (and individually insignificant) incident BW-gap pattern to those gaining only Hispanic members.

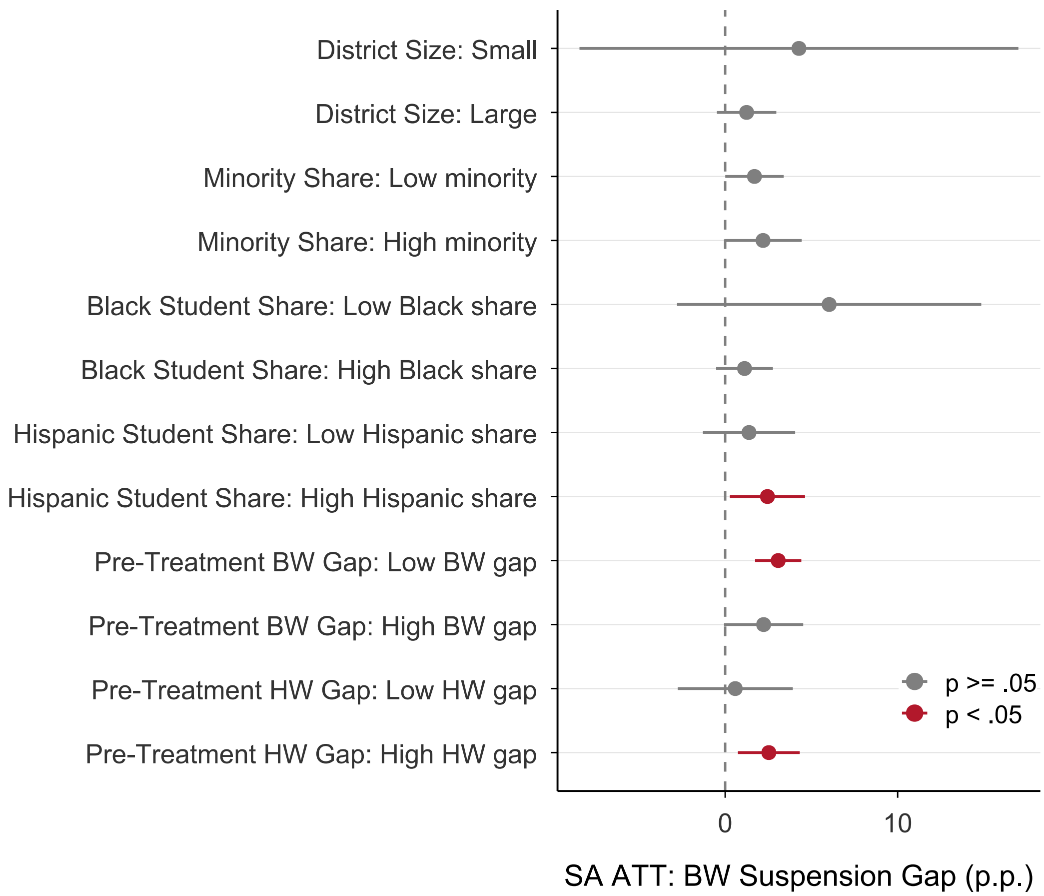

Figure A31: Subgroup Heterogeneity: BW Suspension Gap

Notes: Forest plot of Sun-Abraham ATT estimates for the *incident* BW suspension gap across subgroups defined by district characteristics (exploratory; the primary distinct-student gap is null on average). Points are estimates; whiskers show 95% confidence intervals. The incident BW-gap effect is larger in districts with low Black student shares (+6.0 vs. +1.1 in high-share districts) and smaller in large districts.

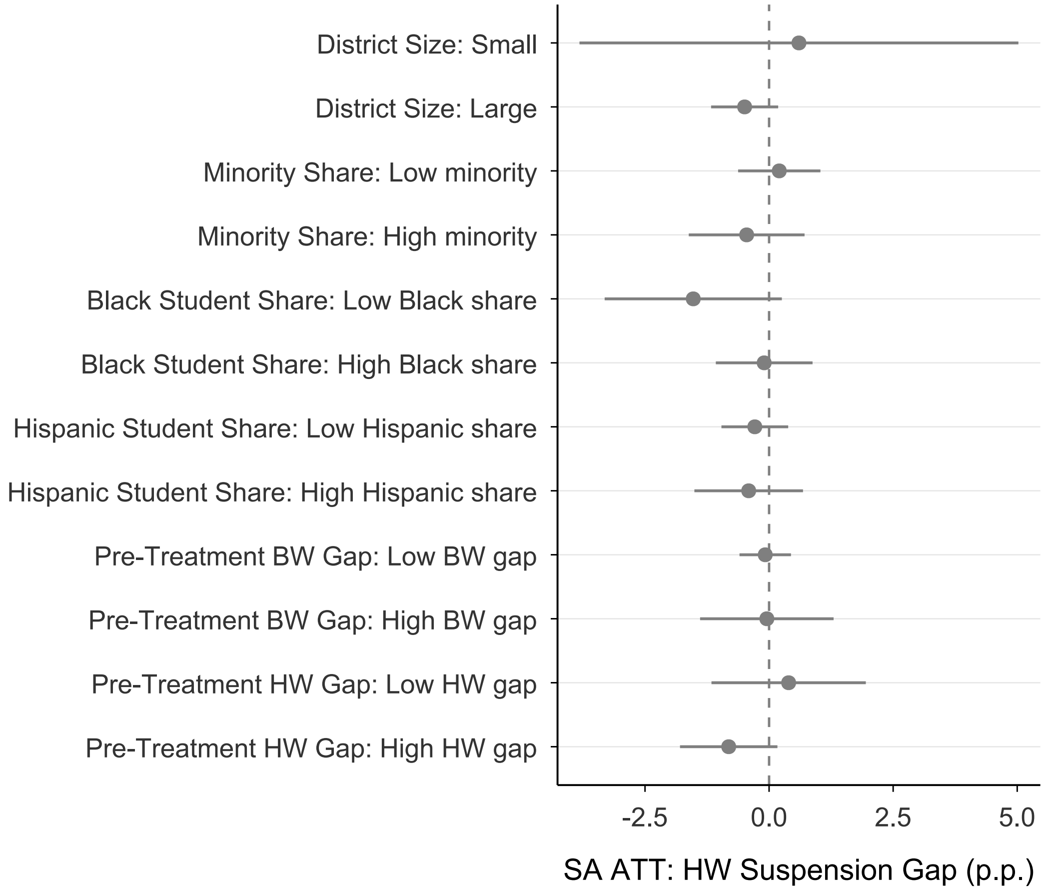

Figure A32: Subgroup Heterogeneity: HW Suspension Gap

Notes: Forest plot of Sun-Abraham ATT estimates for the HW suspension gap across subgroups. All estimates are close to zero and insignificant.

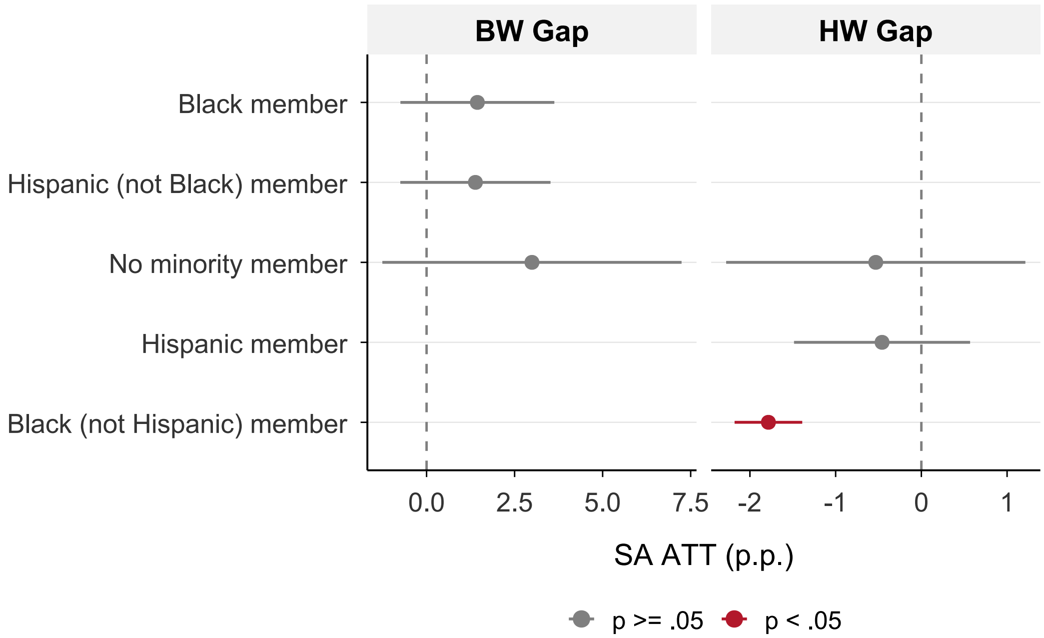

Figure A33: Heterogeneity by Board Racial Composition Gains

Notes: Sun-Abraham ATT estimates on the *incident* BW gap, separately for districts that gained a predicted-Black board member after SMD adoption (18 districts), gained a predicted-Hispanic member but not Black (135 districts), and gained no minority member (42 districts), among treated districts with valid BW gap data and post-treatment board elections. The three subgroup ATTs are similar and individually insignificant (+1.44, +1.39, +2.99), consistent with binary subgrouping being a blunt instrument for the race-match hypothesis; the continuous cumulative-Black-share interaction in Table 4 provides the substantively relevant (exploratory) moderation evidence.

F Teacher Diversity Analysis

This appendix examines whether SMD adoption affects the racial composition of the teaching workforce and whether teacher diversity moderates the discipline gap effects. I construct a harmonized teacher demographics panel from CDE's StaffDemo files (individual-level, 2011–2017) and Staff by Race/Ethnicity files (aggregate, 2021–2024), covering 652 of 721 analysis districts. Figure A34 plots minority teacher share trends for treated and control districts. Figure A35 presents event study estimates for the teacher diversity first stage, showing gradual increases in minority teacher shares after SMD adoption. Figure A36 summarizes the first-stage estimates: SMD adoption increases minority teacher share by approximately

2.34 percentage points (SA, $p = .002$), driven by Hispanic teachers. Figure A37 reports the key *exploratory* moderation pattern: the incident BW discipline-gap effect is smaller in districts with higher Black teacher share (-85 pp contemporaneously, $p < .001$; -49 pp with a one-year lag, $p = .019$), while Hispanic teacher share shows no such association. Because teacher composition is realized after treatment and is itself an outcome of the reform, this is a correlational pattern consistent with—but not establishing—the race-match hypothesis, parallel to the board-composition results in Section 5.

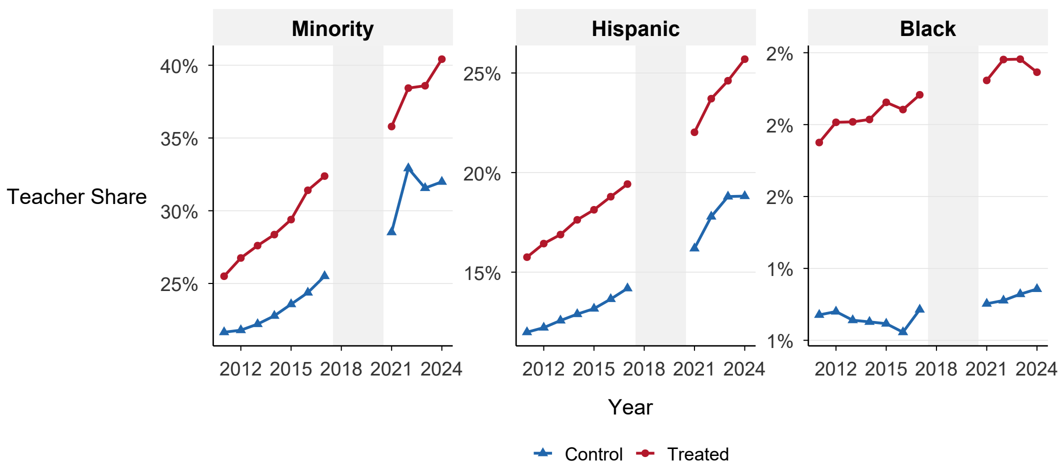

Figure A34: Teacher Workforce Diversity Trends

Notes: Mean teacher racial shares over time, separately for treated and control districts. Both groups show gradual increases in minority teacher shares, with treated districts experiencing slightly faster growth after SMD adoption.

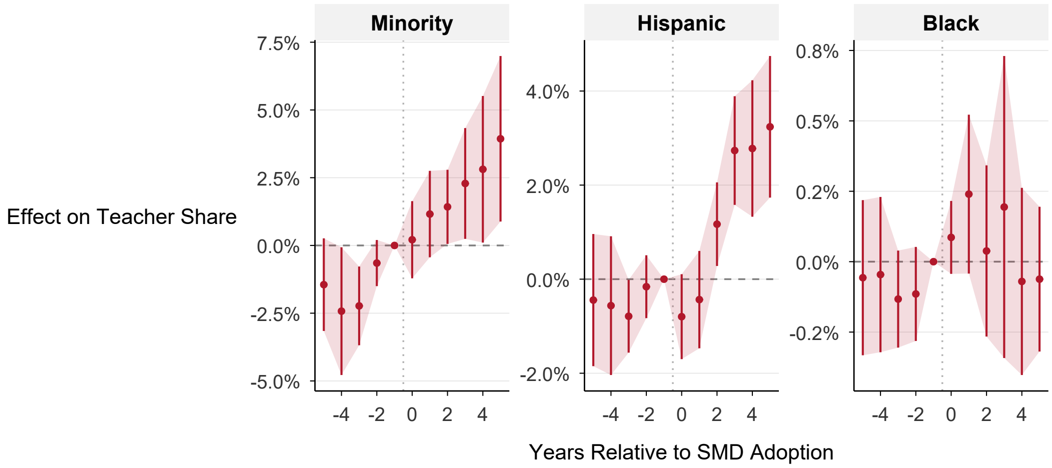

Figure A35: Event Study: Teacher Diversity First Stage

Notes: Event study estimates for the effect of SMD adoption on minority, Hispanic, and Black teacher shares. Effects emerge gradually after treatment, consistent with hiring being a slow-moving process influenced by board priorities.

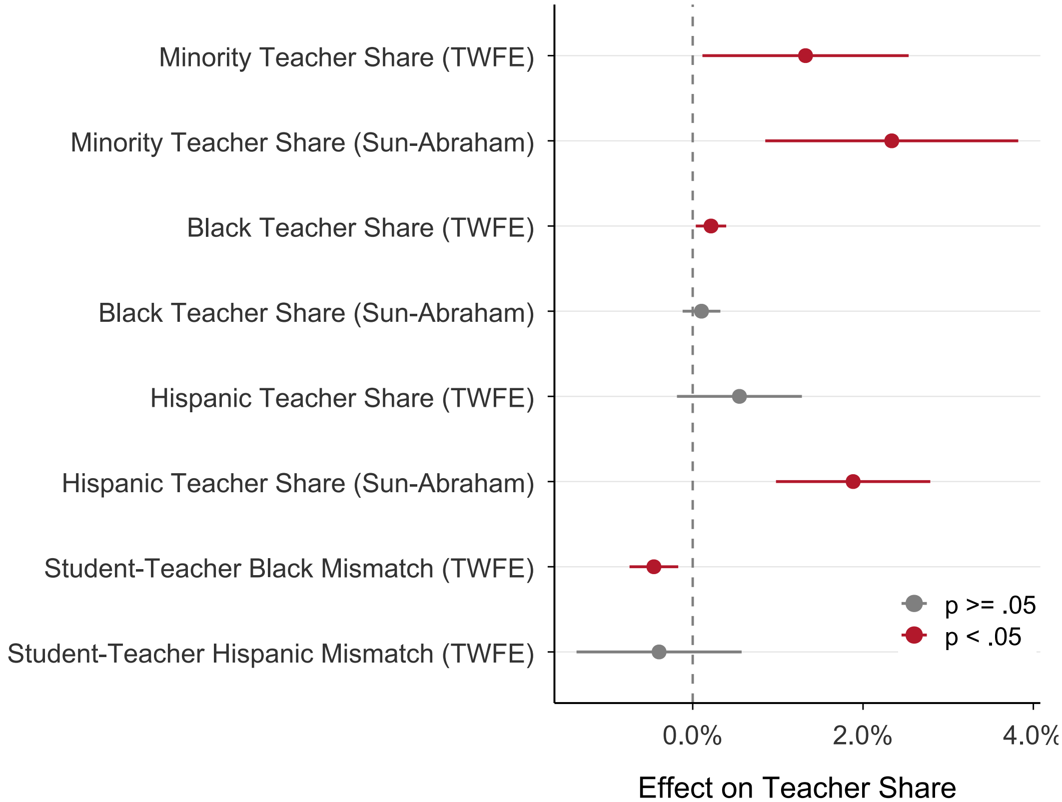

Figure A36: Teacher Diversity First Stage: Summary

Notes: Forest plot of TWFE and Sun-Abraham estimates for the effect of SMD adoption on various teacher diversity measures. Minority teacher share: SA +2.34 pp ($p = .002$); Hispanic teacher share: SA +1.88 pp ($p < .001$); Black teacher share: SA +0.10 pp ($p = .36$, null; TWFE +0.21 pp, $p = .019$).

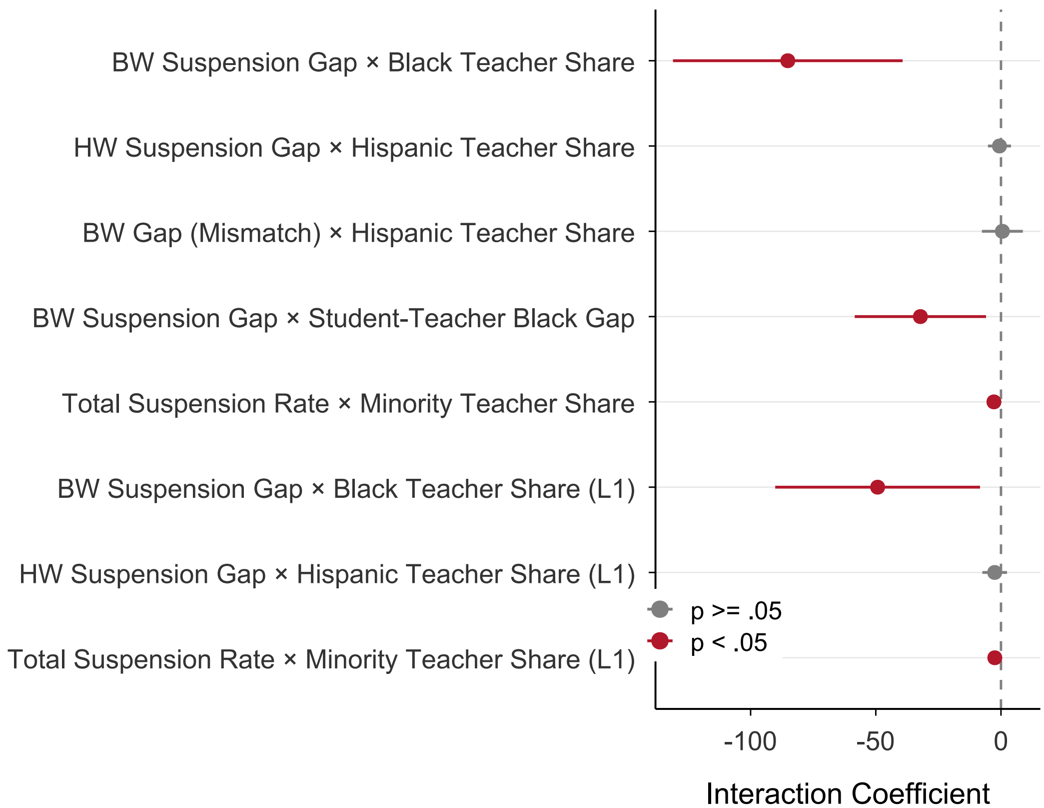

Figure A37: Teacher Diversity as Moderator of Discipline Effects

Notes: Exploratory interaction coefficients from models interacting treat_post with teacher racial shares; the outcome is the incident BW gap. The incident BW-gap effect is smaller where the Black teacher share is higher (-85.15 contemporaneously, $p < .001$; -49.27 with a one-year lag, $p = .019$), while Hispanic teacher share shows no such association. Because teacher composition is post-treatment, these associations are correlational, not causally identified. The student-teacher Black mismatch gap shows a similar association (-32.20, $p = .016$).

G Spending Channel Details

This appendix investigates whether per-pupil spending changes after SMD adoption and whether such changes moderate the discipline gap effects. Figure A38 presents the log-spending event study; because the log estimate is positive while the level estimate is negative, I treat the spending first stage as ambiguous rather than as a clean replication of Fischer (2023). Figure A39 plots raw spending trends for treated and control districts. Figure A40 reports the spending moderation test: the interaction of SMD treatment with log per-pupil spending is not significant for either the BW or HW gap, indicating that spending changes do not drive the discipline effects. Figure A41 provides context by comparing the BISG-predicted racial composition of elected board members before and after SMD adoption.

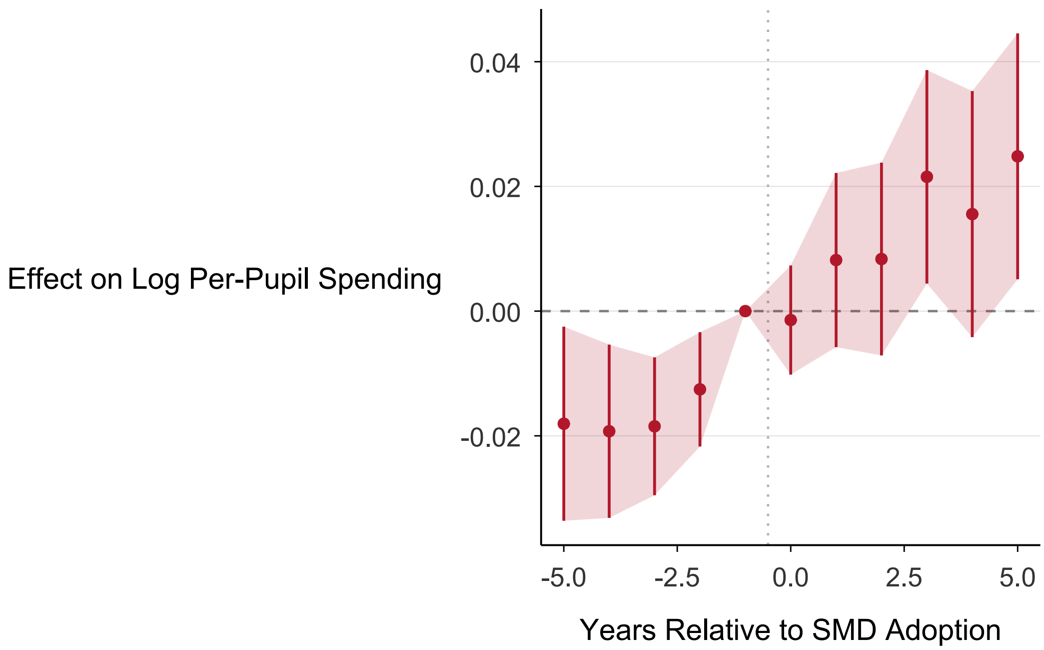

Figure A38: Event Study: Per-Pupil Spending

Notes: Event study estimates for the effect of SMD adoption on log per-pupil expenditure. The log estimate is positive, but the level estimate reported in Section 5.4 is negative, so the spending first stage is ambiguous. Spending does not moderate the BW discipline gap effect.

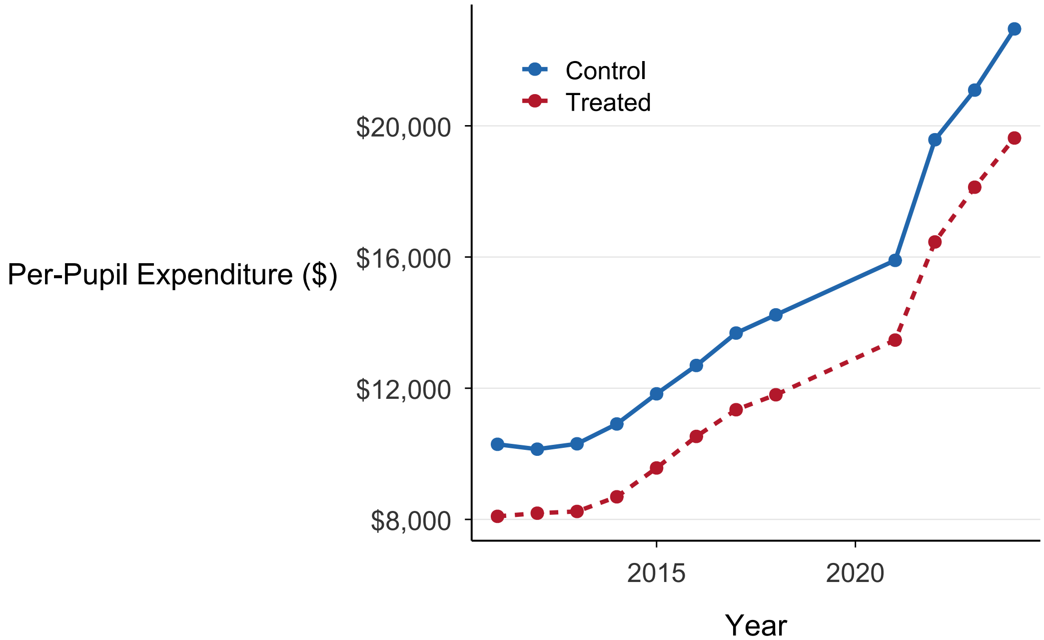

Figure A39: Per-Pupil Spending Trends by Treatment Group

Notes: Mean per-pupil expenditure over time for treated and control districts.

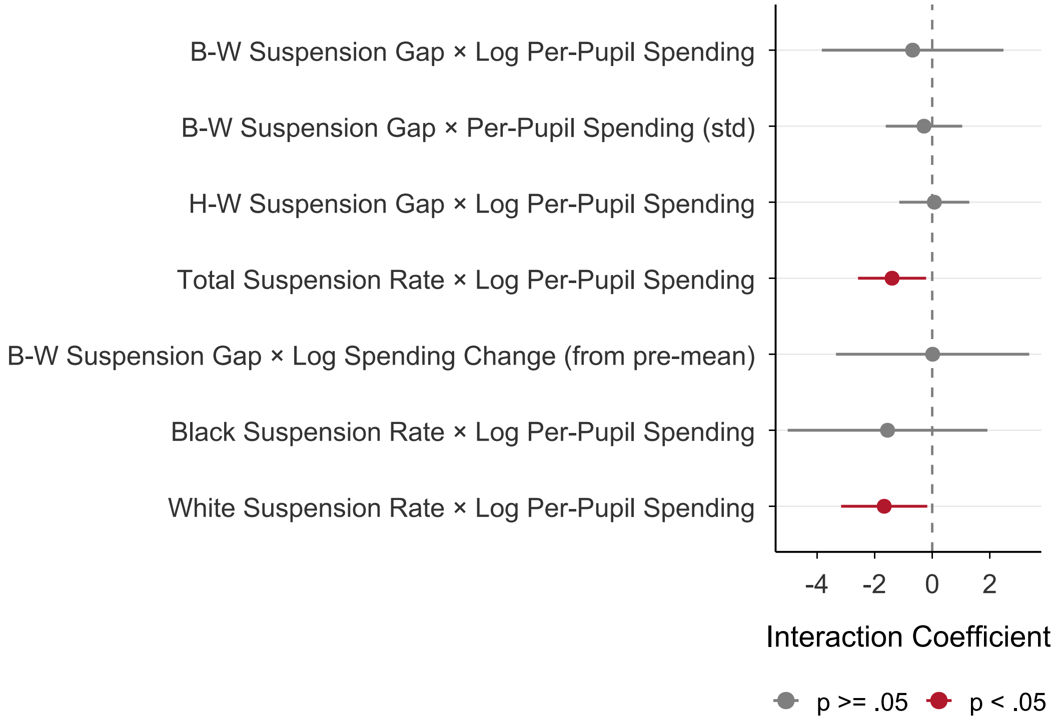

Figure A40: Spending as Moderator of Discipline Effects

Notes: Interaction of SMD treatment with log per-pupil spending on the BW and HW suspension gaps. Neither interaction is significant, indicating that spending does not moderate the discipline gap effects.

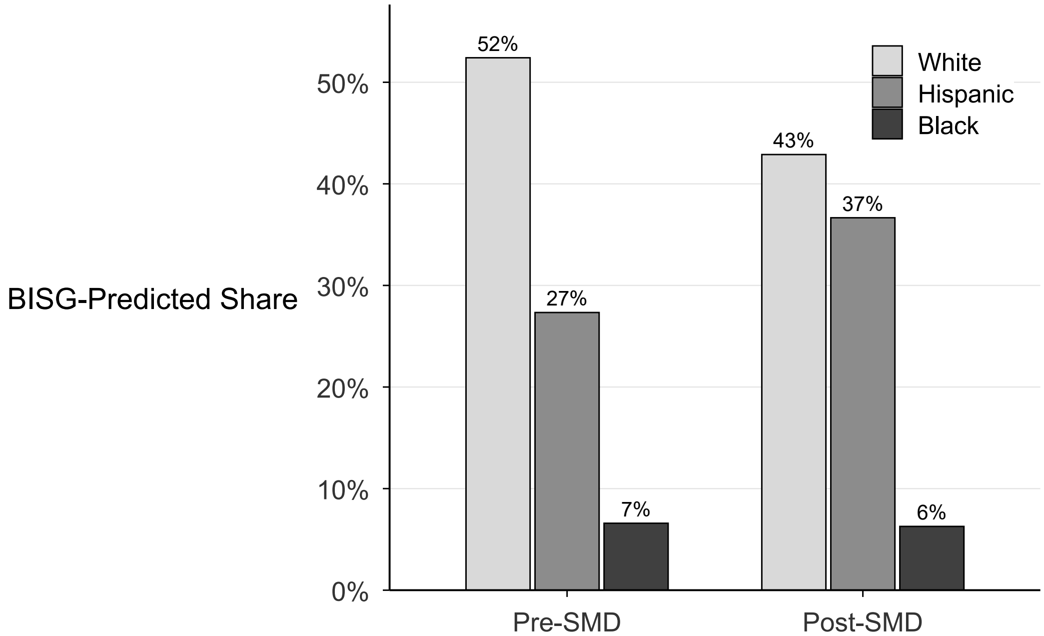

Figure A41: Board Composition: Pre vs. Post SMD Adoption

Notes: Distribution of BISG-predicted racial composition of elected board members before and after SMD adoption, across all treated districts. Hispanic share increases substantially; Black share remains essentially unchanged.

H Achievement Gap Outcomes (SEDA 6.0)

As a domain-specific falsification check, I test whether SMD adoption affects standardized test score gaps using the Stanford Education Data Archive (SEDA) version 6.0 (Reardon et al. 2025). Unlike the placebo outcomes in Section 4.4, achievement is a domain where governance changes could plausibly have effects—if SMD elections improve broad district quality, we would expect improvements in academic outcomes as well. A null on achievement is therefore informative but not guaranteed, and this analysis is properly framed as a domain-specific falsification check rather than a placebo test.

Data. SEDA 6.0 provides district-level, grade-level, subject-specific mean test scores and racial achievement gaps in cohort-scale standard deviation (SD) units, covering 2009–2018 (with 2014 missing due to the CST-to-SBAC transition in California). I aggregate to the district-year level by weighting across grades and subjects by assessment count, then merge with the analysis panel via a CDE–NCES district crosswalk. The resulting sample covers approximately 579 districts (193 treated, 386 control) observed over 8 years, yielding approximately 3,900 district-year observations for overall achievement, 2,400 for the W-H gap, and 1,000 for the W-B gap (which requires sufficient Black enrollment). Treatment does not predict SEDA data availability ($p = .57$), confirming that the merged sample is not selected on treatment.

Results. Table A13 reports the main estimates. Across all three estimators (TWFE, Sun-Abraham, Callaway-Sant'Anna), the effects on achievement gaps are null. The preferred SA estimates are +0.015 SD (SE = 0.013; $p = .26$) for the W-H gap and +0.004 SD (SE = 0.016; $p = .78$) for the W-B gap. The CS estimates, which use a balanced subsample restricted to in-range cohorts, are similarly null. TWFE shows a significant negative effect on overall achievement levels (−0.053 SD; $p < .001$); this attenuates under SA but remains marginally significant (−0.026 SD; $p = .04$), so a modest decline in overall achievement cannot be ruled out—though the achievement *gaps*, the relevant falsification outcome, are null.

Table A14 provides the key falsification test: a side-by-side comparison of achievement and discipline effects on the same SEDA-available district×year cells. Achievement gaps are uniformly null, and the discipline BW suspension gap is positive but insignificant on this restricted sample (+1.53 pp; SE = 1.49; $p = .31$), consistent with the full-sample null. The discipline HW gap is significant and negative on this subsample (−1.84 pp; SE = 0.52; $p < .001$), in contrast to the full-sample null—likely reflecting the SEDA-available districts being larger and more urban, with higher Hispanic enrollment shares and longer post-treatment exposure windows; the full-sample null remains the more conservative and appropriate estimate for the HW gap. The analysis provides no evidence that SMD reform narrowed academic disparities on the cells where discipline gaps are at most marginal.

Figure A42 presents achievement gap trends and event studies. Pre-treatment trends are parallel for both the W-H and W-B gaps, and the event study coefficients are flat both before and after treatment. Pre-trend Wald tests are clean for both gaps (W-H $p = .93$; W-B $p = .74$), though the overall-achievement *level* shows a pre-trend ($p = .002$).

Figure A43 displays the achievement vs. discipline comparison as a forest plot, showing null achievement-gap estimates alongside mixed discipline estimates: an insignificant BW discipline gap, a significant negative HW discipline gap, and a marginal total-rate decline.

Table A15 reports robustness checks including an alternative reference period ($t = -2$), grade-level panel with district×grade fixed effects, separate estimates for math and reading, and SEDA SE-based filtering. All gap estimates remain null across specifications. The final rows report minimum detectable effects (MDEs) at 80% power: 0.037 SD for the W-H gap and 0.046 SD for the W-B gap, both below the discipline-proportional benchmark (0.055 SD), confirming adequate statistical power.

Table A13: Domain-Specific Falsification: Achievement Gap Outcomes (SEDA 6.0)

Outcome / Estimator	ATT	(SE)	N
W-H Achievement Gap			
TWFE	0.010	(0.010)	2,404
Sun-Abraham	0.015	(0.013)	2,404
CS	0.022*	(0.013)	1,575
W-B Achievement Gap			
TWFE	0.012	(0.016)	998
Sun-Abraham	0.004	(0.016)	998
CS	−0.006	(0.020)	546
TWFE [BW-eligible]	0.012	(0.016)	998
Sun-Abraham [BW-eligible]	0.004	(0.016)	998
CS [BW-eligible]	−0.006	(0.020)	546
Overall Achievement			
TWFE	−0.053***	(0.012)	3,904
Sun-Abraham	−0.026**	(0.012)	3,904
CS	−0.043***	(0.013)	3,052

Notes: ATT estimates in cohort-scale standard deviation units (SEDA 6.0). SEDA data cover 2009–2018 (no 2014 due to CST→SBAC transition). CS estimates use a balanced panel restricted to cohorts within the data year range (≥3 districts per cohort). BW-eligible restricts to districts with mean pre-treatment Black enrollment ≥20. SEs clustered at district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A14: Achievement vs. Discipline: Same-Sample Comparison (Sun-Abraham)

Outcome	ATT	(SE)	N
Panel A: Achievement (SD units)			
W-H Achievement Gap	0.015	(0.013)	2,404
Overall Achievement	-0.026^{**}	(0.012)	3,904
W-B Achievement Gap	0.004	(0.016)	998
Panel B: Discipline (p.p., same sample)			
B-W Suspension Gap	1.53	(1.49)	2,500
H-W Suspension Gap	-1.84^{***}	(0.52)	3,771
Total Suspension Rate	-0.89^{*}	(0.52)	3,872

Notes: Sun-Abraham ATT estimates. Panel A: SEDA 6.0 achievement outcomes (SD units). Panel B: CDE suspension outcomes (percentage points) estimated on the same district×year cells where SEDA data are available. SEs clustered at district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

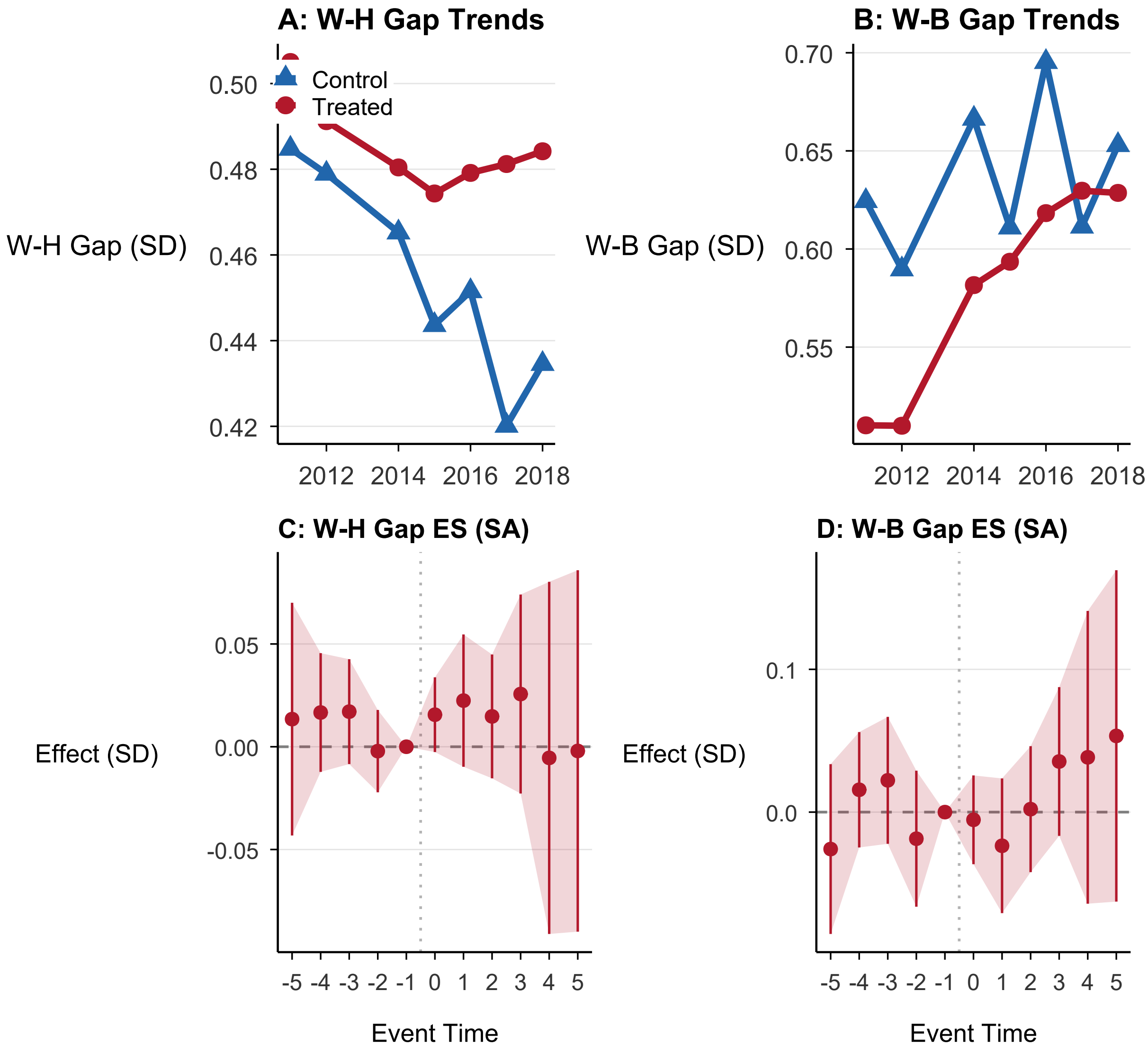

Figure A42: Achievement Gap Trends and Event Studies (SEDA 6.0)

Notes: Panels A–B show mean W-H and W-B standardized test score gaps (cohort-scale SD units) from SEDA 6.0, 2009–2018. Panels C–D show Sun-Abraham event study coefficients with 95% CIs; the omitted period is $t-1$. All post-treatment coefficients are near zero, consistent with no effect of SMD adoption on achievement gaps.

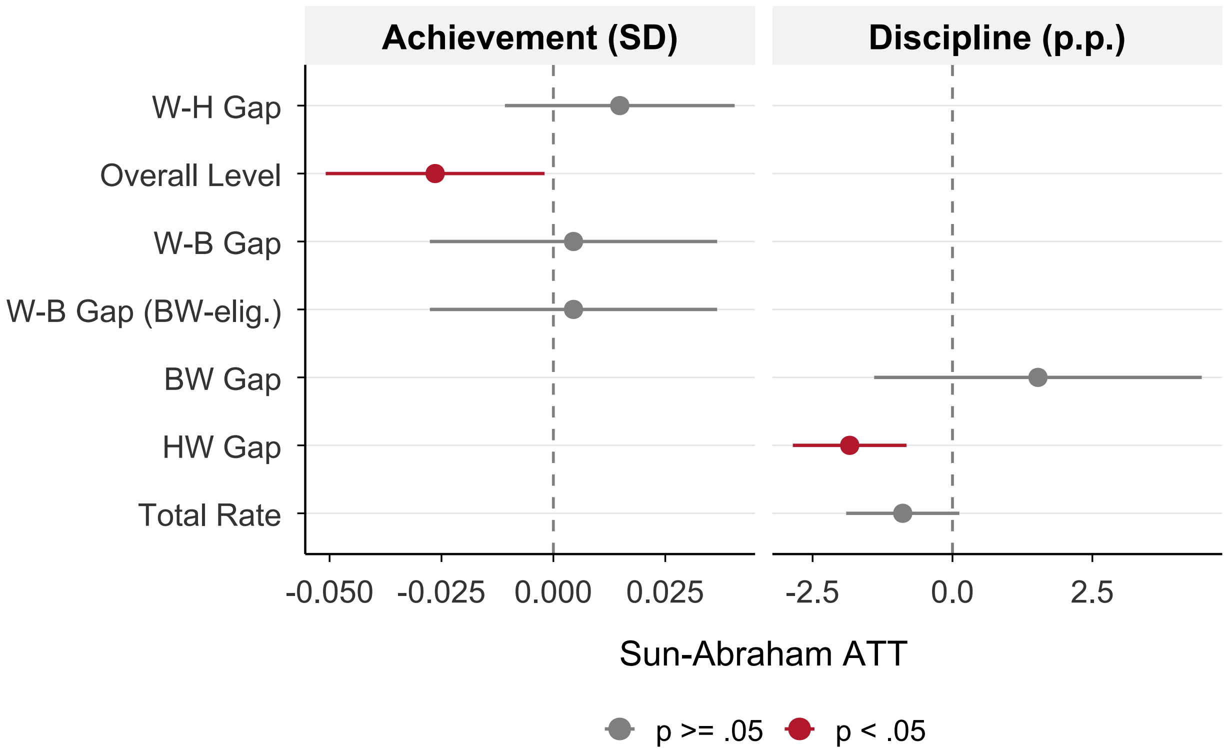

Figure A43: Achievement vs. Discipline: Same-Sample Comparison

Notes: Sun-Abraham ATT estimates with 95% confidence intervals. Left panel: SEDA achievement outcomes (SD units). Right panel: CDE discipline outcomes (percentage points) restricted to the SEDA-eligible district-year sample used in the left panel; estimates therefore differ from the full-sample main-text values. Achievement-gap effects are null; discipline estimates are mixed, with the BW suspension gap insignificant, the HW suspension gap negative and significant, and the total rate marginally negative.

Table A15: Achievement Gap Robustness and Power

	W-H Gap		W-B Gap	
Specification	ATT	(SE)	ATT	(SE)
SA ref.p=-2	0.003	(0.011)	0.004	(0.022)
Grade-level TWFE	0.005	(0.008)	—	—
Grade-level SA	0.003	(0.009)	—	—
Math TWFE	0.010	(0.012)	0.010	(0.018)
Math SA	0.015	(0.016)	0.008	(0.019)
Reading TWFE	0.013	(0.010)	0.011	(0.016)
Reading SA	0.017	(0.012)	0.006	(0.019)
SE<=p75 TWFE	0.006	(0.008)	−0.004	(0.012)
SE<=p75 SA	0.013	(0.011)	−0.011	(0.015)
MDE (80% power)	0.037		0.046	
Disc. proportional benchmark	0.057		0.056	

Notes: Robustness checks for achievement gap outcomes. All estimates in cohort-scale SD units. Grade-level specifications use district×grade fixed effects. SE$\leq$p75 drops observations with SEDA standard errors above the 75th percentile. MDE at 80% power = 2.802× SE. Discipline proportional benchmark = 0.225 × outcome SD. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.